

SOLIDWORKS API (NET VBA C#) — Examples (33/50): Get_Sketch_Contours_Example_VBNET-Ex Get_Template_Sheet_Metal_Feature_Data_Ex

SOLIDWORKS API Documentation (converted from CHM)

Get Sketch Contours Example (VB.NET)

This example shows how to get the sketch contours in a model document.

```
'-----  
' Preconditions:  
' 1. Open a part that contains a Sketch1 feature.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the number of sketch contours in Sketch1 and whether  
'    they are open or closed.  
' 2. Gets the number of sketch segments in Sketch1 and their type.  
' 3. Gets the number of edges Sketch1.  
' 4. Examine the Immediate window.  
'-----  
  
Imports SolidWorks.Interop.sldworks  
  
Imports SolidWorks.Interop.swconst  
  
Imports System.Runtime.InteropServices  
  
Imports System  
  
Imports System.Diagnostics  
  
Partial Class SolidWorksMacro  
  
    Sub main()  
  
        Dim myModel As ModelDoc2  
  
        Dim myPart As PartDoc  
  
        Dim SelMgr As SelectionMgr  
  
        Dim mySelectData As SelectData  
  
        Dim myFeature As Feature  
  
        Dim mySketch As Sketch  
  
        Dim contourCount As Integer  
  
        Dim vSkContours As Object  
  
        Dim skContour As SketchContour  
  
        Dim vEdges As Object, myEdge As Edge  
  
        Dim skSegCount As Integer  
  
        Dim vSkSegments As Object  
  
        Dim skSegment As SketchSegment  
  
        Dim skSegType As Integer, skSegTypesString As String = Nothing  
  
        Dim closed As Boolean, closedString As String  
  
        Dim i As Integer, j As Integer, k As Integer  
  
        Dim boolstatus As Boolean  
  
  
        myModel = swApp.ActiveDoc
```



```

        skSegTypesString = skSegTypesString & "line"

    Case swSketchSegments_e.swSketchARC
        skSegTypesString = skSegTypesString & "arc"

    Case swSketchSegments_e.swSketchELLIPSE
        skSegTypesString = skSegTypesString & "ellipse"

    Case swSketchSegments_e.swSketchPARABOLA
        skSegTypesString = skSegTypesString & "parabola"

    Case swSketchSegments_e.swSketchSPLINE
        skSegTypesString = skSegTypesString & "spline"

    Case swSketchSegments_e.swSketchTEXT
        skSegTypesString = skSegTypesString & "text"

    Case Nothing
        skSegTypesString = skSegTypesString & "unknown"

    End Select

End If

If j = UBound(vSkSegments) Then
    skSegTypesString = skSegTypesString & ")"
Else
    skSegTypesString = skSegTypesString & ","
End If

Next j

Debug.Print("    Sketch
segment count = " & skSegCount & " " & skSegTypesString)

vEdges = skContour.GetEdges()

If IsNothing(vEdges) Then
    Debug.Print("    No
edges.")
Else
    For k = LBound(vEdges) To UBound(vEdges)
        myEdge = vEdges(k)

        If Not myEdge Is Nothing Then
            Debug.Print("    Edge
" & k & ": ")

        End If
    Next k

```

```
End If

boolstatus = skContour.Select2(False, mySelectData)
If boolstatus = 0 Then
    Debug.Print("    Selection
of contour failed.")
End If

Stop

End If

Next i

End If

End Sub

Public swApp As SldWorks

End Class
```

Get Sketch Entities Example (VBA)

This example shows how to get all of the sketch entities (arcs, ellipses, lines, parabolas, splines, and text) in a sketch.

```
'-----  
' 1. Open a part that has a Sketch1 feature that  
'    contains sketch text, sketch lines, sketch parabolas,  
'    sketch arcs, sketch ellipses, and sketch splines.  
' 2. Open the Immediate window.  
'  
' Postconditions: Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
  
Sub main()  
  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSketch As SldWorks.Sketch  
    Dim swSketchMgr As SldWorks.SketchManager  
    Dim vSketchTextSegments As Variant  
    Dim vSketchTextSegment As Variant  
    Dim swSketchText As SldWorks.SketchText  
    Dim vSketchSegments As Variant  
    Dim vSketchSegment As Variant  
    Dim swSketchSegment As SldWorks.SketchSegment  
    Dim bValue As Boolean  
  
    ' Connect to SOLIDWORKS Design  
    Set swApp = Application.SldWorks  
  
    ' Get active document  
    Set swModel = swApp.ActiveDoc  
  
    ' Get SketchManager  
    Set swSketchMgr = swModel.SketchManager  
  
    ' Clear the selection  
    swModel.ClearSelection2 True  
  
    ' Select the sketch  
    bValue = swModel.Extension.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)  
  
    ' Activate the sketch  
    swSketchMgr.InsertSketch False  
  
    ' Get the sketch itself  
    Set swSketch = swModel.GetActiveSketch2  
  
    ' Loop over all SketchText instances  
  
    ' Get SketchText  
    vSketchTextSegments = swSketch.GetSketchTextSegments  
  
    If (Not IsEmpty(vSketchTextSegments)) Then  
        For Each vSketchTextSegment In vSketchTextSegments  
            Set swSketchText = vSketchTextSegment  
            Debug.Print "Text = " & swSketchText.Text  
        Next vSketchTextSegment  
    End If  
  
    Debug.Print  
  
    ' Loop over all sketch segments  
  
    ' Get sketch segments  
    vSketchSegments = swSketch.GetSketchSegments  
  
    If (Not IsEmpty(vSketchSegments)) Then  
        For Each vSketchSegment In vSketchSegments  
            Set swSketchSegment = vSketchSegment  
  
            ' Determine actual type of sketch segment  
            Select Case (swSketchSegment.GetType)
```

```

' SketchText "is-a" sketch segment
Case swSketchSegments_e.swSketchText

    ' Cast sketch segment to sketch text
    Set swSketchText = swSketchSegment

    ' Get sketch text specific property
    Debug.Print "Sketch text = " & swSketchText.Text

Case swSketchSegments_e.swSketchLINE
    Debug.Print "Sketch line"
Case swSketchSegments_e.swSketchELLIPSE
    Debug.Print "Sketch ellipse"
Case swSketchSegments_e.swSketchARC
    Debug.Print "Sketch arc"
Case swSketchSegments_e.swSketchPARABOLA
    Debug.Print "Sketch parabola"
Case swSketchSegments_e.swSketchSPLINE
    Debug.Print "Sketch spline"
Case Else
    Debug.Print "<unknown>"
End Select

Next vSketchSegment

End If

swSketchMgr.InsertSketch True

End Sub

```

Get_Sketch_Point_s_View_Example_VB (Topic)

SOLIDWORKS API Help

Get Sketch Point's View Example (VBA)

This example shows how to get the name of the drawing view for the selected sketch point.

```
'-----  
' Preconditions:  
' 1. Open a drawing in which a sketch point  
'    exists in a drawing view.  
' 2. Select the sketch point.  
' 3. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the name of the sketch point feature  
'    and the name of the drawing view in which  
'    the sketch point exists.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swSkPt As SldWorks.SketchPoint  
    Dim swSketch As SldWorks.Sketch  
    Dim swSketchFeat As SldWorks.Feature  
    Dim swDraw As SldWorks.DrawingDoc  
    Dim swView As SldWorks.View  
    Dim swViewSketch As SldWorks.Sketch  
    Dim bRet As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel  
    Set swSelMgr = swModel.SelectionManager  
    Set swSkPt = swSelMgr.GetSelectedObject5(1)  
    Set swSketch = swSkPt.GetSketch  
    Set swSketchFeat = swSketch  
  
    Debug.Print "File = " & swModel.GetPathName  
    Debug.Print " " & swSketchFeat.Name  
  
    Set swView = swDraw.GetFirstView  
    Do While Not Nothing Is swView  
        Set swViewSketch = swView.GetSketch  
        If swSketch Is swViewSketch Then  
            Debug.Print "    -- > " & swView.Name  
            Exit Do  
        End If  
        Set swView = swView.GetNextView  
    Loop  
  
End Sub
```

Get Sketch Points and Sketch Point IDs Example (VBA)

This example shows how to get all of the sketch points in the selected sketch and their IDs.

```
'-----
' Preconditions:
' 1. Open a part and select a sketch.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Gets the feature name of the selected sketch and
'    each sketch point ID and their coordinates.
' 2. Examine the Immediate window.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swSelData As SldWorks.SelectData
    Dim swSketch As SldWorks.Sketch
    Dim swSketchPt As SldWorks.SketchPoint
    Dim swFeat As SldWorks.Feature
    Dim vSketchUserPt As Variant
    Dim vSketchPt As Variant
    Dim vSketchPtID As Variant
    Dim i As Long
    Dim bRet As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swSelData = swSelMgr.CreateSelectData
    Set swFeat = swSelMgr.GetSelectedObject6(1, 0)
    Set swSketch = swFeat.GetSpecificFeature2

    Debug.Print "Feature name = " + swFeat.Name

    vSketchUserPt = swSketch.GetUserPoints2
    For i = 0 To UBound(vSketchUserPt) / 3 - 1
        ' Coordinates in sketch space
        Debug.Print "  User point (" & i & ") = (" & vSketchUserPt(3 * i + 0) * 1000# & ", " & vSketchUserPt(3 * i + 1)
* 1000# & ", " & vSketchUserPt(3 * i + 2) * 1000# & ") mm"
    Next i

    vSketchPt = swSketch.GetSketchPoints2
    For i = 0 To UBound(vSketchPt)
        Set swSketchPt = vSketchPt(i)

        ' Will fail if point has been deleted
        ' or if point references another entity;
        ' for example, point in sketch or vertex
        bRet = swSketchPt.Select4(True, swSelData)

        ' Dump persistent identifier
        vSketchPtID = swSketchPt.GetID
        Debug.Print "  Sketch point ID (" & i & ") = [" & vSketchPtID(0) & ", " & vSketchPtID(1) & "]"
        Debug.Print "    Sketch point coordinates (" & swSketchPt.X * 1000# & ", " & swSketchPt.Y * 1000# & ", " &
swSketchPt.Z * 1000# & ") mm"
    Next i

End Sub
```

Get Sketch Points Example (VBA)

This example shows how to loop through a selected sketch that has one or more sketch points and get the x and y coordinates of every sketch point.

```
'-----  
' Preconditions:  
' 1. Open a model containing a sketch with one or more  
'    sketch points.  
' 2. Select the sketch.  
' 3. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the value for each sketch point in the sketch.  
' 2. Gets the x and y coordinates of each sketch  
'    point in the sketch.  
' 3. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swSketch As SldWorks.Sketch  
    Dim swFeat As SldWorks.Feature  
    Dim swSketchPoint As SldWorks.SketchPoint  
    Dim sketchPointArray As Variant  
    Dim i As Long  
    Dim xValue As Double  
    Dim yValue As Double  
  
    Set swApp = CreateObject("SldWorks.Application")  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swSketch = swFeat.GetSpecificFeature2  
    Debug.Print "FeatName = " + swFeat.Name  
    Debug.Print " "  
  
    sketchPointArray = swSketch.GetSketchPoints2  
    For i = 0 To UBound(sketchPointArray)  
        ' Get value returned by ISketchPoint::GetCoords  
        Set swSketchPoint = sketchPointArray(i)  
        Debug.Print "Value returned by ISketchPoint::GetCoords: " & swSketchPoint.GetCoords  
  
        ' Get the x & y coordinates  
        xValue = sketchPointArray(i).X  
        yValue = sketchPointArray(i).Y  
        Debug.Print "Sketch point coordinates: "  
        Debug.Print "  x: " & xValue  
        Debug.Print "  y: " & yValue  
        Debug.Print " "  
        ' Do something useful with the data  
    Next i  
  
End Sub
```

Get and Add Sketch Points in Hole Wizard Feature Example (C#)

This example shows how to get and add the sketch points in a Hole Wizard feature.

```
//-----
// Preconditions:
// 1. Verify that the specified part document template exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Creates a new part document.
// 2. Creates Boss-Extrude1 and #0-80 Tapped Hole1 features.
// 3. Selects #8-80 Tapped Hole1; i.e., the Hole Wizard feature.
// 4. Gets the number of sketch points in the Hole Wizard feature.
// 5. Adds two sketch points to the Hole Wizard feature; thus, adds two more
//    holes to the Hole Wizard feature.
// 6. Examine the Immediate window and graphics area.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SketchManager swSketchMgr = default(SketchManager);
            FeatureManager swFeatureMgr = default(FeatureManager);
            Feature swFeature = default(Feature);
            SelectionMgr swSelectionMgr = default(SelectionMgr);
            WizardHoleFeatureData2 swWizardHoleFeatureData = default(WizardHoleFeatureData2);
            SketchPoint swSketchPoint = default(SketchPoint);
            object sketchLines = null;
            bool status = false;
            int count = 0;
            object[] points = null;

            //Create part with Hole Wizard feature
            swModel = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SolidWorks\\SolidWorks 2015\\templates\\Part.prtdot
", 0, 0, 0);
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            status = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstToRectEntity, (
            int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, false);
            status = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstLineDiagonalTyp
            e, (int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, true);
            swSketchMgr = (SketchManager)swModel.SketchManager;
            sketchLines = (object[])swSketchMgr.CreateCornerRectangle(0, 0, 0, 0.0968848174375125, -0.0708129015764598,
            0);

            swModel.ClearSelection2(true);
            status = swModelDocExt.SelectByID2("Line2", "SKETCHSEGMENT", 0, 0, 0, false, 0, null, 0);
            status = swModelDocExt.SelectByID2("Line1", "SKETCHSEGMENT", 0, 0, 0, true, 0, null, 0);
            status = swModelDocExt.SelectByID2("Line4", "SKETCHSEGMENT", 0, 0, 0, true, 0, null, 0);
            status = swModelDocExt.SelectByID2("Line3", "SKETCHSEGMENT", 0, 0, 0, true, 0, null, 0);
            swFeatureMgr = (FeatureManager)swModel.FeatureManager;
            swFeature = (Feature)swFeatureMgr.FeatureExtrusion2(true, false, false, 0, 0, 0.0254, 0.00254, false, false,
            false,
            false, 0.0174532925199433, 0.0174532925199433, false, false, false, false, true, true, true,
            0, 0, false);
            swSelectionMgr = (SelectionMgr)swModel.SelectionManager;
            swSelectionMgr.EnableContourSelection = false;
            status = swModelDocExt.SelectByID2("", "FACE", 0.0471052662929878, -0.0336338467782298, 0.0253999999998769,
            false, 0, null, 0);
            swFeature = (Feature)swFeatureMgr.HoleWizard5(4, 0, 27, "#0-80", 1, 0.00119126, 0.0254, 0.020066, 0, 0,
            0, 0, 0, 0, 1, 0, 0, -1, -1, -1,
            "2B", false, true, true, true, false);
            swModel.ViewZoomtofit2();
            status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
            swFeature = (Feature)swSelectionMgr.GetSelectedObject6(1, -1);
            swModel.ClearSelection2(true);
            swWizardHoleFeatureData = (WizardHoleFeatureData2)swFeature.GetDefinition();
            count = swWizardHoleFeatureData.GetSketchPointCount();
            Debug.Print(" Number of sketch points in original Hole Wizard Feature = " + count);
            points = (object[])swWizardHoleFeatureData.GetSketchPoints();
            foreach (object point in points)
            {
                swSketchPoint = (SketchPoint)point;
                swSketchPoint.Select4(false, null);
            }
        }
    }
}
```

```

    }
    status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, null);
    swModel.ClearSelection2(true);

    //Select sketch point of Hole Wizard feature
    //and add two sketch points to Hole Wizard feature
    status = swModelDocExt.SelectByID2("Sketch3", "SKETCH", 0, 0, 0, false, 0, null, 0);
    swSketchMgr.AddToDB = true;
    swModel.EditSketch();
    swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.04, 0);
    swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.02, 0);
    swSketchMgr.InsertSketch(true);
    swSketchMgr.AddToDB = false;
    swModel.ForceRebuild3(true);

    //Get number of sketch points in modified Hole Wizard feature
    status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
    swFeature = (Feature)swSelectionMgr.GetSelectedObject6(1, -1);
    swModel.ClearSelection2(true);
    swWizardHoleFeatureData = (WizardHoleFeatureData2)swFeature.GetDefinition();
    count = swWizardHoleFeatureData.GetSketchPointCount();
    Debug.Print(" Number of sketch points in modified Hole Wizard Feature = " + count);
    points = (object[])swWizardHoleFeatureData.GetSketchPoints();
    foreach (object point in points)
    {
        swSketchPoint = (SketchPoint)point;
        swSketchPoint.Select4(false, null);
    }
    status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, null);
    swModel.ClearSelection2(true);
}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
}
}

```

Get and Add Sketch Points in Hole Wizard Feature Example (VBA)

This example shows how to get and add the sketch points in a Hole Wizard feature.

```
'-----
' Preconditions:
' 1. Verify that the specified part document template exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Creates a new part document.
' 2. Creates Boss-Extrude1 and #0-80 Tapped Hole1 features.
' 3. Selects #8-80 Tapped Hole1; i.e., the Hole Wizard feature.
' 4. Gets the number of sketch points in the Hole Wizard feature.
' 5. Adds two sketch points to the Hole Wizard feature; thus, adds two more
'    holes to the Hole Wizard feature.
' 6. Examine the Immediate window and graphics area.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swFeature As SldWorks.Feature
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swWizardHoleFeatureData As SldWorks.WizardHoleFeatureData2
Dim swSketchPoint As SldWorks.SketchPoint
Dim sketchLines As Variant
Dim status As Boolean
Dim count As Long
Dim points As Variant
Dim point As Variant

Sub main()

    Set swApp = Application.SldWorks

    'Create part with Hole Wizard feature
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2015\templates\Part.prtdot", 0, 0, 0)
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
    swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
    swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
    Set swSketchMgr = swModel.SketchManager
    sketchLines = swSketchMgr.CreateCornerRectangle(0, 0, 0, 9.68848174375125E-02, -7.08129015764598E-02, 0)
    swModel.ClearSelection2 True
    status = swModelDocExt.SelectByID2("Line2", "SKETCHSEGMENT", 0, 0, 0, False, 0, Nothing, 0)
    status = swModelDocExt.SelectByID2("Line1", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID2("Line4", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID2("Line3", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
    Set swFeatureMgr = swModel.FeatureManager
    Set swFeature = swFeatureMgr.FeatureExtrusion2(True, False, False, 0, 0, 0.0254, 0.00254, False, False, False,
    False, 1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)
    Set swSelectionMgr = swModel.SelectionManager
    swSelectionMgr.EnableContourSelection = False
    status = swModelDocExt.SelectByID2("", "FACE", 4.71052662929878E-02, -3.36338467782298E-02, 2.53999999998769E-02,
    False, 0, Nothing, 0)
    Set swFeature = swFeatureMgr.HoleWizard5(4, 0, 27, "#0-80", 1, 0.00119126, 0.0254, 0.020066, 0, 0, 0, 0, 0, 1,
    0, 0, -1, -1, -1, "2B", False, True, True, True, True, False)
    swModel.ViewZoomtofit2

    status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
    Set swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
    swModel.ClearSelection2 True
    Set swWizardHoleFeatureData = swFeature.GetDefinition
    count = swWizardHoleFeatureData.GetSketchPointCount
    Debug.Print " Number of sketch points in original Hole Wizard Feature = " & count
    points = swWizardHoleFeatureData.GetSketchPoints
    For Each point In points
        Set swSketchPoint = point
    
```

```

        swSketchPoint.Select4 False, Nothing
Next
status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, Nothing)
swModel.ClearSelection2 True

'Select sketch point of Hole Wizard feature
'and add two sketch points to Hole Wizard feature
status = swModelDocExt.SelectByID2("Sketch3", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)
swSketchMgr.AddToDB = True
swModel.EditsSketch
Set swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.04, 0)
Set swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.02, 0)
swSketchMgr.InsertSketch True
swSketchMgr.AddToDB = False
swModel.ForceRebuild3 True

'Get number of sketch points in modified Hole Wizard feature
status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
swModel.ClearSelection2 True
Set swWizardHoleFeatureData = swFeature.GetDefinition
count = swWizardHoleFeatureData.GetSketchPointCount
Debug.Print " Number of sketch points in modified Hole Wizard Feature = " & count
points = swWizardHoleFeatureData.GetSketchPoints
For Each point In points
    Set swSketchPoint = point
    swSketchPoint.Select4 False, Nothing
Next
status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, Nothing)
swModel.ClearSelection2 True

```

End Sub

Get_Sketch_Points_in_Wizard_Hole_Example_VBNET (Topic)

SOLIDWORKS API Help

Get and Add Sketch Points in Hole Wizard Feature Example (VB.NET)

This example shows how to get and add the sketch points in a Hole Wizard feature.

```
-----
' Preconditions:
' 1. Verify that the specified part document template exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Creates a new part document.
' 2. Creates Boss-Extrude1 and #0-80 Tapped Hole1 features.
' 3. Selects #8-80 Tapped Hole1; i.e., the Hole Wizard feature.
' 4. Gets the number of sketch points in the Hole Wizard feature.
' 5. Adds two sketch points to the Hole Wizard feature; thus, adds two more
'    holes to the Hole Wizard feature.
' 6. Examine the Immediate window and graphics area.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swSketchMgr As SketchManager
        Dim swFeatureMgr As FeatureManager
        Dim swFeature As Feature
        Dim swSelectionMgr As SelectionMgr
        Dim swWizardHoleFeatureData As WizardHoleFeatureData2
        Dim swSketchPoint As SketchPoint
        Dim sketchLines As Object
        Dim status As Boolean
        Dim count As Integer
        Dim points() As Object
        Dim point As Object

        'Create part with Hole Wizard feature
        swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2015\templates\Part.prt", 0, 0, 0)
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity, swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
        status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
        swSketchMgr = swModel.SketchManager
        sketchLines = swSketchMgr.CreateCornerRectangle(0, 0, 0, 0.0968848174375125, -0.0708129015764598, 0)
        swModel.ClearSelection2(True)
        status = swModelDocExt.SelectByID2("Line2", "SKETCHSEGMENT", 0, 0, 0, False, 0, Nothing, 0)
        status = swModelDocExt.SelectByID2("Line1", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
        status = swModelDocExt.SelectByID2("Line4", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
        status = swModelDocExt.SelectByID2("Line3", "SKETCHSEGMENT", 0, 0, 0, True, 0, Nothing, 0)
        swFeatureMgr = swModel.FeatureManager
        swFeature = swFeatureMgr.FeatureExtrusion2(True, False, False, 0, 0, 0.0254, 0.00254, False, False, False, False, 0.0174532925199433, 0.0174532925199433, False, False, False, False, True, True, True, 0, 0, False)
        swSelectionMgr = swModel.SelectionManager
        swSelectionMgr.EnableContourSelection = False
        status = swModelDocExt.SelectByID2("", "FACE", 0.0471052662929878, -0.0336338467782298, 0.02539999999998769, False, 0, Nothing, 0)
        swFeature = swFeatureMgr.HoleWizard5(4, 0, 27, "#0-80", 1, 0.00119126, 0.0254, 0.020066, 0, 0, 0, 0, 0, 0, 1, 0, 0, -1, -1, -1, "2B", False, True, True, True, True, False)
        swModel.ViewZoomToFit2()
        status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
        swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
        swModel.ClearSelection2(True)
        swWizardHoleFeatureData = swFeature.GetDefinition
        count = swWizardHoleFeatureData.GetSketchPointCount
        Debug.Print(" Number of sketch points in original Hole Wizard Feature = " & count)
        points = swWizardHoleFeatureData.GetSketchPoints
        For Each point In points
            swSketchPoint = point
            swSketchPoint.Select4(False, Nothing)
        Next
        status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, Nothing)
        swModel.ClearSelection2(True)

        'Select sketch point of Hole Wizard feature
        'and add two sketch points to Hole Wizard feature
        status = swModelDocExt.SelectByID2("Sketch3", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)
        swSketchMgr.AddToDB = True
    End Sub

End Class
```

```

swModel.EditSketch()
swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.04, 0)
swSketchPoint = swSketchMgr.CreatePoint(0.01, -0.02, 0)
swSketchMgr.InsertSketch(True)
swSketchMgr.AddToDB = False
swModel.ForceRebuild3(True)

'Get number of sketch points in modified Hole Wizard feature
status = swModelDocExt.SelectByID2("#0-80 Tapped Hole1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
swModel.ClearSelection2(True)
swWizardHoleFeatureData = swFeature.GetDefinition
count = swWizardHoleFeatureData.GetSketchPointCount
Debug.Print(" Number of sketch points in modified Hole Wizard Feature = " & count)
points = swWizardHoleFeatureData.GetSketchPoints
For Each point In points
    swSketchPoint = point
    swSketchPoint.Select4(False, Nothing)
Next
status = swFeature.ModifyDefinition(swWizardHoleFeatureData, swModel, Nothing)
swModel.ClearSelection2(True)

```

End Sub

```

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

```

End Class

Get Sketch Point's Selection Mark Example (C#)

This example shows how to get the selection mark of a sketch point.

```
//-----
// Preconditions:
// 1. Open a part containing a sketch point.
// 2. Select the sketch containing the sketch point.
// 3. Open the Immediate window.
//
// Postconditions:
// 1. Gets the selection mark of the sketch point.
// 2. Examine the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace GetReferencePointCSharp.csproj
{
    public partial class SolidWorksMacro
    {

        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            SelectionMgr swSelMgr = default(SelectionMgr);
            Feature swFeat = default(Feature);
            Sketch swSketch = default(Sketch);
            object[] sketchPoints = null;
            SketchPoint swSketchPt = default(SketchPoint);
            SelectData swSelData = default(SelectData);
            bool status = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swSketch = (Sketch)swFeat.GetSpecificFeature2();
            sketchPoints = (object[])swSketch.GetSketchPoints2();
            swSketchPt = (SketchPoint)sketchPoints[0];

            // Get selection mark of sketch point
            swSelData = (SelectData)swSelMgr.CreateSelectData();
            status = swSketchPt.Select4(true, swSelData);
            Debug.Print("Selection mark of sketch point: " + swSelData.Mark);

        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Sketch Point's Selection Mark Example (VBA)

This example shows how to get the selection mark of a sketch point.

```
'-----  
' Preconditions:  
' 1. Open a part containing a sketch point.  
' 2. Select the sketch containing the sketch point.  
' 3. Open the Immediate window.  
,  
' Postconditions:  
' 1. Gets the selection mark of the sketch point.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swFeat As SldWorks.Feature  
    Dim swSketch As Sketch  
    Dim sketchPoints As Variant  
    Dim swSketchPoint As SldWorks.SketchPoint  
    Dim swSelData As SldWorks.SelectData  
    Dim status As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swSketch = swFeat.GetSpecificFeature2  
    sketchPoints = swSketch.GetSketchPoints2  
    Set swSketchPoint = sketchPoints(0)  
  
    ' Get selection mark of sketch point  
    Set swSelMgr = swModel.SelectionManager  
    Set swSelData = swSelMgr.CreateSelectData  
    status = swSketchPoint.Select4(True, swSelData)  
    Debug.Print "Selection mark of sketch point: " & swSelData.Mark  
  
End Sub
```

Get Sketch Point's Selection Mark Example (VB.NET)

This example shows how to get the selection mark of a sketch point.

```
'-----  
' Preconditions:  
' 1. Open a part containing a sketch point.  
' 2. Select the sketch containing the sketch point.  
' 3. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the selection mark of the sketch point.  
' 2. Examine the Immediate window.  
'-----  
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System.Runtime.InteropServices  
Imports System  
Imports System.Diagnostics  
  
Partial Class SolidWorksMacro  
  
    Public Sub Main()  
  
        Dim swModel As ModelDoc2  
        Dim swSelMgr As SelectionMgr  
        Dim swFeat As Feature  
        Dim swSketch As Sketch  
        Dim sketchPoints As Object  
        Dim swSketchPt As SketchPoint  
        Dim swSelData As SelectData  
        Dim status As Boolean  
  
        swModel = swApp.ActiveDoc  
        swSelMgr = swModel.SelectionManager  
        swFeat = swSelMgr.GetSelectedObject6(1, -1)  
        swSketch = swFeat.GetSpecificFeature2  
        sketchPoints = swSketch.GetSketchPoints2  
        swSketchPt = sketchPoints(0)  
  
        ' Get selection mark of sketch point  
        swSelMgr = swModel.SelectionManager  
        swSelData = swSelMgr.CreateSelectData  
        status = swSketchPt.Select4(True, swSelData)  
        Debug.Print("Selection mark of sketch point: " & swSelData.Mark)  
  
    End Sub  
  
    ''' <summary>  
    ''' The SldWorks swApp variable is pre-assigned for you.  
    ''' </summary>  
    Public swApp As SldWorks  
  
End Class
```

Get Sketch Regions Example (C#)

This example shows how to get the sketch regions in a sketch.

```
//-----  
// Preconditions:  
// 1. Open a model document that contains a Sketch1 feature with  
//    one or more sketch regions.  
// 2. Open the Immediate window.  
//  
// Postconditions:  
// 1. Gets each sketch region.  
// 2. Press F5 at each System.Diagnostics.Debugger.Break statement.  
// 3. Examine the Immediate window.  
//-----  
using SolidWorks.Interop.sldworks;  
using SolidWorks.Interop.swconst;  
using System.Runtime.InteropServices;  
using System;  
using System.Diagnostics;  
  
namespace SelectSketchRegion_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
  
        public void Main()  
        {  
            ModelDoc2 myModel = default(ModelDoc2);  
            PartDoc myPart = default(PartDoc);  
            SelectionMgr SelMgr = default(SelectionMgr);  
            SelectData mySelectData = default(SelectData);  
            Feature myFeature = default(Feature);  
            Sketch mySketch = default(Sketch);  
            int regionCount = 0;  
            object[] vSkRegions = null;  
            SketchRegion skRegion = default(SketchRegion);  
            Loop2 myLoop = default(Loop2);  
            int edgeCount = 0;  
            int vertexCount = 0;  
            object[] vEdges = null;  
            Edge myEdge = default(Edge);  
            object[] vVertices = null;  
            Vertex myVertex = default(Vertex);  
            double[] vPoint = null;  
            double X = 0;  
            double Y = 0;  
            double Z = 0;  
            bool outer = false;  
            string strOuter = null;  
            int i = 0;  
            int j = 0;  
            int k = 0;  
            bool boolstatus = false;  
  
            myModel = (ModelDoc2)swApp.ActiveDoc;  
            SelMgr = (SelectionMgr)myModel.SelectionManager;  
            mySelectData = (SelectData)SelMgr.CreateSelectData();  
            myPart = (PartDoc)myModel;  
            myFeature = (Feature)myPart.FeatureByName("Sketch1");  
            mySketch = (Sketch)myFeature.GetSpecificFeature2();  
            //      or  
            //      Set mySketch = myModel.GetActiveSketch2()  
            //      Set myFeature = mySketch  
  
            if ((mySketch != null))  
            {
```

```

regionCount = mySketch.GetSketchRegionCount();
Debug.Print("");
Debug.Print(regionCount + " regions in sketch " + myFeature.Name);

vSkRegions = (object[])mySketch.GetSketchRegions();

for (i = vSkRegions.GetLowerBound(0); i <= vSkRegions.GetUpperBound(0); i++)
{
    skRegion = (SketchRegion)vSkRegions[i];

    if ((skRegion != null))
    {
        Debug.Print(" Region " + i + ":");
        j = 0;
        myLoop = skRegion.GetFirstLoop();
        while ((myLoop != null))
        {
            edgeCount = myLoop.GetEdgeCount();
            vertexCount = myLoop.GetVertexCount();
            outer = myLoop.IsOuter();

            if (outer == false)
            {
                strOuter = "Inner loop";
            }
            else
            {
                strOuter = "Outer loop";
            }

            Debug.Print(" Loop " + j + ": " + edgeCount + " edges, " + vertexCount
+ " vertices, " + strOuter);

            vEdges = (object[])myLoop.GetEdges();

            for (k = vEdges.GetLowerBound(0); k <= vEdges.GetUpperBound(0); k++)
            {
                myEdge = (Edge)vEdges[k];
                if ((myEdge != null))
                {
                    Debug.Print(" Edge " + k + ": ");
                }
            }

            vVertices = (object[])myLoop.GetVertices();

            for (k = vVertices.GetLowerBound(0); k <= vVertices.GetUpperBound(0); k++)
            {
                myVertex = (Vertex)vVertices[k];
                if ((myVertex != null))
                {
                    vPoint = (double[])myVertex.GetPoint();
                    X = vPoint[0];
                    Y = vPoint[1];
                    Z = vPoint[2];

                    Debug.Print(" Vertex " + k + ": " + "(" + X + ", " + Y + ", "
+ Z + ")");
                }
            }

            myLoop = (Loop2)myLoop.GetNext();
            j = j + 1;
        }

        boolstatus = skRegion.Select2(false, mySelectData);
        if (boolstatus == false)
        {
            Debug.Print(" Selection of region failed.");
        }

        System.Diagnostics.Debugger.Break();
    }
}

```

```
        }  
    }  
}  
  
    public SldWorks swApp;  
}  
}
```

Get Sketch Regions Example (VBA)

This example shows how to get the sketch regions in a sketch.

```
'-----  
' Preconditions:  
' 1. Open a model document that contains a Sketch1 feature with  
'    one or more sketch regions.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets each sketch region.  
' 2. Press F5 at each Stop statement.  
' 3. Examine the Immediate window.  
'-----
```

Option Explicit

```
Sub main()  
    Dim swApp As SldWorks.SldWorks  
    Dim myModel As SldWorks.ModelDoc2  
    Dim myPart As SldWorks.PartDoc  
    Dim SelMgr As SldWorks.SelectionMgr  
    Dim mySelectData As SldWorks.SelectData  
    Dim myFeature As SldWorks.Feature  
    Dim mySketch As SldWorks.Sketch  
    Dim regionCount As Long  
    Dim vSkRegions As Variant  
    Dim skRegion As SketchRegion  
    Dim myLoop As Loop2  
    Dim edgeCount As Long, vertexCount As Long  
    Dim vEdges As Variant, myEdge As SldWorks.Edge  
    Dim vVertices As Variant, myVertex As SldWorks.Vertex  
    Dim vPoint As Variant, X As Double, Y As Double, Z As Double  
    Dim outer As Boolean, strOuter As String  
    Dim i As Integer, j As Integer, k As Integer  
    Dim boolstatus As Boolean  
  
    Set swApp = Application.SldWorks  
    Set myModel = swApp.ActiveDoc  
    Set SelMgr = myModel.SelectionManager  
    Set mySelectData = SelMgr.CreateSelectData  
    Set myPart = myModel  
    Set myFeature = myPart.FeatureByName("Sketch1")  
    Set mySketch = myFeature.GetSpecificFeature2()  
'    or  
'    Set mySketch = myModel.GetActiveSketch2()  
'    Set myFeature = mySketch  
  
    If Not mySketch Is Nothing Then  
        regionCount = mySketch.GetSketchRegionCount()  
        Debug.Print ""  
        Debug.Print regionCount & " regions in sketch " & myFeature.Name  
  
        vSkRegions = mySketch.GetSketchRegions()  
  
        For i = LBound(vSkRegions) To UBound(vSkRegions)  
            Set skRegion = vSkRegions(i)  
  
            If Not skRegion Is Nothing Then  
                Debug.Print "  Region " & i & ":"  
                j = 0  
                Set myLoop = skRegion.GetFirstLoop()  
                While Not myLoop Is Nothing  
                    edgeCount = myLoop.GetEdgeCount()  
                    vertexCount = myLoop.GetVertexCount()  
                    outer = myLoop.IsOuter()  
  
                    If outer = 0 Then  
                        strOuter = "Inner loop"  
                    End If  
                End While  
            End If  
        Next i  
    End If  
End Sub
```

```

Else
    strOuter = "Outer loop"
End If
Debug.Print "    Loop " & j & ": " & edgeCount & " edges, " & vertexCount & "
vertices, " & strOuter
vEdges = myLoop.GetEdges()
For k = LBound(vEdges) To UBound(vEdges)
    Set myEdge = vEdges(k)
    If Not myEdge Is Nothing Then
        Debug.Print "        Edge " & k & ": "
    End If
Next k
vVertices = myLoop.GetVertices()
For k = LBound(vVertices) To UBound(vVertices)
    Set myVertex = vVertices(k)
    If Not myVertex Is Nothing Then
        vPoint = myVertex.GetPoint()
        X = vPoint(0)
        Y = vPoint(1)
        Z = vPoint(2)

        Debug.Print "        Vertex " & k & ": " & "(" & X & ", " & Y & ", " & Z &
    End If
Next k
Set myLoop = myLoop.GetNext()
j = j + 1
Wend
boolstatus = skRegion.Select2(False, mySelectData)
If boolstatus = 0 Then
    Debug.Print "    Selection of region failed."
End If
Stop
End If
Next i
End If
End Sub

```

Get Sketch Regions Example (VB.NET)

This example shows how to get the sketch regions in a sketch.

```
'-----  
' Preconditions:  
' 1. Open a model document that contains a Sketch1 feature with  
'    one or more sketch regions.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets each sketch region.  
' 2. Press F5 at each Stop statement.  
' 3. Examine the Immediate window.  
'-----  
  
Imports SolidWorks.Interop.swconst  
  
Imports System.Runtime.InteropServices  
  
Imports System  
  
Imports System.Diagnostics  
  
  
Partial Class SolidWorksMacro  
  
    Sub main()  
  
        Dim myModel As ModelDoc2  
  
        Dim myPart As PartDoc  
  
        Dim SelMgr As SelectionMgr  
  
        Dim mySelectData As SelectData  
  
        Dim myFeature As Feature  
  
        Dim mySketch As Sketch  
  
        Dim regionCount As Integer  
  
        Dim vSkRegions As Object  
  
        Dim skRegion As SketchRegion  
  
        Dim myLoop As Loop2  
  
        Dim edgeCount As Integer, vertexCount As Integer  
  
        Dim vEdges As Object, myEdge As Edge  
  
        Dim vVertices As Object, myVertex As Vertex  
  
        Dim vPoint As Object, X As Double, Y As Double, Z As Double  
  
        Dim outer As Boolean, strOuter As String  
  
        Dim i As Integer, j As Integer, k As Integer  
  
        Dim boolstatus As Boolean  
  
  
        myModel = swApp.ActiveDoc  
  
        SelMgr = myModel.SelectionManager  
  
        mySelectData = SelMgr.CreateSelectData
```

```

myPart = myModel

myFeature = myPart.FeatureByName("Sketch1")

mySketch = myFeature.GetSpecificFeature2()

'           or

'   Set mySketch = myModel.GetActiveSketch2()
'   Set myFeature = mySketch

If Not mySketch Is Nothing Then

    regionCount = mySketch.GetSketchRegionCount()

    Debug.Print("")

    Debug.Print(regionCount & "
regions in sketch " & myFeature.Name)

    vSkRegions = mySketch.GetSketchRegions()

    For i = LBound(vSkRegions) To UBound(vSkRegions)

        skRegion = vSkRegions(i)

        If Not skRegion Is Nothing Then

            Debug.Print("   Region

" & i & ":")

            j = 0

            myLoop = skRegion.GetFirstLoop()

            While Not myLoop Is Nothing

                edgeCount = myLoop.GetEdgeCount()

                vertexCount = myLoop.GetVertexCount()

                outer = myLoop.IsOuter()

                If outer = 0 Then

                    strOuter = "Inner loop"

                Else

                    strOuter = "Outer loop"

                End If

                Debug.Print("       Loop

" & j & ": " & edgeCount & " edges, " & vertexCount & " vertices, " & strOuter)

                vEdges = myLoop.GetEdges()

                For k = LBound(vEdges) To UBound(vEdges)

                    myEdge = vEdges(k)

                    If Not myEdge Is Nothing Then

                        Debug.Print("           Edge

" & k & ": ")

                    End If

```

```

        Next k

        vVertices = myLoop.GetVertices()

        For k = LBound(vVertices) To UBound(vVertices)
            myVertex = vVertices(k)

            If Not myVertex Is Nothing Then
                vPoint = myVertex.GetPoint()

                X = vPoint(0)
                Y = vPoint(1)
                Z = vPoint(2)

                Debug.Print("      Vertex
" & k & ": " & "(" & X & ", " & Y & ", " & Z
& ")")

            End If
        Next k

        myLoop = myLoop.GetNext()

        j = j + 1

    End While

    boolstatus = skRegion.Select2(False, mySelectData)

    If boolstatus = 0 Then
        Debug.Print("      Selection
of region failed.")
    End If

    Stop

End If

Next i

End If

End Sub

Public swApp As SldWorks

End Class

```

Get Sketch Relations Example (C#)

This example shows how to get all of the sketch relations in a sketch.

```
//-----
// Preconditions:
// 1. Open a part or assembly document.
// 2. Select the sketch whose relations you want to see.
//
// Postconditions: Inspect the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;
namespace GetDefinitionEntities2_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            SelectionMgr swSelMgr = default(SelectionMgr);
            SelectData swSelData = default(SelectData);
            Feature swFeat = default(Feature);
            Sketch swSketch = default(Sketch);
            SketchRelationManager swSkRelMgr = default(SketchRelationManager);
            SketchRelation swSkRel = default(SketchRelation);
            DisplayDimension dispDim = default(DisplayDimension);
            object[] vSkRelArr = null;
            int[] vEntTypeArr = null;
            object[] vEntArr = null;
            object[] vDefEntArr = null;
            SketchSegment swSkSeg = default(SketchSegment);
            SketchPoint swSkPt = default(SketchPoint);
            int i = 0;
            int j = 0;
            bool bRet = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swSelData = (SelectData)swSelMgr.CreateSelectData();
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swSketch = (Sketch)swFeat.GetSpecificFeature2();
            swSkRelMgr = swSketch.RelationManager;

            swModel.ClearSelection2(true);

            Debug.Print("File = " + swModel.GetPathName());
            Debug.Print("Feat = " + swFeat.Name);

            vSkRelArr = (object[])swSkRelMgr.GetRelations((int)swSketchRelationFilterType_e.swAll);
            if (vSkRelArr == null)
                return;
            foreach (SketchRelation vRel in vSkRelArr) {
                swSkRel = (SketchRelation)vRel;

                Debug.Print("Relation(" + i + ")");
                Debug.Print("Type = " + swSkRel.GetRelationType());

                dispDim = (DisplayDimension)swSkRel.GetDisplayDimension();
                if (dispDim != null) {
                    Debug.Print("Display dimension = " + dispDim.GetNameforSelection());
                }

                vEntTypeArr = (int[])swSkRel.GetEntitiesType();
```

```

vEntArr = (object[])swSkRel.GetEntities();

vDefEntArr = (object[])swSkRel.GetDefinitionEntities2();
if ((vDefEntArr == null)) {
} else {
    Debug.Print("    Number of definition entities in this relation: " +
vDefEntArr.GetUpperBound(0));
}

if ((vEntTypeArr != null) & (vEntArr != null)) {

    if (vEntTypeArr.GetUpperBound(0) == vEntArr.GetUpperBound(0)) {
        j = 0;

        foreach (swSketchRelationEntityTypes_e vType in vEntTypeArr) {
            Debug.Print("    EntityType = " + vType);

            switch (vType) {
                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Unknown:
                    Debug.Print("    Not known");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_SubSketch:
                    Debug.Print("SubSketch");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Point:
                    swSkPt = (SketchPoint)vEntArr[j];
                    Debug.Assert((swSkPt != null));

                    Debug.Print("    SkPoint ID = [" + ((int[])(swSkPt.GetID()))[0]
+ ", " + ((int[])(swSkPt.GetID()))[1] + "]);

                    bRet = swSkPt.Select4(true, swSelData);
                    //
                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Line:
                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Arc:
                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Ellipse:
                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Parabola:
                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Spline:

                    swSkSeg = (SketchSegment)vEntArr[j];

                    Debug.Print("    SkSeg ID = [" + ((int[])(swSkSeg.GetID()))[0]
+ ", " + ((int[])(swSkSeg.GetID()))[1] + "]);

                    bRet = swSkSeg.Select4(true, swSelData);

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Hatch:
                    Debug.Print("Hatch");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Text:
                    Debug.Print("Text");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Plane:
                    Debug.Print("Plane");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Cylinder:
                    Debug.Print("Cylinder");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Sphere:
                    Debug.Print("Sphere");

                    break;

                case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Surface:
                    Debug.Print("Surface");

                    break;
            }
        }
    }
}

```

```

        case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Dimension:
            Debug.Print("Dimension");

            break;
        default:
            Debug.Print("Something else");

            break;
    }

    j = j + 1;
}

}

}

i = i + 1;

}

}

public SldWorks swApp;

}

}

```

Get Sketch Relations Example (VBA)

This example shows how to get all of the sketch relations in a sketch.

```
'-----  
' Preconditions:  
' 1. Open a part or assembly document.  
' 2. Select the sketch whose relations you want to see.  
'  
' Postconditions: Inspect the Immediate window.  
'-----
```

Option Explicit

Sub main()

```
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swSelData As SldWorks.SelectData  
    Dim swFeat As SldWorks.Feature  
    Dim swSketch As SldWorks.Sketch  
    Dim swSkRelMgr As SldWorks.SketchRelationManager  
    Dim swSkRel As SldWorks.SketchRelation  
    Dim dispDim As SldWorks.DisplayDimension  
    Dim vSkRelArr As Variant  
    Dim vSkRel As Variant  
    Dim vEntTypeArr As Variant  
    Dim vEntType As Variant  
    Dim vEntArr As Variant  
    Dim vEnt As Variant  
    Dim vDefEntArr As Variant  
    Dim swSkSeg As SldWorks.SketchSegment  
    Dim swSkPt As SldWorks.SketchPoint  
    Dim i As Long  
    Dim j As Long  
    Dim bRet As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swSelData = swSelMgr.CreateSelectData  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swSketch = swFeat.GetSpecificFeature2  
    Set swSkRelMgr = swSketch.RelationManager  
  
    swModel.ClearSelection2 True  
  
    Debug.Print "File = " & swModel.GetPathName  
    Debug.Print " Feat = " & swFeat.Name  
  
    vSkRelArr = swSkRelMgr.GetRelations(swAll): If IsEmpty(vSkRelArr) Then Exit Sub  
    For Each vSkRel In vSkRelArr  
        Set swSkRel = vSkRel  
  
        Debug.Print "    Relation(" & i & ")"  
        Debug.Print "        Type = " & swSkRel.GetRelationType  
  
        Set dispDim = swSkRel.GetDisplayDimension  
        If Not dispDim Is Nothing Then  
            Debug.Print "            Display dimension = " & dispDim.GetNameForSelection  
        End If  
  
        vEntTypeArr = swSkRel.GetEntitiesType  
        vEntArr = swSkRel.GetEntities  
  
        vDefEntArr = swSkRel.GetDefinitionEntities2  
        If IsEmpty(vDefEntArr) Then  
        Else  
            Debug.Print "        Number of definition entities in this relation: " & UBound(vDefEntArr)
```

End If

If Not IsEmpty(vEntTypeArr) And Not IsEmpty(vEntArr) Then
If UBound(vEntTypeArr) = UBound(vEntArr) Then

j = 0

For Each vEntType In vEntTypeArr
Debug.Print "EntityType = " & vEntType

Select Case vEntType
Case swSketchRelationEntityType_Unknown
Debug.Print "Not known"

Case swSketchRelationEntityType_SubSketch
Debug.Print "SubSketch"

Case swSketchRelationEntityType_Point
Set swSkPt = vEntArr(j): Debug.Assert Not swSkPt Is Nothing

Debug.Print "SkPoint ID = [" & swSkPt.GetID(0) & ", " &
swSkPt.GetID(1) & "]"

bRet = swSkPt.Select4(True, swSelData) '

Case swSketchRelationEntityType_Line, _
swSketchRelationEntityType_Arc, _
swSketchRelationEntityType_Ellipse, _
swSketchRelationEntityType_Parabola, _
swSketchRelationEntityType_Spline

Set swSkSeg = vEntArr(j)

Debug.Print "SkSeg ID = [" & swSkSeg.GetID(0) & ", " &
swSkSeg.GetID(1) & "]"

bRet = swSkSeg.Select4(True, swSelData)

Case swSketchRelationEntityType_Hatch
Debug.Print "Hatch"

Case swSketchRelationEntityType_Text
Debug.Print "Text"

Case swSketchRelationEntityType_Plane
Debug.Print "Plane"

Case swSketchRelationEntityType_Cylinder
Debug.Print "Cylinder"

Case swSketchRelationEntityType_Sphere
Debug.Print "Sphere"

Case swSketchRelationEntityType_Surface
Debug.Print "Surface"

Case swSketchRelationEntityType_Dimension
Debug.Print "Dimension"

Case Else
Debug.Print "Something else"

End Select

j = j + 1

Next
End If
End If

i = i + 1

Next
End Sub

Get Sketch Relations Example (VB.NET)

This example shows how to get all of the sketch relations in a sketch.

```
'-----
' Preconditions:
' 1. Open a part or assembly document.
' 2. Select the sketch whose relations you want to see.
'
' Postconditions: Inspect the Immediate window.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Sub main()

        Dim swModel As ModelDoc2
        Dim swSelMgr As SelectionMgr
        Dim swSelData As SelectData
        Dim swFeat As Feature
        Dim swSketch As Sketch
        Dim swSkRelMgr As SketchRelationManager
        Dim swSkRel As SketchRelation
        Dim dispDim As DisplayDimension
        Dim vSkRelArr As Object
        Dim vSkRel As Object
        Dim vEntTypeArr As Object
        Dim vEntType As swSketchRelationEntityTypes_e
        Dim vEntArr As Object
        Dim vDefEntArr As Object
        Dim swSkSeg As SketchSegment
        Dim swSkPt As SketchPoint
        Dim i As Integer
        Dim j As Integer
        Dim bRet As Boolean

        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager
        swSelData = swSelMgr.CreateSelectData
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
        swSketch = swFeat.GetSpecificFeature2
        swSkRelMgr = swSketch.RelationManager

        swModel.ClearSelection2(True)

        Debug.Print("File = " & swModel.GetPathName)
        Debug.Print("  Feat = " & swFeat.Name)

        vSkRelArr = swSkRelMgr.GetRelations(swSketchRelationFilterType_e.swAll) : If
IsNothing(vSkRelArr) Then Exit Sub
        For Each vSkRel In vSkRelArr
            swSkRel = vSkRel

            Debug.Print("    Relation(" & i & ")")
            Debug.Print("      Type          = " & swSkRel.GetRelationType)

            dispDim = swSkRel.GetDisplayDimension
            If Not dispDim Is Nothing Then
                Debug.Print("        Display dimension      = " & dispDim.GetNameForSelection)
            End If

            vEntTypeArr = swSkRel.GetEntitiesType
            vEntArr = swSkRel.GetEntities
```

```

vDefEntArr = swSkRel.GetDefinitionEntities2
If IsNothing(vDefEntArr) Then
Else
    Debug.Print("    Number of definition entities in this relation: " &
UBound(vDefEntArr))
End If

If Not IsNothing(vEntTypeArr) And Not IsNothing(vEntArr) Then
    If UBound(vEntTypeArr) = UBound(vEntArr) Then

        j = 0

        For Each vEntType In vEntTypeArr
            Debug.Print("        EntType    = " & vEntType)

            Select Case vEntType
                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Unknown
                    Debug.Print("            Not known")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_SubSketch
                    Debug.Print("SubSketch")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Point
                    swSkPt = vEntArr(j) : Debug.Assert(Not swSkPt Is Nothing)

                    Debug.Print("            SkPoint ID = [" & swSkPt.GetID(0) & ", " &
swSkPt.GetID(1) & "]"")

                    bRet = swSkPt.Select4(True, swSelData) '

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Line, _
                    swSketchRelationEntityTypes_e.swSketchRelationEntityType_Arc, _
                    swSketchRelationEntityTypes_e.swSketchRelationEntityType_Ellipse,
                    swSketchRelationEntityTypes_e.swSketchRelationEntityType_Parabola,
                    swSketchRelationEntityTypes_e.swSketchRelationEntityType_Spline
                    swSkSeg = vEntArr(j)

                    Debug.Print("            SkSeg    ID = [" & swSkSeg.GetID(0) & ", " &
swSkSeg.GetID(1) & "]"")

                    bRet = swSkSeg.Select4(True, swSelData)

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Hatch
                    Debug.Print("Hatch")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Text
                    Debug.Print("Text")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Plane
                    Debug.Print("Plane")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Cylinder
                    Debug.Print("Cylinder")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Sphere
                    Debug.Print("Sphere")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Surface
                    Debug.Print("Surface")

                Case swSketchRelationEntityTypes_e.swSketchRelationEntityType_Dimension
                    Debug.Print("Dimension")

                Case Else
                    Debug.Print("Something else")

            End Select

            j = j + 1

        Next

```

```
        End If  
    End If
```

```
    i = i + 1
```

```
Next
```

```
End Sub
```

```
Public swApp As SldWorks
```

```
End Class
```

Get Sketch Segment and Curve Data Example (VBA)

This example shows how to get data about a sketch segment and curve.

```
'-----
' Preconditions:
' 1. Open a part document.
' 2. Edit a sketch and select a sketch segment.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Gets sketch segment and curve data.
' 2. Examine the Immediate window.
'-----

Option Explicit

' Define two types
Type DoubleRec
    dValue As Double
End Type

Type Long2Rec
    iLower As Long
    iUpper As Long
End Type

' Extract two integer values from of a single double value
' by assigning a DoubleRec to the double value and
' copying the value to a Long2Rec and
' extracting the integer values
Function ExtractFields(ByVal dValue As Double, iLower As Long, iUpper As Long)
    Dim dr As DoubleRec
    Dim i2r As Long2Rec

    ' Set the double value
    dr.dValue = dValue

    ' Copy the values
    LSet i2r = dr

    ' Extract the values
    iLower = i2r.iLower
    iUpper = i2r.iUpper
End Function

Sub ProcessCurve(swApp As SldWorks.SldWorks, swModel As SldWorks.ModelDoc2, swCurve As SldWorks.Curve)
    Dim nStartParam As Double
    Dim nEndParam As Double
    Dim bIsClosed As Boolean
    Dim bIsPeriodic As Boolean
    Dim vStartEval As Variant
    Dim vEndEval As Variant
    Dim nSuccessStart As Long
    Dim nEndStart As Long
    Dim nDummy As Long
    Dim bRet As Boolean

    bRet = swCurve.GetEndParams(nStartParam, nEndParam, bIsClosed, bIsPeriodic): Debug.Assert bRet
    vStartEval = swCurve.Evaluate(nStartParam)
    vEndEval = swCurve.Evaluate(nEndParam)

    ExtractFields vStartEval(6), nSuccessStart, nDummy
    ExtractFields vEndEval(6), nEndStart, nDummy

    Debug.Print "    IsClosed      = " & bIsClosed
    Debug.Print "    IsPeriodic   = " & bIsPeriodic
    Debug.Print "    IsTrimmed    = " & swCurve.IsTrimmedCurve
    Debug.Print "    Start"
    Debug.Print "    Point        = (" & vStartEval(0) * 1000# & ", " & vStartEval(1) * 1000# & ", " & vStartEval(2)
* 1000# & ") mm"
    Debug.Print "    Tangent      = (" & vStartEval(3) & ", " & vStartEval(4) & ", " & vStartEval(5) & ")"
    Debug.Print "    Success      = " & nSuccessStart
```

```

        Debug.Print "    Finish"
        Debug.Print "    Point      = (" & vEndEval(0) * 1000# & ", " & vEndEval(1) * 1000# & ", " & vEndEval(2) *
1000# & ") mm"
        Debug.Print "    Tangent    = (" & vEndEval(3) & ", " & vEndEval(4) & ", " & vEndEval(5) & ")"
        Debug.Print "    Success    = " & nEndStart
End Sub

Sub main()

    Dim sSkSegmentsName(5) As String
    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swSkSeg As SldWorks.SketchSegment
    Dim swCurve As SldWorks.Curve
    Dim swSkFeat As SldWorks.Feature
    Dim swSketch As SldWorks.Sketch
    Dim vID As Variant
    Dim bRet As Boolean

    sSkSegmentsName(swSketch.LINE) = "swSketchLINE"
    sSkSegmentsName(swSketch.ARC) = "swSketchARC"
    sSkSegmentsName(swSketch.ELLIPSE) = "swSketchELLIPSE"
    sSkSegmentsName(swSketch.SPLINE) = "swSketchSPLINE"
    sSkSegmentsName(swSketch.TEXT) = "swSketchTEXT"
    sSkSegmentsName(swSketch.PARABOLA) = "swSketchPARABOLA"

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swSkSeg = swSelMgr.GetSelectedObject6(1, -1)
    Set swCurve = swSkSeg.GetCurve
    Set swSkFeat = swSkSeg.GetSketch
    Set swSketch = swSkFeat.GetSpecificFeature2

    vID = swSkSeg.GetID
    Debug.Print "Feature = " & swSkFeat.Name & " [" & swSketch.Is3D & "]"
    Debug.Print "    Sketch segment data"
    Debug.Print "        ID = [" & vID(0) & ", " & vID(1) & "]"
    Debug.Print "        Type      = " & sSkSegmentsName(swSkSeg.GetType)
    Debug.Print "        ConstGeom  = " & swSkSeg.ConstructionGeometry

    Debug.Print "    Curve data"
    ProcessCurve swApp, swModel, swCurve

End Sub

```

Get Sketch Segment Constraints Example (C++ COM)

This example shows how to get the constraints for the selected sketch segment.

```
// -----
void APITestFunction()
{
    HRESULT                hr = S_OK;
    VARIANT_BOOL           bRet = VARIANT_FALSE;
    long                   nRetVal = -1;
    CComPtr <ISldWorks>     pSldWorks;
    CComPtr <IModelDoc2>    pModel;
    CComPtr <ISelectionMgr> pSelMgr;
    CComPtr <IUnknown>      pUnk;
    CComQIPtr <ISketchSegment> pSkSeg;
    long                   nNumConstr = -1;
    BSTR*                  pConstrArr = NULL;
    long                   i = -1;

    // connect to SW
    pSldWorks = TheApplication->GetSWApp();
    ASSERT(pSldWorks);
    hr = pSldWorks->get_IActiveDoc2(&pModel);
    ASSERT(pModel);
    hr = pModel->get_ISelectionManager(&pSelMgr);
    ASSERT(pSelMgr);
    hr = pSelMgr->IGetSelectedObject5(1, &pUnk);
    ASSERT(pUnk);
    pSkSeg = pUnk;
    ASSERT(pSkSeg);
    hr = pSkSeg->IGetConstraintsCount(&nNumConstr);
    ASSERT (nNumConstr > 0);
    pConstrArr = new BSTR[nNumConstr];
    ASSERT(pConstrArr);
    ZeroMemory(pConstrArr, nNumConstr * sizeof(BSTR));
    hr = pSkSeg->IGetConstraints(pConstrArr);
    ASSERT(pConstrArr[0]);
    for (i = 0; i < nNumConstr; i++)
    {
        CComBSTR            sConstrStr;
        sConstrStr.Attach(pConstrArr[i]);
        OutputDebugString(sConstrStr);
        OutputDebugString(_T("\n"));
    }
    delete [] pConstrArr;
    return;
} //void APITestFunction()
// -----
```

Get Sketch Segment Length Example (C++ COM)

This example shows how to get the length of a sketch segment.

```
double GetLength(LPMODELDOC Part,LPSELECTIONMGR SelectionManager,LPTSTR objectName)
{
    LPTSTR segType = _T("EXTSKETCHSEGMENT");
    VARIANT_BOOL success;
    double Length(-1.0); // neg number == failure
    Part->SelectByID(objectName, segType, 0.0, 0.0, 0,&success);
    if(Part->SelectByID(objectName, segType, 0.0, 0.0, 0,&success) == S_OK && success)
    {
        LPSKETCHSEGMENT SketchSegment;
        LPDISPATCH disp;
        if(SelectionManager->GetSelectedObject2(1,&disp) == S_OK &&
            disp != NULL &&
            disp->QueryInterface(IID_ISketchSegment,(void **)&SketchSegment) == S_OK &&
            SketchSegment != NULL)
            SketchSegment->GetLength(&Length);
    }
    return Length;
}
```

Get_Sketch_Segment_Length_Example_VB (Topic)

SOLIDWORKS API Help

Get Sketch Segment Length Example (VBA)

This example shows how to get the length of a sketch segment.

```
'-----  
' Preconditions:  
' 1. Open a part that has a Sketch1 feature of a parabola.  
' 2. Select the sketch.  
' 3. Open the Immediate window.  
,  
' Postconditions:  
' 1. Gets the selected sketch.  
' 2. Gets the length of the parabola.  
' 3. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim Part As SldWorks.PartDoc  
Dim SelectionManager As SldWorks.SelectionMgr  
Dim SketchSegment As SldWorks.SketchSegment  
Dim Length As Double  
  
Sub main()  
    Set swApp = CreateObject("SldWorks.Application")  
    Set Part = swApp.ActiveDoc  
    Set SelectionManager = Part.SelectionManager()  
    Part.SelectByID "Parabola1@Sketch1", "EXTSKETCHSEGMENT", 0, 0, 0  
    Set SketchSegment = SelectionManager.GetSelectedObject2(1)  
    Length = SketchSegment.GetLength()  
    Debug.Print "Length = " & Length * 1000 & " mm"  
  
End Sub
```

Get Sketch Segment Names Example (C#)

This example shows how to get the names of selected sketch segments.

```
//-----
// Preconditions:
// 1. Verify that the specified part document template exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens a new part document.
// 2. Inserts a sketch of a rectangle.
// 3. Selects two sketch segments and prints their names
//    to the Immediate window.
// 4. Examine the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SelectionMgr swSelMgr = default(SelectionMgr);
            SketchManager swSketchManager = default(SketchManager);
            object[] sketchLines = null;
            SketchSegment swSketchSegHoriz = default(SketchSegment);
            SketchSegment swSketchSegVert = default(SketchSegment);
            bool ret = false;

            swModel = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SolidWorks\\SolidWorks 2016\\templates\\Part.prtdot", 0, 0, 0);
            swModelDocExt = (ModelDocExtension)swModel.Extension;

            ret = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstToRectEntity, (int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, false);
            ret = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, (int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, true);
            swSketchManager = (SketchManager)swModel.SketchManager;
            sketchLines = (object[])swSketchManager.CreateCornerRectangle(0, 0, 0, 0.110951010058045, -0.066328380491143, 0);
            ret = swModelDocExt.SelectByID2("Line3", "SKETCHSEGMENT", 0.00443505736694483, -0.012832795562811, 0.00637809258389225, false, 0, null, 0);
            ret = swModelDocExt.SelectByID2("Line4", "SKETCHSEGMENT", 0.095835993249203, -0.0306185999393385, -0.0297695225643872, true, 0, null, 0);
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swSketchSegHoriz = (SketchSegment)swSelMgr.GetSelectedObject6(1, -1);
            Debug.Print("Name of selected horizontal sketch segment = " + swSketchSegHoriz.GetName());
            swSketchSegVert = (SketchSegment)swSelMgr.GetSelectedObject6(2, -1);
            Debug.Print("Name of selected vertical sketch segment = " + swSketchSegVert.GetName());

            swSketchManager.InsertSketch(true);
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Sketch Segment Names Example (VBA)

This example shows how to get the names of selected sketch segments.

```
'-----  
' Preconditions:  
' 1. Verify that the specified part document template exists.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Opens a new part document.  
' 2. Inserts a sketch of a rectangle.  
' 3. Selects two sketch segments and prints their names  
'    to the Immediate window.  
' 4. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swModelDocExt As SldWorks.ModelDocExtension  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swSketchManager As SldWorks.SketchManager  
    Dim sketchLines As Variant  
    Dim swSketchSegHoriz As SldWorks.SketchSegment  
    Dim swSketchSegVert As SldWorks.SketchSegment  
    Dim ret As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2016\templates\Part.prtdot", 0, 0, 0)  
    Set swModelDocExt = swModel.Extension  
  
    ret = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,  
swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)  
    ret = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,  
swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)  
    Set swSketchManager = swModel.SketchManager  
    sketchLines = swSketchManager.CreateCornerRectangle(0, 0, 0, 0.110951010058045, -0.066328380491143, 0)  
    ret = swModelDocExt.SelectByID("Line3", "SKETCHSEGMENT", 4.43505736694483E-03, -0.012832795562811,  
6.37809258389225E-03, False, 0, Nothing, 0)  
    ret = swModelDocExt.SelectByID("Line4", "SKETCHSEGMENT", 0.095835993249203, -3.06185999393385E-02,  
-2.97695225643872E-02, True, 0, Nothing, 0)  
    Set swSelMgr = swModel.SelectionManager  
    Set swSketchSegHoriz = swSelMgr.GetSelectedObject6(1, -1)  
    Debug.Print "Name of selected horizontal sketch segment = " & swSketchSegHoriz.GetName  
    Set swSketchSegVert = swSelMgr.GetSelectedObject6(2, -1)  
    Debug.Print "Name of selected vertical sketch segment = " & swSketchSegVert.GetName  
  
    swSketchManager.InsertSketch True  
  
End Sub
```

Get_Sketch_Segment_Names_Example_VBNET (Topic)

SOLIDWORKS API Help

Get Sketch Segment Names Example (VB.NET)

This example shows how to get the names of selected sketch segments.

```
'-----
' Preconditions:
' 1. Verify that the specified part document template exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens a new part document.
' 2. Inserts a sketch of a rectangle.
' 3. Selects two sketch segments and prints their names
'    to the Immediate window.
' 4. Examine the Immediate window.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swSelMgr As SelectionMgr
        Dim swSketchManager As SketchManager
        Dim sketchLines As Object
        Dim swSketchSegHoriz As SketchSegment
        Dim swSketchSegVert As SketchSegment
        Dim ret As Boolean

        swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2016\templates\Part.prtdot", 0, 0, 0)
        swModelDocExt = swModel.Extension

        ret = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity, swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
        ret = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
        swSketchManager = swModel.SketchManager
        sketchLines = swSketchManager.CreateCornerRectangle(0, 0, 0, 0.110951010058045, -0.066328380491143, 0)
        ret = swModelDocExt.SelectByID2("Line3", "SKETCHSEGMENT", 0.00443505736694483, -0.012832795562811, 0.00637809258389225, False, 0, Nothing, 0)
        ret = swModelDocExt.SelectByID2("Line4", "SKETCHSEGMENT", 0.095835993249203, -0.0306185999393385, -0.0297695225643872, True, 0, Nothing, 0)
        swSelMgr = swModel.SelectionManager
        swSketchSegHoriz = swSelMgr.GetSelectedObject6(1, -1)
        Debug.Print("Name of selected horizontal sketch segment = " & swSketchSegHoriz.GetName)
        swSketchSegVert = swSelMgr.GetSelectedObject6(2, -1)
        Debug.Print("Name of selected vertical sketch segment = " & swSketchSegVert.GetName)

        swSketchManager.InsertSketch(True)

    End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class
```

Get Sketch Slot Data (VBA)

This example shows how to get sketch slot data.

```
' -----  
' Preconditions: Model document is active and contains  
'               a Sketch3 feature with one or more  
'               sketch slots.  
'  
' Postconditions: None  
' -----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swPart As SldWorks.PartDoc  
Dim longstatus As Long, longwarnings As Long  
Dim swSketchMgr As SldWorks.SketchManager  
Dim swSketch As SldWorks.Sketch  
Dim iSlotCount As Integer  
Dim vSketchSlots As Variant  
Dim vSketchSlot As Variant  
Dim swSketchSlot As SldWorks.SketchSlot  
Dim i As Integer  
Dim swFeature As SldWorks.Feature  
Dim swMathPoint As SldWorks.MathPoint  
Dim vArray As Variant  
Dim swSketchPoint As SldWorks.SketchPoint  
Dim vSketchPtID As Variant  
  
Sub main()  
  
Set swApp = Application.SldWorks  
  
Set swPart = swApp.OpenDoc6("C:\test\Slots.SLDPRT", 1, 0, "", longstatus, longwarnings)  
Set swSketchMgr = swPart.SketchManager  
  
Set swFeature = swPart.FeatureByName("Sketch3")  
Set swSketch = swFeature.GetSpecificFeature2()  
iSlotCount = swSketch.GetSketchSlotCount  
Debug.Print "Number of slots: " & iSlotCount  
Debug.Print " "  
vSketchSlots = swSketch.GetSketchSlots  
  
For Each vSketchSlot In vSketchSlots  
    Set swSketchSlot = vSketchSlot  
    Debug.Print "Length: " & swSketchSlot.Length
```

```

Debug.Print "Width: " & swSketchSlot.Width
Debug.Print "LengthType: " & swSketchSlot.LengthType
Set swMathPoint = swSketchSlot.GetCenterPoint
vArray = swMathPoint.ArrayData
Debug.Print "CenterPoint (x,y,z): " & vArray(0) & "," & vArray(1) & "," & vArray(2)
Set swSketchPoint = swSketchSlot.GetCenterPointHandle
Debug.Print "CenterPointHandle ISketchPoint(x,y,z): " & swSketchPoint.X & "," & swSketchPoint.Y &
"," & swSketchPoint.Z & vbCrLf
vSketchPtID = swSketchPoint.GetID
Debug.Print "  IDPt(" & i & ") = [" & vSketchPtID(0) & ", " & vSketchPtID(1) & "]" & vbCrLf
Next

End Sub

```

Get Sketch Slot Using Sketch Point and Segment Example (C#)

This example shows how to get a sketch slot using a sketch point and a sketch segment.

```
//-----
//Preconditions:
// 1. Open a new part document.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Creates a sketch slot.
// 2. Gets the length of the sketch slot.
// 3. Selects a sketch point on the sketch slot
//    and accesses the sketch slot using that
//    sketch point.
// 4. Gets the length of the sketch slot.
// 5. Selects a sketch segment on the sketch slot
//    and accesses the sketch slot using that
//    sketch segment.
// 6. Gets the length of the sketch slot.
// 7. Examine the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
using System.Diagnostics;

namespace Macro1.csproj
{
    public partial class SolidWorksMacro
    {
        ModelDoc2 swModel;
        ModelDocExtension swExt;
        SelectionMgr swSelMgr;
        bool boolstatus;
        SketchManager swSketchManager;
        SketchSlot swSketchSlot;
        SketchPoint swSketchPoint;
        SketchSegment swSketchSegment;
        public void Main()
        {
            swModel = (ModelDoc2)swApp.ActiveDoc;
            swExt = (ModelDocExtension)swModel.Extension;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swSketchManager = (SketchManager)swModel.SketchManager;

            //Select a plane and open a sketch
            boolstatus = swExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, false, 0, null, 0);
            swSketchManager.InsertSketch(true);

            // Create a sketch slot
            swSketchSlot =
            (SketchSlot)swSketchManager.CreateSketchSlot((int)swSketchSlotCreationType_e.swSketchSlotCreationType_line,
            (int)swSketchSlotLengthType_e.swSketchSlotLengthType_CenterCenter,
            0.05, -0.05, 0, 0, 0.05, 0, 0, 0, 0, 0, 1, false);
            Debug.Print("Length: " + swSketchSlot.Length);
            Debug.Print(" ");
            swSketchManager.InsertSketch(true);

            // Get a sketch point on the sketch slot
            boolstatus = swExt.SelectByID2("Point1@Sketch1", "EXTSKETCHPOINT", 0.05, 0.025, 0, false, 0, null, 0);
            swSketchPoint = (SketchPoint)swSelMgr.GetSelectedObject6(1, -1);
            // Get sketch slot
            swSketchSlot = (SketchSlot)swSketchPoint.GetSketchSlot();
            Debug.Print("Length: " + swSketchSlot.Length);
            Debug.Print(" ");

            // Get a sketch segment on the sketch slot
            boolstatus = swExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.03969355327396, -0.025, 0, false, 0,
            null, 0);
            swSketchSegment = (SketchSegment)swSelMgr.GetSelectedObject6(1, -1);
            // Get sketch slot
            swSketchSlot = (SketchSlot)swSketchSegment.GetSketchSlot();
```

```
        Debug.Print("Length: " + swSketchSlot.Length);
        Debug.Print(" ");
    }
    /// <summary>
    /// The SldWorks swApp variable is pre-assigned for you.
    /// </summary>
    public SldWorks swApp;
}
}
```

Get Sketch Slot Using Sketch Point and Segment Example_VB (Topic)

SOLIDWORKS API Help

Get Sketch Slot Using Sketch Point and Segment Example (VBA)

This example shows you how to get a sketch slot using a sketch point and a sketch segment.

```
'-----  
' Preconditions:  
' 1. Open a new part document.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Creates a sketch slot.  
' 2. Gets the length of the sketch slot.  
' 3. Selects a sketch point on the sketch slot  
'    and accesses the sketch slot using that  
'    sketch point.  
' 4. Gets the length of the sketch slot.  
' 5. Selects a sketch segment on the sketch slot  
'    and accesses the sketch slot using that  
'    sketch segment.  
' 6. Gets the length of the sketch slot.  
' 7. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swExt As SldWorks.ModelDocExtension  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim boolstatus As Boolean  
Dim swSketchManager As SldWorks.SketchManager  
Dim swSketchSlot As SldWorks.SketchSlot  
Dim swSketchPoint As SldWorks.SketchPoint  
Dim swSketchSegment As SldWorks.SketchSegment  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swExt = swModel.Extension  
    Set swSelMgr = swModel.SelectionManager  
    Set swSketchManager = swModel.SketchManager  
  
    ' Select a plane and open a sketch  
    boolstatus = swExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)  
    swModel.InsertSketch  
  
    ' Create a sketch slot  
    Set swSketchSlot = swSketchManager.CreateSketchSlot(swSketchSlotCreationType_e.swSketchSlotCreationType_line,  
swSketchSlotLengthType_e.swSketchSlotLengthType_CenterCenter, 0.05, -0.05, 0, 0, 0.05, 0, 0, 0, 0, 0, 1, False)  
    Debug.Print "Length: " & swSketchSlot.Length  
    Debug.Print " "  
    swModel.InsertSketch  
  
    ' Get a sketch point on the sketch slot  
    boolstatus = swExt.SelectByID2("Point1@Sketch1", "EXTSKETCHPOINT", 0.05, 0.025, 0, False, 0, Nothing, 0)  
    Set swSketchPoint = swSelMgr.GetSelectedObject6(1, -1)  
  
    ' Get sketch slot  
    Set swSketchSlot = swSketchPoint.GetSketchSlot  
    Debug.Print "Length: " & swSketchSlot.Length  
    Debug.Print " "  
  
    ' Get a sketch segment on the sketch slot  
    boolstatus = swExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0, -0, 0, False, 0, Nothing, 0)  
    Set swSketchSegment = swSelMgr.GetSelectedObject6(1, -1)  
  
    ' Get sketch slot  
    Set swSketchSlot = swSketchSegment.GetSketchSlot  
    Debug.Print "Length: " & swSketchSlot.Length  
  
End Sub
```

Get Sketch Slot Using Sketch Point and Segment Example (VB.NET)

This example shows you how to get a sketch slot using a sketch point and a sketch segment.

```
'-----  
' Preconditions:  
' 1. Open a new part document.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Creates a sketch slot.  
' 2. Gets the length of the sketch slot.  
' 3. Selects a sketch point on the sketch slot  
'    and accesses the sketch slot using that  
'    sketch point.  
' 4. Gets the length of the sketch slot.  
' 5. Selects a sketch segment on the sketch slot  
'    and accesses the sketch slot using that  
'    sketch segment.  
' 6. Gets the length of the sketch slot.  
' 7. Examine the Immediate window.  
'-----
```

```
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System  
Imports System.Diagnostics
```

Partial Class SolidWorksMacro

```
    Dim swModel As ModelDoc2  
    Dim swExt As ModelDocExtension  
    Dim swSelMgr As SelectionMgr  
    Dim boolstatus As Boolean  
    Dim swSketchManager As SketchManager  
    Dim swSketchSlot As SketchSlot  
    Dim swSketchPoint As SketchPoint  
    Dim swSketchSegment As SketchSegment  
  
    Public Sub main()  
        swModel = swApp.ActiveDoc  
        swExt = swModel.Extension  
        swSelMgr = swModel.SelectionManager  
        swSketchManager = swModel.SketchManager  
  
        ' Select a plane and open a sketch  
        boolstatus = swExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)  
        swModel.InsertSketch()  
  
        ' Insert a sketch of sketch slot  
        swSketchSlot =  
        swSketchManager.CreateSketchSlot(swSketchSlotCreationType_e.swSketchSlotCreationType_line,  
        swSketchSlotLengthType_e.swSketchSlotLengthType_CenterCenter, 0.05, -0.05, 0, 0, 0.05, 0, 0, 0, 0,  
        1, False)  
        Debug.Print("Length: " & swSketchSlot.Length)  
        Debug.Print(" ")  
        swModel.InsertSketch()  
  
        ' Get a sketch point on the sketch slot  
        boolstatus = swExt.SelectByID2("Point1@Sketch1", "EXTSKETCHPOINT", 0.05, 0.025, 0, False, 0,  
        Nothing, 0)  
        swSketchPoint = swSelMgr.GetSelectedObject6(1, -1)  
        ' Get sketch slot  
        swSketchSlot = swSketchPoint.GetSketchSlot  
        Debug.Print("Length: " & swSketchSlot.Length)  
        Debug.Print(" ")
```

```

        ' Get a sketch segment on the sketch slot
        boolstatus = swExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.03969355327396, -0.025,
0, False, 0, Nothing, 0)
        swSketchSegment = swSelMgr.GetSelectedObject6(1, -1)
        ' Get sketch slot
        swSketchSlot = swSketchSegment.GetSketchSlot
        Debug.Print("Length: " & swSketchSlot.Length)
        Debug.Print(" ")
    End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Get Sketches Example (C++ COM)

If you use a sketch to create a feature on your solid body, that sketch is a subfeature of that solid feature. This is illustrated by expanding and examining the FeatureManager design tree of a SOLIDWORKS Design feature.

This example shows how to traverse the FeatureManager design tree and use subfeature traversal to grab the ISketch objects used by each feature.

// Precompiled header

```
#include "stdafx.h"
```

```
#include <mbstring.h>
```

// SOLIDWORKS includes

```
#include <amapp.h>
```

// User Dll includes

```
#include "userapp.h"
```

```
void GetSketchesCOM()
```

```
{
```

```
LPMODELDOC pModelDoc;
```

```
HRESULT hres = UserApp->getSWApp()->get_IActiveDoc( &pModelDoc );
```

```
LPFEATURE pFeature = NULL;
```

// Get first feature in document

```
hres = pModelDoc->IFirstFeature(&pFeature);
```

```
int count = 0;
```

// While features exist

```
while (pFeature)
```

```
{
```

```
    BSTR bFeatName;
```

```
        // Get the feature name
```

```
hres = pFeature->get_Name( &bFeatName );
```

```
CString featName( bFeatName );
```

```
CString message;
```

```
message.Format( _T("\n\nFeature: %s"), bFeatName);
```

```
LPFEATURE pSubFeat;
```

```
        // Get the subfeature
```

```
hres = pFeature->IGetFirstSubFeature( &pSubFeat );
```

```
if (pSubFeat != NULL)
```

```
    // Print to Debug Output Window
```

```
TRACE1( "%s", message );
```

```
        // Popup dialog for Release builds
```

```
        // AfxMessageBox( message );
```

```

        // While subfeatures exist
while ( pSubFeat )
{

    BSTR bFeatureTypeName;

    // Get the subfeature type
    hres = pSubFeat->GetTypeName(&bFeatureTypeName);
    CString featureTypeName(bFeatureTypeName);

    // If the sub-feature is a Sketch
    if (featureTypeName == "ProfileFeature")
    {
        LPUNKNOWN iUnk = NULL;

        // Get the Sketch interface
        hres = pSubFeat->IGetSpecificFeature( &iUnk );

        LPSKETCH pSketch;
        hres = iUnk->QueryInterface( IID_ISketch, (LPVOID*)&pSketch);

        if ( (hres == S_OK) && (pSketch != NULL) )
        {
            BSTR bSubFeatName;

            // Get the sub-feature name
            hres = pSubFeat->get_Name( &bSubFeatName );
            CString subFeatName( bSubFeatName );
            CString message;
            message.Format( _T("\n\tSketch:\t%s\n"), bSubFeatName );

            // Print to Debug Output Window
            TRACE1("%s",message);

            // Popup dialog for Release builds
            // AfxMessageBox( message );

            // Now that you have the Sketch used by this feature (pFeature) and
            // you have its name, you can Select the sketch by calling
            // SelectByID and passing the sketch name, or you can use the
            // m_Sketch object to call the methods and properties available
            // with the Sketch class.

            // < YOUR CODE >

            pSketch->Release();
        }

        // End if subfeature is a sketch
    }

    LPFEATURE pNextSubFeature;

```

```
        // Get next subfeature
        hres = pSubFeat->IGetNextSubFeature( & pNextSubFeature);
        pSubFeat->Release();
        pSubFeat = pNextSubFeature;

        // End while subfeatures exist
    }

    LPFEATURE pNextFeature;

    // Get next feature
    hres = pFeature->IGetNextFeature(&pNextFeature);
    pFeature->Release();
    pFeature = pNextFeature;

// End while features exist
}

pModelDoc->Release();

// End function
}
```

Get Solid-fill Hatch Information Example (C#)

This example shows how to get information about solid-fill hatches in a detail view in the current drawing sheet.

```
//-----
// Preconditions:
// 1. Open public_documents\introsw\bolt-assembly.slddrw.
// 2. Verify that c:\temp exists.
// 3. Create a detail view of Section View A-A:
//    a. Click Insert > Drawing View > Detail.
//    b. Sketch the profile for the detail view of Section View A-A.
//    c. Move the pointer while dragging drawing view. When the view
//       is where you want it to be, click to place the view.
//    d. Click OK to close the Detail View PropertyManager page.
// 4. Right-click the detail view in the drawing to open the
//    Area Hatch/Fill PropertyManager page.
//    a. Clear the Material crosshatch check box.
//    b. Select Solid.
//    c. Click OK.
//
// Postconditions:
// 1. Traverses the drawing views.
// 2. Gets data about the solid-fill hatches in the detail view.
// 3. Open c:\temp\SolidFillData.txt in a text editor and examine the
//    contents of the file.
//
// NOTE: Because the drawing is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using Microsoft.VisualBasic;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            DrawingDoc swDrawing = default(DrawingDoc);
            Sheet swSheet = default(Sheet);
            View swView = default(View);
            long nbrSolidFillHatches = 0;
            int ArraySize = 0;
            long i = 0;
            Double[] boundaryData = new Double[4];
            System.IO.StreamWriter file = new System.IO.StreamWriter("c:\\temp\\SolidFillData.txt", false);

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swDrawing = (DrawingDoc)swModel;

            // Get drawing sheet
            swSheet = (Sheet)swDrawing.GetCurrentSheet();

            // Get name of drawing sheet
            file.WriteLine(" Number of drawing views on drawing sheet: " + swDrawing.GetViewCount());

            // First view is the current drawing sheet
            swView = (View)swDrawing.GetFirstView();
            file.WriteLine(" First drawing view is the current drawing sheet, so...");

            // Get first drawing view on drawing sheet
            swView = (View)swView.GetNextView();
            nbrSolidFillHatches = 0;
            ArraySize = 0;
            while ((swView != null))
            {
                file.WriteLine(" Get next drawing view on drawing sheet...");
                file.WriteLine(" View name: " + swView.Name);
                nbrSolidFillHatches = swView.GetSolidHatchCount(out ArraySize);
                file.WriteLine(" Number of solid-fill hatches in this view: " + nbrSolidFillHatches);
                file.WriteLine(" Size of array for the boundary data for the solid-fill hatch(es): " + ArraySize);
                if (ArraySize > 0)
                {
                    boundaryData = (Double[])swView.GetSolidHatchInfo();
                }
            }
        }
    }
}
```

```

        file.WriteLine(" Is the loop an outer loop (first)? " + boundaryData[0]);
        file.WriteLine(" Number of polylines in loop: " + boundaryData[1]);
        file.WriteLine(" Type ( 0 = polyline; 1 = arc or circle): " + boundaryData[2]);
        file.WriteLine(" Size of geometric data array (will be 0 if Type = 0): " + boundaryData[3]);
        file.WriteLine(" See IView::GetSolidHatchInfo's API Help topic for descriptions of these array elements: ");

        for (i = 4; i <= ArraySize - 1; i++)
        {
            file.WriteLine(" Boundary data, array element " + i + ": " + boundaryData[i]);
        }
    }
    // Get next drawing view
    swView = (View)swView.GetNextView();
}
file.Close();

}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
}
}

```

Get Solid-fill Hatch Information Example (VBA)

This example shows how to get information about solid-fill hatches in a detail view in the current drawing sheet.

```
'-----
' Preconditions:
' 1. Open public_documents\introsw\bolt-assembly.slddrw.
' 2. Verify that c:\temp exists.
' 3. Create a detail view of Section View A-A:
'   a. Click Insert > Drawing View > Detail.
'   b. Sketch the profile for the detail view of Section View A-A.
'   c. Move the pointer while dragging drawing view. When the view
'      is where you want it to be, click to place the view.
'   d. Click OK to close the Detail View PropertyManager page.
' 4. Right-click the detail view in the drawing to open the
'   Area Hatch/Fill PropertyManager page.
'   a. Clear the Material crosshatch check box.
'   b. Select Solid.
'   c. Click OK.
'
' Postconditions:
' 1. Traverses the drawing views.
' 2. Gets data about the solid-fill hatches in the detail view.
' 3. Open c:\temp\SolidFillData.txt in a text editor and examine the
'   contents of the file.
'
' NOTE: Because the drawing is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swDrawing As SldWorks.DrawingDoc
Dim swSheet As SldWorks.Sheet
Dim swView As SldWorks.View
Dim nbrSolidFillHatches As Long
Dim ArraySize As Long
Dim i As Long
Dim boundaryData As Variant

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swDrawing = swModel

    ' Open output file for solid-fill data
    Open "c:\temp\SolidFillData.txt" For Output As #1

    ' Get drawing sheet
    Set swSheet = swDrawing.GetCurrentSheet

    ' Get name of drawing sheet
    Write #1, "    Number of drawing views on drawing sheet: " & swDrawing.GetViewCount

    ' First view is the current drawing sheet
    Set swView = swDrawing.GetFirstView
    Write #1, "    First drawing view is the current drawing sheet, so..."

    ' Get first drawing view on drawing sheet
    Set swView = swView.GetNextView
    nbrSolidFillHatches = 0
    ArraySize = 0

    While Not swView Is Nothing
        Write #1, "        Get next drawing view on drawing sheet..."
        Write #1, "        View name: " & swView.Name
        nbrSolidFillHatches = swView.GetSolidHatchCount(ArraySize)
        Write #1, "        Number of solid-fill hatches in this view: " & nbrSolidFillHatches
        Write #1, "        Size of array for the boundary data for the solid-fill hatches: " & ArraySize
        If ArraySize > 0 Then
            boundaryData = swView.GetSolidHatchInfo
            Write #1, "            Is the loop an outer loop (first)? " & boundaryData(0)
            Write #1, "            Number of polylines in loop: " & boundaryData(1)
        End If
        Set swView = swView.GetNextView
    End While
End Sub
```

```

        Write #1, "
        Write #1, "
boundaryData(3)
        Write #1, "
these array elements: "
        For i = 4 To ArraySize - 1
            Write #1, "
boundaryData(i)
            Next i
        End If
        ' Get next drawing view
        Set swView = swView.GetNextView
Wend

        ' Close c:\temp\SolidFillData.txt
        Close #1

End Sub

```

```

Type ( 0 = polyline; 1 = arc or circle): " & boundaryData(2)
Size of geometric data array (will be 0 if Type = 0): " &

```

See IView::GetSolidHatchInfo's API Help topic for descriptions of

Boundary data, array element " & i & ": " &

Get_Solid-fill_Hatch_Information_Example_VBNET (Topic)

SOLIDWORKS API Help

Get Solid-fill Hatch Information Example (VB.NET)

This example shows how to get information about solid-fill hatches in a detail view in the current drawing sheet.

```
'-----  
' Preconditions:  
' 1. Open public_documents\introsw\bolt-assembly.slddrw.  
' 2. Verify that c:\temp exists.  
' 3. Create a detail view of Section View A-A:  
'   a. Click Insert > Drawing View > Detail.  
'   b. Sketch the profile for the detail view of Section View A-A.  
'   c. Move the pointer while dragging drawing view. When the view  
'      is where you want it to be, click to place the view.  
'   d. Click OK to close the Detail View PropertyManager page.  
' 4. Right-click the detail view in the drawing to open the  
'    Area Hatch/Fill PropertyManager page.  
'   a. Clear the Material crosshatch check box.  
'   b. Select Solid.  
'   c. Click OK.  
'  
' Postconditions:  
' 1. Traverses the drawing views.  
' 2. Gets data about the solid-fill hatches in the detail view.  
' 3. Open c:\temp\SolidFillData.txt in a text editor and examine the  
'    contents of the file.  
'  
' NOTE: Because the drawing is used elsewhere, do not save changes.  
'-----
```

```
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System.Runtime.InteropServices  
Imports System
```

Partial Class SolidWorksMacro

```
Public Sub main()  
  
    Dim swModel As ModelDoc2  
    Dim swDrawing As DrawingDoc  
    Dim swSheet As Sheet  
    Dim swView As View  
    Dim nbrSolidFillHatches As Integer  
    Dim ArraySize As Integer  
    Dim i As Integer  
    Dim boundaryData As Object  
    Dim file As System.IO.StreamWriter  
  
    swModel = swApp.ActiveDoc  
    swDrawing = swModel  
  
    ' Open output file for solid-fill data  
    file = My.Computer.FileSystem.OpenTextFileWriter("c:\temp\SolidFillData.txt", False)  
  
    ' Get drawing sheet  
    swSheet = swDrawing.GetCurrentSheet  
  
    ' Get name of drawing sheet  
    file.WriteLine(" Number of drawing views on drawing sheet: " & swDrawing.GetViewCount)  
  
    ' First view is the current drawing sheet  
    swView = swDrawing.GetFirstView  
    file.WriteLine(" First drawing view is the current drawing sheet, so...")  
  
    ' Get first drawing view on drawing sheet  
    swView = swView.GetNextView  
    nbrSolidFillHatches = 0  
    ArraySize = 0  
    While Not swView Is Nothing  
        file.WriteLine(" Get next drawing view on drawing sheet...")  
        file.WriteLine(" View name: " & swView.Name)  
        nbrSolidFillHatches = swView.GetSolidHatchCount(ArraySize)  
        file.WriteLine(" Number of solid-fill hatches in this view: " & nbrSolidFillHatches)  
        file.WriteLine(" Size of array for the boundary data for the solid-fill hatch(es): " & ArraySize)  
        If ArraySize > 0 Then  
            boundaryData = swView.GetSolidHatchInfo  
            file.WriteLine(" Is the loop an outer loop (first)? " & boundaryData(0))  
            file.WriteLine(" Number of polylines in loop: " & boundaryData(1))  
            file.WriteLine(" Type ( 0 = polyline; 1 = arc or circle): " & boundaryData(2))  
            file.WriteLine(" Size of geometric data array (will be 0 if Type = 0): " & boundaryData(3))  
            file.WriteLine(" See IView::GetSolidHatchInfo's API Help topic for descriptions of these array elements: ")  
            For i = 4 To ArraySize - 1  
                file.WriteLine(" Boundary data, array element " & i & ": " & boundaryData(i))  
            Next i  
        End If  
    End While  
End Sub
```

```
' Get next drawing view  
swView = swView.GetNextView  
End While
```

```
file.Close()
```

```
End Sub
```

```
''' <summary>  
''' The SldWorks swApp variable is pre-assigned for you.  
''' </summary>  
Public swApp As SldWorks
```

```
End Class
```

Get Solid Bodies from Cut-list Folders and Get Custom Properties Example (C#)

This example shows how to get the solid bodies from cut-list folders and how to get the custom properties for the solid bodies.

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
// 2. Click Tools > Options > Document Properties > Weldments >
//    Rename cut list folders with Description property value > OK.
// 3. Right-click Cut list(31) in the FeatureManager design tree
//    and click Update.
// 4. Add a reference to Microsoft.VisualBasic (click Project >
//    Add Reference > Microsoft.VisualBasic).
// 5. Open the Immediate window.
//
// Postconditions:
// 1. Traverses the FeatureManager design tree.
// 2. Examine the Immediate window.
//
// NOTE: Because this part is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using Microsoft.VisualBasic;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        ModelDoc2 swPart;
        Feature swFeat;

        public void Main()
        {
            swPart = (ModelDoc2)swApp.ActiveDoc;
            Debug.Print("File: " + swPart.GetPathName());
            string ConfigName = null;
            ConfigName = swPart.ConfigurationManager.ActiveConfiguration.Name;
            Debug.Print("Active configuration name: " + ConfigName);
            swFeat = (Feature)swPart.FirstFeature();
            TraverseFeatures(swFeat, true, "Root Feature");
        }

        public void GetFeatureCustomProps(Feature thisFeat)
        {
            CustomPropertyManager CustomPropMgr = default(CustomPropertyManager);
            CustomPropMgr = (CustomPropertyManager)thisFeat.CustomPropertyManager;
            string[] vCustomPropNames = null;
            vCustomPropNames = (string[])CustomPropMgr.GetNames();
            if ((vCustomPropNames != null))
            {
                Debug.Print("                Cut-list custom properties:");
                int i = 0;
                for (i = 0; i <= (vCustomPropNames.Length - 1); i++)
                {
                    string CustomPropName = null;
                    CustomPropName = (string)vCustomPropNames[i];
                    int CustomPropType = 0;
                    CustomPropType = CustomPropMgr.GetType2(CustomPropName);
                    string CustomPropVal = "";
                    string CustomPropResolvedVal = "";
                    CustomPropMgr.Get2(CustomPropName, out CustomPropVal, out CustomPropResolvedVal);
                    Debug.Print("                Name: " + CustomPropName);
                    Debug.Print("                Value: " + CustomPropVal);
                    Debug.Print("                Resolved value: " + CustomPropResolvedVal);
                }
            }
        }

        bool static_DoTheWork_InBodyFolder;
        readonly Microsoft.VisualBasic.CompilerServices.StaticLocalInitFlag static_DoTheWork_BodyFolderType_Init = new M
icrosoft.VisualBasic.CompilerServices.StaticLocalInitFlag();
        string[] static_DoTheWork_BodyFolderType;
        bool static_DoTheWork_BeenHere;
        public void DoTheWork(Feature thisFeat, string ParentName)
    }
}
```

```

{
    lock (static_DoTheWork_BodyFolderType_Init)
    {
        try
        {
            if (InitStaticVariableHelper(static_DoTheWork_BodyFolderType_Init))
            {
                static_DoTheWork_BodyFolderType = new string[6];
            }
        }
        finally
        {
            static_DoTheWork_BodyFolderType_Init.State = 1;
        }
    }
    bool bAllFeatures = false;
    bool bCutListCustomProps = false;
    object vSuppressed = null;
    int BodyCount = 0;

    if (!static_DoTheWork_BeenHere)
    {
        static_DoTheWork_BodyFolderType[0] = "dummy";
        static_DoTheWork_BodyFolderType[1] = "swSolidBodyFolder";
        static_DoTheWork_BodyFolderType[2] = "swSurfaceBodyFolder";
        static_DoTheWork_BodyFolderType[3] = "swBodySubFolder";
        static_DoTheWork_BodyFolderType[4] = "swWeldmentSubFolder";
        static_DoTheWork_BodyFolderType[5] = "swWeldmentCutListFolder";
        static_DoTheWork_InBodyFolder = false;
        static_DoTheWork_BeenHere = true;
        bAllFeatures = false;
        bCutListCustomProps = false;
    }

    //Comment out next line to print information for just BodyFolders
    bAllFeatures = true;
    //True to print information about all features

    //Comment out next line if you do not want cut list's custom properties
    bCutListCustomProps = true;

    string FeatType = null;
    FeatType = thisFeat.GetTypeName();
    if ((FeatType == "SolidBodyFolder") & (ParentName == "Root Feature"))
    {
        static_DoTheWork_InBodyFolder = true;
    }
    if ((FeatType != "SolidBodyFolder") & (ParentName == "Root Feature"))
    {
        static_DoTheWork_InBodyFolder = false;
    }

    //Only consider the CutListFolders that are under SolidBodyFolder
    if ((static_DoTheWork_InBodyFolder == false) & (FeatType == "CutListFolder"))
    {
        //Skip the second occurrence of the CutListFolders during the feature traversal
        return;
    }

    //Only consider the SubWeldFolder that are under the SolidBodyFolder
    if ((static_DoTheWork_InBodyFolder == false) & (FeatType == "SubWeldFolder"))
    {
        //Skip the second occurrence of the SubWeldFolders during the feature traversal
        return;
    }

    bool IsBodyFolder = false;
    if (FeatType == "SolidBodyFolder" | FeatType == "SurfaceBodyFolder" | FeatType == "CutListFolder" |
FeatType == "SubWeldFolder" | FeatType == "SubAtomFolder")
    {
        IsBodyFolder = true;
    }
    else
    {
        IsBodyFolder = false;
    }
    if (bAllFeatures & (!IsBodyFolder))
    {
        Debug.Print("Feature name: " + thisFeat.Name);
        Debug.Print("Feature type: " + FeatType);
        vSuppressed = thisFeat.IsSuppressed2((int)swInConfigurationOpts_e.swThisConfiguration, null);
        if ((vSuppressed == null))
        {
            Debug.Print("Suppression failed");
        }
        else
    }
}

```

```

        {
            Debug.Print("          Suppressed");
        }
    }
    if (IsBodyFolder)
    {
        BodyFolder BodyFolder = default(BodyFolder);
        BodyFolder = (BodyFolder)thisFeat.GetSpecificFeature2();
        BodyCount = BodyFolder.GetBodyCount();
        if ((FeatType == "CutListFolder") & (BodyCount < 1))
        {
            //When BodyCount = 0, this cut list folder is not displayed in the
            //FeatureManager design tree, so skip it
            return;
        }
        else
        {
            Debug.Print("Feature name: " + thisFeat.Name);
            Debug.Print("  Feature type: " + FeatType);
            vSuppressed = thisFeat.IsSuppressed2((int)swInConfigurationOpts_e.swThisConfiguration, null);
            if ((vSuppressed == null))
            {
                Debug.Print("          Suppression failed");
            }
            else
            {
                Debug.Print("          Suppressed");
            }
        }
    }
    if (!bAllFeatures)
    {
        Debug.Print("Feature name: " + thisFeat.Name);
        Debug.Print("  Feature type: " + FeatType);
        vSuppressed = thisFeat.IsSuppressed2((int)swInConfigurationOpts_e.swThisConfiguration, null);
        if ((vSuppressed == null))
        {
            Debug.Print("          Suppression failed");
        }
        else
        {
            Debug.Print("          Suppressed");
        }
    }
    int BodyFolderTypeE = 0;
    BodyFolderTypeE = BodyFolder.Type;
    Debug.Print("          Body folder: " + static_DoTheWork_BodyFolderType[BodyFolderTypeE]);
    Debug.Print("          Body folder type: " + BodyFolderTypeE);
    Debug.Print("          Body count: " + BodyCount);
    object[] vBodies = null;
    vBodies = (object[])BodyFolder.GetBodies();
    int i = 0;
    if ((vBodies != null))
    {
        for (i = 0; i <= (vBodies.Length - 1); i++)
        {
            Body2 Body = default(Body2);
            Body = (Body2)vBodies[i];
            Debug.Print("          Body name: " + Body.Name);
        }
    }
    }
    else
    {
        if (bAllFeatures)
        {
            Debug.Print("");
        }
    }
    if ((FeatType == "CutListFolder"))
    {
        //When BodyCount = 0, this cut-list folder is not displayed
        //in the FeatureManager design tree, so skip it
        if (BodyCount > 0)
        {
            if (bCutListCustomProps)
            {
                //Comment out this line if you do not want to
                //print the cut-list folder's custom properties
                GetFeatureCustomProps(thisFeat);
            }
        }
    }
}

public void TraverseFeatures(Feature thisFeat, bool isTopLevel, string ParentName)
{

```

```

Feature curFeat = default(Feature);
curFeat = thisFeat;
while ((curFeat != null))
{
    DoTheWork(curFeat, ParentName);
    Feature subFeat = default(Feature);
    subFeat = (Feature)curFeat.GetFirstSubFeature();
    while ((subFeat != null))
    {
        TraverseFeatures(subFeat, false, curFeat.Name);
        Feature nextSubFeat = default(Feature);
        nextSubFeat = (Feature)subFeat.GetNextSubFeature();
        subFeat = (Feature)nextSubFeat;
        nextSubFeat = null;
    }
    subFeat = null;
    Feature nextFeat = default(Feature);
    if (isTopLevel)
    {
        nextFeat = (Feature)curFeat.GetNextFeature();
    }
    else
    {
        nextFeat = null;
    }
    curFeat = (Feature)nextFeat;
    nextFeat = null;
}
}

static bool InitStaticVariableHelper(Microsoft.VisualBasic.CompilerServices.StaticLocalInitFlag flag)
{
    if (flag.State == 0)
    {
        flag.State = 2;
        return true;
    }
    else if (flag.State == 2)
    {
        throw new Microsoft.VisualBasic.CompilerServices.IncompleteInitialization();
    }
    else
    {
        return false;
    }
}

}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
}
}

```

Get Solid Bodies from Cut-list Folders and Get Custom Properties Example (VBA)

This example shows how to get the solid bodies from cut-list folders and how to get the custom properties for the solid bodies.

```
'-----
' Preconditions:
' 1. Verify that c:\temp\outputFile.txt does not exist; this macro
'    creates and writes to this file.
' 2. Add a reference to Microsoft Scripting Runtime (click
'    Tools > References > Browse > C:\windows\system32\scrrun.dll).
' 3. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 4. Click Tools > Options > Document Properties > Weldments >
'    Rename cut list folders with Description property value > OK.
' 5. Right-click Cut list(31) in the FeatureManager design tree
'    and click Update.
' 6. Open the Immediate window.
'
' Postconditions:
' 1. Traverses the FeatureManager design tree.
' 2. Writes Done! to the Immediate window when
'    the macro finishes executing.
' 3. Open and examine c:\temp\outputFile.txt.
'
' NOTE: Because this part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swPart As SldWorks.ModelDoc2
Dim swFeat As SldWorks.Feature
Dim fs As Scripting.FileSystemObject
Dim a As Scripting.TextStream

Sub GetFeatureCustomProps(thisFeat As SldWorks.Feature)

    Dim CustomPropMgr As SldWorks.CustomPropertyManager
    Set CustomPropMgr = thisFeat.CustomPropertyManager
    Dim vCustomPropNames As Variant

    vCustomPropNames = CustomPropMgr.GetNames
    If Not IsEmpty(vCustomPropNames) Then
        a.WriteLine "          Cut-list custom properties:"
        Dim i As Long
        For i = LBound(vCustomPropNames) To UBound(vCustomPropNames)
            Dim CustomPropName As String
            CustomPropName = vCustomPropNames(i)
            Dim CustomPropType As Long
            CustomPropType = CustomPropMgr.GetType2(CustomPropName)
            Dim CustomPropVal As String
            Dim CustomPropResolvedVal As String
            CustomPropMgr.Get2 CustomPropName, CustomPropVal, CustomPropResolvedVal
            a.WriteLine "          Name: " & CustomPropName
            a.WriteLine "          Value: " & CustomPropVal
            a.WriteLine "          Resolved value: " & CustomPropResolvedVal
        Next i
    End If
End Sub

Sub DoTheWork(thisFeat As SldWorks.Feature, ParentName As String)

    Static InBodyFolder As Boolean
    Static BodyFolderType(5) As String
    Static BeenHere As Boolean
    Dim bAllFeatures As Boolean
    Dim bCutListCustomProps As Boolean
    Dim vSuppressed As Variant

    If Not BeenHere Then
        BodyFolderType(0) = "dummy"
        BodyFolderType(1) = "swSolidBodyFolder"
        BodyFolderType(2) = "swSurfaceBodyFolder"
        BodyFolderType(3) = "swBodySubFolder"
        BodyFolderType(4) = "swWeldmentSubFolder"
        BodyFolderType(5) = "swWeldmentCutListFolder"
```

```

        InBodyFolder = False
        BeenHere = True
        bAllFeatures = False
        bCutListCustomProps = False
    End If

    'Comment out next line to print information for just BodyFolders
    bAllFeatures = True 'True to print information about all features

    'Comment out next line if you do not want cut list's custom properties
    bCutListCustomProps = True
    Dim FeatType As String
    FeatType = thisFeat.GetTypeName
    If (FeatType = "SolidBodyFolder") And (ParentName = "Root Feature") Then
        InBodyFolder = True
    End If
    If (FeatType <> "SolidBodyFolder") And (ParentName = "Root Feature") Then
        InBodyFolder = False
    End If

    'Only consider the CutListFolders that are under SolidBodyFolder
    If (InBodyFolder = False) And (FeatType = "CutListFolder") Then
        'Skip the second occurrence of the CutListFolders during the feature traversal
        Exit Sub
    End If

    'Only consider the SubWeldFolder that are under the SolidBodyFolder
    If (InBodyFolder = False) And (FeatType = "SubWeldFolder") Then
        'Skip the second occurrence of the SubWeldFolders during the feature traversal
        Exit Sub
    End If

    Dim IsBodyFolder As Boolean
    If FeatType = "SolidBodyFolder" Or FeatType = "SurfaceBodyFolder" Or FeatType = "CutListFolder" Or FeatType =
"SubWeldFolder" Or FeatType = "SubAtomFolder" Then
        IsBodyFolder = True
    Else
        IsBodyFolder = False
    End If

    If bAllFeatures And (Not IsBodyFolder) Then
        a.WriteLine "Feature name: " & thisFeat.Name
        a.WriteLine "    Feature type: " & FeatType
        vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Nothing)
        If IsEmpty(vSuppressed) Then
            a.WriteLine "        Suppression failed"
        Else
            a.WriteLine "        Suppressed"
        End If
    End If

    If IsBodyFolder Then
        Dim BodyFolder As SldWorks.BodyFolder
        Set BodyFolder = thisFeat.GetSpecificFeature2
        Dim BodyCount As Long
        BodyCount = BodyFolder.GetBodyCount

        If (FeatType = "CutListFolder") And (BodyCount < 1) Then
            'When BodyCount = 0, this cut list folder is not displayed in the
            'FeatureManager design tree, so skip it
            Exit Sub
        Else
            a.WriteLine "Feature name: " & thisFeat.Name
            vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Empty)
            If IsEmpty(vSuppressed) Then
                a.WriteLine "        Suppression failed"
            Else
                a.WriteLine "        Suppressed"
            End If
        End If

        If Not bAllFeatures Then
            a.WriteLine "Feature name: " & thisFeat.Name
            vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Empty)
            If IsEmpty(vSuppressed) Then
                a.WriteLine "        Suppression failed"
            Else
                a.WriteLine "        Suppressed"
            End If
        End If
    End If

```

```

Dim BodyFolderTypeE As Long
BodyFolderTypeE = BodyFolder.Type
a.WriteLine "      Body folder: " & BodyFolderType(BodyFolderTypeE)
a.WriteLine "      Body folder type: BodyFolderTypeE"
a.WriteLine "      Body count: " & BodyCount

Dim vBodies As Variant
vBodies = BodyFolder.GetBodies
Dim i As Long
If Not IsEmpty(vBodies) Then
    For i = LBound(vBodies) To UBound(vBodies)
        Dim Body As SldWorks.Body2
        Set Body = vBodies(i)
        a.WriteLine "      Body name: " & Body.Name
    Next i
End If
Else
    If bAllFeatures Then
        a.WriteLine ""
    End If
End If

If (FeatType = "CutListFolder") Then
    'When BodyCount = 0, this cut list folder is not displayed
    'in the FeatureManager design tree, so skip it
    If BodyCount > 0 Then
        If bCutListCustomProps Then
            'Comment out this line if you do not want to
            'print the cut list folder's custom properties
            GetFeatureCustomProps thisFeat
        End If
    End If
End If

End Sub

Sub TraverseFeatures(thisFeat As SldWorks.Feature, isTopLevel As Boolean, ParentName As String)

Dim curFeat As SldWorks.Feature
Set curFeat = thisFeat

While Not curFeat Is Nothing
    DoTheWork curFeat, ParentName
    Dim subfeat As SldWorks.Feature
    Set subfeat = curFeat.GetFirstSubFeature
    While Not subfeat Is Nothing
        TraverseFeatures subfeat, False, curFeat.Name
        Dim nextSubFeat As SldWorks.Feature
        Set nextSubFeat = subfeat.GetNextSubFeature
        Set subfeat = nextSubFeat
        Set nextSubFeat = Nothing
    Wend
    Set subfeat = Nothing
    Dim nextFeat As SldWorks.Feature
    If isTopLevel Then
        Set nextFeat = curFeat.GetNextFeature
    Else
        Set nextFeat = Nothing
    End If
    Set curFeat = nextFeat
    Set nextFeat = Nothing
Wend

End Sub

Sub main()

Set swApp = Application.SldWorks
Set swPart = swApp.ActiveDoc
Set fs = CreateObject("Scripting.FileSystemObject")
Set a = fs.CreateTextFile("c:\temp\outputFile.txt", True)
a.WriteLine "File: " & swPart.GetPathName
Dim ConfigName As String
ConfigName = swPart.ConfigurationManager.ActiveConfiguration.Name
a.WriteLine "Active configuration name: " & ConfigName
Set swFeat = swPart.FirstFeature
TraverseFeatures swFeat, True, "Root Feature"
a.Close
Debug.Print "Done!"

End Sub

```

Get Solid Bodies from Cut-list Folders and Get Custom Properties Example (VB.NET)

This example shows how to get the solid bodies from cut-list folders and how to get the custom properties for the solid bodies.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Click Tools > Options > Document Properties > Weldments >
'    Rename cut list folders with Description property value > OK.
' 3. Right-click Cut list(31) in the FeatureManager design tree
'    and click Update.
' 4. Open the Immediate window.
'
' Postconditions:
' 1. Traverses the FeatureManager design tree.
' 2. Examine the Immediate window.
'
' NOTE: Because this part is used elsewhere, do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim swPart As ModelDoc2
    Dim swFeat As Feature

    Sub GetFeatureCustomProps(ByVal thisFeat As Feature)
        Dim CustomPropMgr As CustomPropertyManager
        CustomPropMgr = thisFeat.CustomPropertyManager
        Dim vCustomPropNames As Object
        vCustomPropNames = CustomPropMgr.GetNames
        If Not IsNothing(vCustomPropNames) Then
            Debug.Print("          Cut-list custom properties:")
            Dim i As Integer
            For i = LBound(vCustomPropNames) To UBound(vCustomPropNames)
                Dim CustomPropName As String
                CustomPropName = vCustomPropNames(i)
                Dim CustomPropType As Integer
                CustomPropType = CustomPropMgr.GetType2(CustomPropName)
                Dim CustomPropVal As String = ""
                Dim CustomPropResolvedVal As String = ""
                CustomPropMgr.Get2(CustomPropName, CustomPropVal, CustomPropResolvedVal)
                Debug.Print("          Name: " & CustomPropName)
                Debug.Print("          Value: " & CustomPropVal)
                Debug.Print("          Resolved value: " & CustomPropResolvedVal)
            Next i
        End If
    End Sub

    Sub DoTheWork(ByVal thisFeat As Feature, ByVal ParentName As String)
        Static InBodyFolder As Boolean
        Static BodyFolderType(5) As String
        Static BeenHere As Boolean
        Dim bAllFeatures As Boolean
        Dim bCutListCustomProps As Boolean
        Dim vSuppressed As Object
        Dim BodyCount As Integer

        If Not BeenHere Then
            BodyFolderType(0) = "dummy"
            BodyFolderType(1) = "swSolidBodyFolder"
            BodyFolderType(2) = "swSurfaceBodyFolder"
            BodyFolderType(3) = "swBodySubFolder"
            BodyFolderType(4) = "swWeldmentSubFolder"
            BodyFolderType(5) = "swWeldmentCutListFolder"
            InBodyFolder = False
            BeenHere = True
            bAllFeatures = False
            bCutListCustomProps = False
        End If

        'Comment out next line to print information for just BodyFolders
        bAllFeatures = True 'True to print information about all features

        'Comment out next line if you do not want cut list's custom properties
        bCutListCustomProps = True

        Dim FeatType As String
        FeatType = thisFeat.GetTypeName
        If (FeatType = "SolidBodyFolder") And (ParentName = "Root Feature") Then

```

```

        InBodyFolder = True
    End If
    If (FeatType <> "SolidBodyFolder") And (ParentName = "Root Feature") Then
        InBodyFolder = False
    End If

    'Only consider the CutListFolders that are under SolidBodyFolder
    If (InBodyFolder = False) And (FeatType = "CutListFolder") Then
        'Skip the second occurrence of the CutListFolders during the feature traversal
        Exit Sub
    End If

    'Only consider the SubWeldFolder that are under the SolidBodyFolder
    If (InBodyFolder = False) And (FeatType = "SubWeldFolder") Then
        'Skip the second occurrence of the SubWeldFolders during the feature traversal
        Exit Sub
    End If

    Dim IsBodyFolder As Boolean
    If FeatType = "SolidBodyFolder" Or FeatType = "SurfaceBodyFolder" Or FeatType = "CutListFolder" Or FeatType = "SubWeldFolder" Or FeatType = "SubAtomFolder" Then
        IsBodyFolder = True
    Else
        IsBodyFolder = False
    End If
    If bAllFeatures And (Not IsBodyFolder) Then
        Debug.Print("Feature name: " & thisFeat.Name)
        Debug.Print("  Feature type: " & FeatType)
        vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Nothing)
        If IsNothing(vSuppressed) Then
            Debug.Print("    Suppression failed")
        Else
            Debug.Print("    Suppressed")
        End If
    End If
    If IsBodyFolder Then
        Dim BodyFolder As BodyFolder
        BodyFolder = thisFeat.GetSpecificFeature2
        BodyCount = BodyFolder.GetBodyCount
        If (FeatType = "CutListFolder") And (BodyCount < 1) Then
            'When BodyCount = 0, this cut list folder is not displayed in the
            'FeatureManager design tree, so skip it
            Exit Sub
        Else
            Debug.Print("Feature name: " & thisFeat.Name)
            Debug.Print("  Feature type: " & FeatType)
            vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Nothing)
            If IsNothing(vSuppressed) Then
                Debug.Print("    Suppression failed")
            Else
                Debug.Print("    Suppressed")
            End If
        End If
    End If
    If Not bAllFeatures Then
        Debug.Print("Feature name: " & thisFeat.Name)
        Debug.Print("  Feature type: " & FeatType)
        vSuppressed = thisFeat.IsSuppressed2(swInConfigurationOpts_e.swThisConfiguration, Nothing)
        If IsNothing(vSuppressed) Then
            Debug.Print("    Suppression failed")
        Else
            Debug.Print("    Suppressed")
        End If
    End If
    Dim BodyFolderTypeE As Integer
    BodyFolderTypeE = BodyFolder.Type
    Debug.Print("    Body folder: " & BodyFolderType(BodyFolderTypeE))
    Debug.Print("    Body folder type: " & BodyFolderTypeE)
    Debug.Print("    Body count: " & BodyCount)
    Dim vBodies As Object
    vBodies = BodyFolder.GetBodies
    Dim i As Integer
    If Not IsNothing(vBodies) Then
        For i = LBound(vBodies) To UBound(vBodies)
            Dim Body As Body2
            Body = vBodies(i)
            Debug.Print("      Body name: " & Body.Name)
        Next i
    End If
Else
    If bAllFeatures Then
        Debug.Print("")
    End If
End If
If (FeatType = "CutListFolder") Then
    'When BodyCount = 0, this cut-list folder is not displayed
    'in the FeatureManager design tree, so skip it
    If BodyCount > 0 Then
        If bCutListCustomProps Then
            'Comment out this line if you do not want to
            'print the cut-list folder's custom properties
            GetFeatureCustomProps(thisFeat)
        End If
    End If
End If

```

```

End If

End Sub
Sub TraverseFeatures(ByVal thisFeat As Feature, ByVal isTopLevel As Boolean, ByVal ParentName As String)
    Dim curFeat As Feature
    curFeat = thisFeat
    While Not curFeat Is Nothing
        DoTheWork(curFeat, ParentName)
        Dim subFeat As Feature
        subFeat = curFeat.GetFirstSubFeature
        While Not subFeat Is Nothing
            TraverseFeatures(subFeat, False, curFeat.Name)
            Dim nextSubFeat As Feature
            nextSubFeat = subFeat.GetNextSubFeature
            subFeat = nextSubFeat
            nextSubFeat = Nothing
        End While
        subFeat = Nothing
        Dim nextFeat As Feature
        If isTopLevel Then
            nextFeat = curFeat.GetNextFeature
        Else
            nextFeat = Nothing
        End If
        curFeat = nextFeat
        nextFeat = Nothing
    End While
End Sub

Sub main()
    swPart = swApp.ActiveDoc
    Debug.Print("File: " & swPart.GetPathName)
    Dim ConfigName As String
    ConfigName = swPart.ConfigurationManager.ActiveConfiguration.Name
    Debug.Print("Active configuration name: " & ConfigName)
    swFeat = swPart.FirstFeature
    TraverseFeatures(swFeat, True, "Root Feature")

End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Get SOLIDWORKS Search Third-party Keywords Example (VBA)

This example shows how to get the SOLIDWORKS Search third-party keywords for a model document.

```
'-----  
' Preconditions:  
' 1. Open a model document that was previously assigned  
'    SOLIDWORKS Search third-party keywords  
'    and which have been indexed in SOLIDWORKS Design  
'    - or -  
'    run Add Third-party Application Keywords to SOLIDWORKS Search and Model Example \(VBA\).  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the number of Search third-party keywords  
'    assigned to the open model document and the keywords  
'    and values.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim vAppName As Variant, vKeywordNames As Variant, vKeywordValues As Variant  
Dim i As Long, lNbrKeywords As Long  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swModelDocExt = swModel.Extension  
  
    lNbrKeywords = swModelDocExt.GetSearchData("COSMOSWorks", vAppName, vKeywordNames, vKeywordValues)  
    Debug.Print "Number of keywords: " & lNbrKeywords  
    For i = 0 To lNbrKeywords - 1  
        Debug.Print "Keyword: " & vKeywordNames(i)  
        Debug.Print "    Value: " & vKeywordValues(i)  
    Next i  
  
End Sub
```

Get Version Number in which Feature Was Created Example (VBA)

This example shows how to get the version number in which the selected feature was created.

```
'-----  
' Preconditions:  
' 1. Open a part and select a feature.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the version of SOLIDWORKS Design in which the selected  
'    feature was created.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swFeat As SldWorks.Feature  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject5(1)  
    Debug.Print "Version of SOLIDWORKS Design in which feature was created = " & swFeat.GetCreatedVersion  
  
End Sub
```

Get Version Number in Which Feature Was Last Modified Example (VBA)

This example shows how to get the version number in which the selected feature was last modified.

```
'-----  
' Preconditions:  
' 1. Open a part and select a feature.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the name of the selected feature, its type, and the version  
'    of SOLIDWORKS Design in which the selected feature was last modified.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swFeat As SldWorks.Feature  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject5(1)  
    Debug.Print "Feature = " + swFeat.Name + "[" + swFeat.GetTypeName + "]"  
    Debug.Print "Version of SOLIDWORKS Design in which feature was last modified = " & swFeat.GetModifiedVersion  
  
End Sub
```

Get Version of Display Dimension (C#)

This example shows how to find out if a display dimension in a drawing was created in SOLIDWORKS 2011 or later.

```
//-----
// Preconditions:
// 1. Open a drawing document in which a display
//    dimension exists.
// 2. Open the Immediate window.
// 3. Select the display dimension.
//
// Postconditions: Examine the Immediate window
// to see if the selected display dimension
// was created in SOLIDWORKS 2011 or later.
//-----
using System;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Diagnostics;

namespace GetSupportGenericTextDisplayDimensionCSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            SelectionMgr swSelMgr = default(SelectionMgr);
            DisplayDimension swDispDim = default(DisplayDimension);
            object swSelObj = null;
            int selCount = 0;
            int selType = 0;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;

            selCount = swSelMgr.GetSelectedObjectCount2(-1);
            if (selCount < 1)
            {
                Debug.Print("Select a display dimension and rerun the macro.");
                return;
            }

            selType = swSelMgr.GetSelectedObjectType3(1, 0);
            swSelObj = (object)swSelMgr.GetSelectedObject6(1, 0);
            switch (selType)
            {
                case (int)swSelectType_e.swSelDIMENSIONS:
                    swDispDim = (DisplayDimension)swSelObj;
                    Debug.Print("Was display dimension created in SOLIDWORKS 2011 or later? " +
swDispDim.GetSupportsGenericText());
                    break;
            }
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>

        public SldWorks swApp;
    }
}
```

Get Version of Display Dimension (VBA)

This example shows how to find out if a display dimension in a drawing was created in SOLIDWORKS 2011 or later.

```
'-----  
' Preconditions:  
' 1. Open a drawing document in which a display  
'    dimension exists.  
' 2. Open the Immediate window.  
' 3. Select the display dimension.  
'  
' Postconditions: Examine the Immediate window  
' to see if the selected display dimension  
' was created in SOLIDWORKS 2011 or later.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swDispDim As SldWorks.DisplayDimension  
Dim swSelObj As Object  
Dim selCount As Long  
Dim selType As Long  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
  
    selCount = swSelMgr.GetSelectedObjectCount2(-1)  
    If selCount < 1 Then  
        Debug.Print "Select a display dimension and rerun the macro."  
        Exit Sub  
    End If  
  
    selType = swSelMgr.GetSelectedObjectType3(1, 0)  
    Set swSelObj = swSelMgr.GetSelectedObject6(1, 0)  
    Select Case selType  
        Case swSelDIMENSIONS  
            Set swDispDim = swSelObj  
            Debug.Print "Was display dimension created in SOLIDWORKS 2011 or later? " &  
swDispDim.GetSupportsGenericText  
    End Select  
  
End Sub
```

Get Version of Display Dimension (VB.NET)

This example shows how to find out if a display dimension in a drawing was created in SOLIDWORKS 2011 or later.

```
' -----  
' Preconditions:  
' 1. Open a drawing document in which a display  
'    dimension exists.  
' 2. Open the Immediate window.  
' 3. Select the display dimension.  
'  
' Postconditions: Examine the Immediate window  
' to see if the selected display dimension  
' was created in SOLIDWORKS 2011 or later.  
' -----  
  
Imports  
SolidWorks.Interop.sldworks  
Imports  
SolidWorks.Interop.swconst  
Imports  
System  
Imports  
System.Diagnostics  
Partial  
  
Class SolidWorksMacro  
    Public Sub Main()  
        Dim swModel As ModelDoc2  
  
        Dim swSelMgr As SelectionMgr  
        Dim swDispDim As DisplayDimension  
        Dim swSelObj As Object  
        Dim selCount As Long  
        Dim selType As Long  
  
        swModel = swApp.ActiveDoc  
        swSelMgr = swModel.SelectionManager  
        selCount = swSelMgr.GetSelectedObjectCount2(-1)  
        If selCount < 1 Then  
            Debug.Print("Select a display dimension and rerun the macro.")  
            Exit Sub  
        End If  
  
        selType = swSelMgr.GetSelectedObjectType3(1, 0)  
        swSelObj = swSelMgr.GetSelectedObject6(1, 0)  
        Select Case selType  
            Case swSelectType_e.swSelDIMENSIONS  
                swDispDim = swSelObj  
                Debug.Print("Was display dimension created in SOLIDWORKS 2011 or later? " &  
                    swDispDim.GetSupportsGenericText)  
        End Select  
    End Sub
```

```
''' <summary>  
''' The SldWorks swApp variable is pre-assigned for you.  
''' </summary>
```

```
Public swApp As SldWorks
```

```
End
```

```
Class
```

Get Spline's Parameters Example (C#)

This example shows how to get spline parameterization data for a selected sketch.

```
//-----  
// Preconditions:  
// 1. Open a new part document.  
// 2. Rename the namespace of this macro to match the name of your C# project.  
// 3. Open an Immediate Window.  
//  
// Postconditions:  
// 1. Create and selects a sketch with a spline.  
// 2. Inspect the Immediate Window to view the spline's parameterization data.  
//-----  
using SolidWorks.Interop.sldworks;  
using SolidWorks.Interop.swconst;  
using System;  
using System.Diagnostics;  
namespace GetSplineParameterData_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
        ModelDoc2 swModel;  
        SelectionMgr swSelMgr;  
        Feature swFeat;  
        Sketch swSketch;  
        SplineParamData swSplineParaData;  
        object varCtrlPoints;  
        double[] ctrlPoints;  
        object varKnotPoints;  
        double[] knotPoints;  
        bool boolStatus;  
        int i;  
        SketchManager swSketchMgr;  
        double[] arrCtrlPts = new double[18];  
        double[] arrKnotPts = new double[10];  
        object varSeg;  
  
        public void Main()  
        {  
            swModel = (ModelDoc2)swApp.ActiveDoc;  
            swSketchMgr = swModel.SketchManager;  
            swSplineParaData = swSketchMgr.CreateSplineParamData();  
  
            swSplineParaData.Dimension = 3;  
            swSplineParaData.Order = 4;  
            swSplineParaData.Periodic = 0;  
            swSplineParaData.ControlPointsCount = 6;  
  
            arrCtrlPts[0] = -0.0508;  
            arrCtrlPts[1] = 0.0254;  
            arrCtrlPts[2] = 0;  
  
            arrCtrlPts[3] = -0.0467151082143202;  
            arrCtrlPts[4] = 0.0102856278912272;  
            arrCtrlPts[5] = 0;  
  
            arrCtrlPts[6] = -0.00128736279059822;  
            arrCtrlPts[7] = -0.0242718277078294;  
            arrCtrlPts[8] = 0;  
  
            arrCtrlPts[9] = 0.0314466792227059;  
            arrCtrlPts[10] = 0.0612396847740687;  
            arrCtrlPts[11] = 0;  
  
            arrCtrlPts[12] = 0.060476660388647;  
            arrCtrlPts[13] = -0.0433198423870343;  
            arrCtrlPts[14] = 0;
```

```

arrCtrlPts[15] = 0.0762;
arrCtrlPts[16] = 0.0254;
arrCtrlPts[17] = 0;

boolStatus = swSplineParaData.SetControlPoints(arrCtrlPts);

arrKnotPts[0] = 0;
arrKnotPts[1] = 0;
arrKnotPts[2] = 0;
arrKnotPts[3] = 0;
arrKnotPts[4] = 0.362665828616751;
arrKnotPts[5] = 0.575110552411167;
arrKnotPts[6] = 1;
arrKnotPts[7] = 1;
arrKnotPts[8] = 1;
arrKnotPts[9] = 1;
boolStatus = swSplineParaData.SetKnotPoints(arrKnotPts);
varSeg = swSketchMgr.CreateSplinesByEqnParams2(swSplineParaData);

swSketchMgr.InsertSketch(true);

boolStatus = swModel.Extension.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, false, 0, null,
(int)swSelectOption_e.swSelectOptionDefault);

swSelMgr = (SelectionMgr)swModel.SelectionManager;
swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
swSketch = (Sketch)swFeat.GetSpecificFeature2();
Debug.Print(swFeat.Name);
Debug.Print("");

swSplineParaData = swSketch.GetSplineParams4(true);

Debug.Print("The dimension is: " + swSplineParaData.Dimension);
Debug.Print("The order is: " + swSplineParaData.Order);
Debug.Print("The periodicity is: " + swSplineParaData.Periodic);
Debug.Print("The control point count is: " + swSplineParaData.ControlPointsCount);
boolStatus = swSplineParaData.GetControlPoints(out varCtrlPoints);
Debug.Print("Control Points:");
ctrlPoints = (double[])varCtrlPoints;
for (i = 0; i <= ctrlPoints.GetUpperBound(0); i++)
{
    Debug.Print(ctrlPoints[i].ToString());
}
Debug.Print("The knot point count is: " + swSplineParaData.KnotPointsCount);
boolStatus = swSplineParaData.GetKnotPoints(out varKnotPoints);
Debug.Print("Knot Points:");
knotPoints = (double[])varKnotPoints;
for (i = 0; i <= knotPoints.GetUpperBound(0); i++)
{
    Debug.Print(knotPoints[i].ToString());
}
Debug.Print("The color is: " + swSplineParaData.Color);
Debug.Print("The line style is: " + swSplineParaData.LineStyle);
Debug.Print("The line width is: " + swSplineParaData.LineWidth);
Debug.Print("The layer is: " + swSplineParaData.Layer);

Debug.Print("The layer override is: " + swSplineParaData.LayerOverride);
}
public SldWorks swApp;
}
}

```

Get Spline's Parameters Example (VBA)

This example shows how to get spline parameterization data for a selected sketch.

```
'-----  
' Preconditions:  
' 1. Open a new part document.  
' 2. Open an Immediate Window.  
'  
' Postconditions:  
' 1. Creates and selects a sketch with a spline.  
' 2. Inspect the Immediate Window to view the spline's parameterization data.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swFeat As SldWorks.Feature  
Dim swSketch As SldWorks.Sketch  
Dim swSplineParaData As SldWorks.SplineParamData  
Dim varCtrlPoints As Variant  
Dim varKnotPoints As Variant  
Dim boolStatus As Boolean  
Dim i As Integer  
Dim swSketchMgr As SketchManager  
Dim arrCtrlPts(17) As Double  
Dim arrKnotPts(9) As Double  
Dim varSeg As Variant  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSketchMgr = swModel.SketchManager  
    Set swSplineParaData = swSketchMgr.CreateSplineParamData  
  
    swSplineParaData.Dimension = 3  
    swSplineParaData.Order = 4  
    swSplineParaData.Periodic = 0  
    swSplineParaData.ControlPointsCount = 6  
  
    arrCtrlPts(0) = -0.0508  
    arrCtrlPts(1) = 0.0254  
    arrCtrlPts(2) = 0  
  
    arrCtrlPts(3) = -4.67151082143202E-02  
    arrCtrlPts(4) = 1.02856278912272E-02  
    arrCtrlPts(5) = 0  
  
    arrCtrlPts(6) = -1.28736279059822E-03  
    arrCtrlPts(7) = -2.42718277078294E-02  
    arrCtrlPts(8) = 0  
  
    arrCtrlPts(9) = 3.14466792227059E-02  
    arrCtrlPts(10) = 6.12396847740687E-02  
    arrCtrlPts(11) = 0  
  
    arrCtrlPts(12) = 0.060476660388647  
    arrCtrlPts(13) = -4.33198423870343E-02  
    arrCtrlPts(14) = 0  
  
    arrCtrlPts(15) = 0.0762  
    arrCtrlPts(16) = 0.0254  
    arrCtrlPts(17) = 0
```

```

boolStatus = swSplineParaData.SetControlPoints(arrCtrlPts)

arrKnotPts(0) = 0
arrKnotPts(1) = 0
arrKnotPts(2) = 0
arrKnotPts(3) = 0
arrKnotPts(4) = 0.362665828616751
arrKnotPts(5) = 0.575110552411167
arrKnotPts(6) = 1
arrKnotPts(7) = 1
arrKnotPts(8) = 1
arrKnotPts(9) = 1
boolStatus = swSplineParaData.SetKnotPoints(arrKnotPts)
varSeg = swSketchMgr.CreateSplinesByEqnParams2(swSplineParaData)

swSketchMgr.InsertSketch (True)

boolStatus = swModel.Extension.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, False, 0, Nothing,
swSelectOptionDefault)

Set swSelMgr = swModel.SelectionManager
Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
Set swSketch = swFeat.GetSpecificFeature2
Debug.Print swFeat.Name
Debug.Print ""

Set swSplineParaData = swSketch.GetSplineParams4(True)

Debug.Print "The dimension is: " & swSplineParaData.Dimension
Debug.Print "The order is: " & swSplineParaData.Order
Debug.Print "The periodicity is: " & swSplineParaData.Periodic
Debug.Print "The control point count is: " & swSplineParaData.ControlPointsCount
boolStatus = swSplineParaData.GetControlPoints(varCtrlPoints)
Debug.Print "Control Points:"
For i = 0 To UBound(varCtrlPoints)
    Debug.Print varCtrlPoints(i)

Next i
Debug.Print "The knot point count is: " & swSplineParaData.KnotPointsCount
boolStatus = swSplineParaData.GetKnotPoints(varKnotPoints)
Debug.Print "Knot Points:"
For i = 0 To UBound(varKnotPoints)
    Debug.Print varKnotPoints(i)

Next i
Debug.Print "The color is: " & swSplineParaData.Color
Debug.Print "The line style is: " & swSplineParaData.LineStyle
Debug.Print "The line width is: " & swSplineParaData.LineWidth
Debug.Print "The layer is: " & swSplineParaData.Layer
Debug.Print "The layer override is: " & swSplineParaData.LayerOverride

End Sub

```

Get Spline's Parameters Example (VB.NET)

This example shows how to get spline parameterization data for a selected sketch.

```
'-----  
' Preconditions:  
' 1. Open a new part document.  
' 2. Open an Immediate Window.  
'  
' Postconditions:  
' 1. Creates and selects a sketch with a spline.  
' 2. Inspect the Immediate Window to view the spline's parameterization data.  
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System
```

```
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2  
    Dim swSelMgr As SelectionMgr  
    Dim swFeat As Feature  
    Dim swSketch As Sketch  
    Dim swSplineParaData As SplineParamData  
    Dim varCtrlPoints As Object  
    Dim varKnotPoints As Object  
    Dim boolStatus As Boolean  
    Dim i As Integer  
    Dim swSketchMgr As SketchManager  
    Dim arrCtrlPts(17) As Double  
    Dim arrKnotPts(9) As Double  
    Dim varSeg As Object
```

```
Sub main()
```

```
    swModel = swApp.ActiveDoc  
    swSketchMgr = swModel.SketchManager  
    swSplineParaData = swSketchMgr.CreateSplineParamData
```

```
    swSplineParaData.Dimension = 3  
    swSplineParaData.Order = 4  
    swSplineParaData.Periodic = 0  
    swSplineParaData.ControlPointsCount = 6
```

```
    arrCtrlPts(0) = -0.0508  
    arrCtrlPts(1) = 0.0254  
    arrCtrlPts(2) = 0
```

```
    arrCtrlPts(3) = -0.0467151082143202  
    arrCtrlPts(4) = 0.0102856278912272  
    arrCtrlPts(5) = 0
```

```
    arrCtrlPts(6) = -0.00128736279059822  
    arrCtrlPts(7) = -0.0242718277078294  
    arrCtrlPts(8) = 0
```

```
    arrCtrlPts(9) = 0.0314466792227059  
    arrCtrlPts(10) = 0.0612396847740687  
    arrCtrlPts(11) = 0
```

```
    arrCtrlPts(12) = 0.060476660388647  
    arrCtrlPts(13) = -0.0433198423870343  
    arrCtrlPts(14) = 0
```

```
    arrCtrlPts(15) = 0.0762
```

```

arrCtrlPts(16) = 0.0254
arrCtrlPts(17) = 0

boolStatus = swSplineParaData.SetControlPoints(arrCtrlPts)

arrKnotPts(0) = 0
arrKnotPts(1) = 0
arrKnotPts(2) = 0
arrKnotPts(3) = 0
arrKnotPts(4) = 0.362665828616751
arrKnotPts(5) = 0.575110552411167
arrKnotPts(6) = 1
arrKnotPts(7) = 1
arrKnotPts(8) = 1
arrKnotPts(9) = 1
boolStatus = swSplineParaData.SetKnotPoints(arrKnotPts)
varSeg = swSketchMgr.CreateSplinesByEqnParams2(swSplineParaData)

swSketchMgr.InsertSketch(True)

boolStatus = swModel.Extension.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, False, 0, Nothing,
swSelectOption_e.swSelectOptionDefault)

swSelMgr = swModel.SelectionManager
swFeat = swSelMgr.GetSelectedObject6(1, -1)
swSketch = swFeat.GetSpecificFeature2
Debug.Print(swFeat.Name)
Debug.Print("")

swSplineParaData = swSketch.GetSplineParams4(True)

Debug.Print("The dimension is: " & swSplineParaData.Dimension)
Debug.Print("The order is: " & swSplineParaData.Order)
Debug.Print("The periodicity is: " & swSplineParaData.Periodic)
Debug.Print("The control point count is: " & swSplineParaData.ControlPointsCount)
boolStatus = swSplineParaData.GetControlPoints(varCtrlPoints)
Debug.Print("Control Points:")
For i = 0 To UBound(varCtrlPoints)
    Debug.Print(varCtrlPoints(i))

Next i
Debug.Print("The knot point count is: " & swSplineParaData.KnotPointsCount)
boolStatus = swSplineParaData.GetKnotPoints(varKnotPoints)
Debug.Print("Knot Points:")
For i = 0 To UBound(varKnotPoints)
    Debug.Print(varKnotPoints(i))

Next i
Debug.Print("The color is: " & swSplineParaData.Color)
Debug.Print("The line style is: " & swSplineParaData.LineStyle)
Debug.Print("The line width is: " & swSplineParaData.LineWidth)
Debug.Print("The layer is: " & swSplineParaData.Layer)
Debug.Print("The layer override is: " & swSplineParaData.LayerOverride)

End Sub

Public swApp As SldWorks

End Class

```

Get Spline Parameters Example (VBA)

This example shows how to get the parameters of a spline.

```
'-----
' Preconditions:
' 1. Open a part containing a sketch of a spline.
' 2. Select the sketch of the spline.
'
' Postconditions:
' 1. Gets the parameters of the spline.
' 2. Examine the Immediate window.
'-----
Option Explicit

Public Enum swSketchSegments_e
    swSketchLINE = 0
    swSketchARC = 1
    swSketchELLIPSE = 2
    swSketchSPLINE = 3
    swSketchTEXT = 4
    swSketchPARABOLA = 5
End Enum

' Define two types
Type DoubleRec
    dValue As Double
End Type

Type Int2Rec
    'Assume C int has 4 bytes
    iLower As Long
    iUpper As Long
End Type

'Extract two integer values from a single double value
'by assigning a DoubleRec to the double value and then
'copying the value to Int2Rec and
'extracting the integer values
Function ExtractFields(dValue As Double, iLower As Long, iUpper As Long)

    Dim dr                As DoubleRec
    Dim i2r                As Int2Rec

    'Set the double value
    dr.dValue = dValue

    'Copy the values
    LSet i2r = dr

    'Extract the values
    iLower = i2r.iLower
    iUpper = i2r.iUpper

End Function

Sub main()

    Dim swApp                As SldWorks.SldWorks
    Dim swModel               As SldWorks.ModelDoc2
    Dim swSelMgr              As SldWorks.SelectionMgr
    Dim swFeat                As SldWorks.Feature
    Dim swSketch              As SldWorks.Sketch
    Dim vSkSeg                As Variant
    Dim swSkSeg               As SldWorks.SketchSegment
    Dim swSkSpline            As SldWorks.SketchSpline
    Dim vSplineParam          As Variant
    Dim vSplinePt             As Variant
    Dim swSkPt                As SldWorks.SketchPoint
    Dim nSplineParam0         As Double
    Dim nSplineParam1         As Double
    Dim nDim                  As Long
    Dim nOrder                As Long
    Dim nCtrlPoints           As Long
```

```

Dim nPeriodic           As Long
Dim nNumKnots           As Long
Dim nStyleLayerParam0   As Double
Dim nStyleLayerParam1   As Double
Dim nStyleLayerParam2   As Double
Dim nColor              As Long
Dim nLineStyle          As Long
Dim nLineWidth          As Long
Dim nLayer              As Long
Dim nLayerOverride      As Long
Dim nDummy              As Long
Dim nIndex              As Long
Dim nOffset             As Long
Dim i                   As Long
Dim j                   As Long

Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swSelMgr = swModel.SelectionManager
Set swFeat = swSelMgr.GetSelectedObject5(1)
Set swSketch = swFeat.GetSpecificFeature2

Debug.Print swFeat.Name

vSplineParam = swSketch.GetSplineParams2
nIndex = 0

Do While nIndex < UBound(vSplineParam)
    nOffset = nIndex

    ' Get spline data from first two packed elements
    nSplineParam0 = vSplineParam(nOffset + 0): nIndex = nIndex + 1
    nSplineParam1 = vSplineParam(nOffset + 1): nIndex = nIndex + 1
    ExtractFields nSplineParam0, nDim, nOrder
    ExtractFields nSplineParam1, nCtrlPoints, nPeriodic

    If 1 = nPeriodic Then
        nNumKnots = nCtrlPoints + 1
    Else
        nNumKnots = nCtrlPoints + nOrder
    End If

    Debug.Print " Dim          = " & nDim
    Debug.Print " Order        = " & nOrder
    Debug.Print " CtrlPoints    = " & nCtrlPoints
    Debug.Print " Periodic      = " & nPeriodic
    Debug.Print " NumKnots      = " & nNumKnots
    Debug.Print ""

    ' Get control point data
    ' Dimension is always 3 or 4
    Debug.Assert 3 = nDim Or 4 = nDim
    For i = 2 + nOffset To 2 + nOffset + nCtrlPoints * nDim - 3 Step nDim
        Debug.Print " CtrlPt[" & i & "] = (" & vSplineParam(i + 0) * 1000# & ", " & vSplineParam(i + 1) * 1000# &
        ", " & vSplineParam(i + 2) * 1000# & ") mm"
        If 4 = nDim Then
            Debug.Print " Weight[" & i & "] = " & vSplineParam(i + 3)
        End If
        nIndex = nIndex + nDim
    Next i
    Debug.Print ""
    For i = 2 + nOffset + nCtrlPoints * nDim To 2 + nOffset + nCtrlPoints * nDim + nNumKnots - 1
        ' Knot weights must be non-descending
        Debug.Print " Knot[" & i & "] = " & vSplineParam(i)
        nIndex = nIndex + 1
    Next i
    Debug.Print ""

    ' Get style from packed elements - only supported for drawings
    nStyleLayerParam0 = vSplineParam(2 + nOffset + nNumKnots + nCtrlPoints * nDim + 0): nIndex = nIndex + 0
    nStyleLayerParam1 = vSplineParam(2 + nOffset + nNumKnots + nCtrlPoints * nDim + 1): nIndex = nIndex + 1
    nStyleLayerParam2 = vSplineParam(2 + nOffset + nNumKnots + nCtrlPoints * nDim + 2): nIndex = nIndex + 1
    ExtractFields nStyleLayerParam0, nColor, nLineStyle
    ExtractFields nStyleLayerParam1, nLineWidth, nLayer
    ExtractFields nStyleLayerParam2, nLayerOverride, nDummy

    Debug.Print " Color          = " & nColor
    Debug.Print " LineStyle       = " & nLineStyle
    Debug.Print " LineWidth       = " & nLineWidth
    Debug.Print " Layer           = " & nLayer
    Debug.Print " LayerOverride   = " & nLayerOverride
    Debug.Print " "

```

```

        nIndex = nIndex + 1
    Loop

    vSkSeg = swSketch.GetSketchSegments
    For i = 0 To UBound(vSkSeg)
        Set swSkSeg = vSkSeg(i)

        If swSketchSPLINE = swSkSeg.GetType Then
            Set swSkSpline = swSkSeg

            ' Spline passes through these points
            ' first, and the last spline points are the same
            ' as the first and last control points
            vSplinePt = swSkSpline.GetPoints2
            For j = 0 To UBound(vSplinePt)
                Set swSkPt = vSplinePt(j)
                Debug.Print " SketchSplinePt[" & j & "] = (" & swSkPt.X * 1000# & ", " & swSkPt.Y * 1000# & ", " &
swSkPt.Z * 1000# & ") mm"
            Next j
        End If
        Debug.Print ""
    Next i

End Sub

```

Get Spline Points Example (C++)

This example shows how to get the points on a spline.

```
//-----
// Preconditions:
// 1. Start up SOLIDWORKS Design and open a new part document.
// 2. Sketch a spline.
// 3. Select the Sketch1 feature of the spline in the FeatureManager design tree.
//
// NOTE: Do not select the spline in the graphics area. This
// causes an assert, which results in you having to abort and
// terminate the debug session.
//
// 4. Start Microsoft Visual Studio 2019 or later.
// a. Click File > New > Project > (C++ Windows) Console App.
// b. Click Next.
// c. Type the name of your project in Project Name.
// d. Click Create.
// e. Right-click project, select Add, New Item, Select ATL > ATL Support, Click Add.
// f. Click Project > projectname Properties > Configuration Properties >
// C/C++ and type the path to sldworks.tlb and swconst.tlb,
// typically C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS,
// in Additional Include Directories.
// g. Click Apply or OK.
// 5. Replace the code in the code window with This code.
// 6. Click Debug > Start Debugging.
//
// Postconditions:
// 1. Prints to the Output Window the number of spline points followed by
// the x, y, z values of each point in the spline.
// 2. Examine the Output Window.
//-----

//This code
#include "framework.h"
#import "sldworks.tlb" raw_interfaces_only, raw_native_types, no_namespace, named_guids // SOLIDWORKS type library
#import "swconst.tlb" raw_interfaces_only, raw_native_types, no_namespace, named_guids // SOLIDWORKS constants type library
#include <iostream>
#include <sstream>

int _tmain(int argc, _TCHAR* argv[])
{
    // Initialize COM
    // Do this before using ATL smart pointers so that
    // COM is available
    ::CoInitialize(NULL);

    {
        //Use ATL smart pointers
        CComPtr<ISldWorks> pSwApp;
        if (pSwApp.CoCreateInstance(L"Sldworks.Application", NULL, CLSCTX_LOCAL_SERVER) != S_OK) {
            return(0);
        }

        CComPtr<IModelDoc2> pSwModel;
        pSwApp->get_IActiveDoc2(&pSwModel);

        //Get the selected sketch feature
        CComPtr<ISelectionMgr> pSwSelMgr;
        pSwModel->get_ISelectionManager(&pSwSelMgr);
        IDispatch* pDisp;
        pSwSelMgr->GetSelectedObject6(1, -1, &pDisp);

        CComPtr<IFeature> pSwFeat;
        if (S_OK == pDisp->QueryInterface(IID_IFeature, (void**)&pSwFeat);
        pDisp = NULL;

        //Get the sketch feature
        pSwFeat->GetSpecificFeature2(&pDisp);
        CComPtr<ISketch> pSwSketch;
        if (S_OK == pDisp->QueryInterface(IID_ISketch, (void**)&pSwSketch);
        pDisp = NULL;

        //Gets the spline points by interpolation
        VARIANT vSplines;
        ::VariantInit(&vSplines);
        pSwSketch->GetSplinesInterpolate(&vSplines);

        CComPtr<IDispatch> pDispatchSafeArray = NULL;
        CComPtr<ISafeArrayUtility> pSwSafeArray = NULL;
        HRESULT hres;
        hres = pSwApp->GetSafeArrayUtility(&pDispatchSafeArray);
        hres = pDispatchSafeArray->QueryInterface<ISafeArrayUtility>(&pSwSafeArray);
```

```

//First value in vSplines array is the number of
//sketch (or spline) points, followed by the x, y, z
//coordinates of each point in the spline, so calculate
//the size of the array
long nbrSketchPoints;
pSwSketch->GetSketchPointsCount2(&nbrSketchPoints);
long splineArraySize = 1 + (nbrSketchPoints * 3);

//Print to the Immediate Window the
//number of spline points followed by the
//x, y, z values of each of point in the spline
for (int idx = 0; idx < splineArraySize; idx++)
{
double sketchPointValue;
pSwSafeArray->GetDouble(vSplines, idx, &sketchPointValue);
_bstr_t s;
s = (double)sketchPointValue;

OutputDebugString(s + "\n");
}

}

//Uninitialize COM
::CoUninitialize();

return(0);
}

```

Get Start and End Points of Spline Example (VBA)

This example shows how to get the start and end points of a spline.

```
'-----
' Preconditions:
' 1. Open a sketch that has a spline.
' 2. Select the spline.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Gets the start and end points of the spline.
' 2. Examine the Immediate window.
'-----
Option Explicit

' Define two types
Type DoubleRec
    dValue As Double
End Type

Type Long2Rec
    iLower As Long
    iUpper As Long
End Type

' Extract two integer values from a single double value
' by assigning a DoubleRec to the double value,
' copying the value to a Long2Rec, and
' extracting the integer values
Function ExtractFields(ByVal dValue As Double, iLower As Long, iUpper As Long)
    Dim dr As DoubleRec
    Dim i2r As Long2Rec
    ' Set the double value
    dr.dValue = dValue
    ' Copy the values
    LSet i2r = dr
    ' Extract the values
    iLower = i2r.iLower
    iUpper = i2r.iUpper
End Function

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swSketchSeg As SldWorks.SketchSegment
    Dim swCurve As SldWorks.Curve
    Dim nStartParam As Double
    Dim nEndParam As Double
    Dim bIsClosed As Boolean
    Dim bIsPeriodic As Boolean
    Dim vStart As Variant
    Dim vEnd As Variant
    Dim nDummy As Long
    Dim nStartSuccess As Long
    Dim nEndSuccess As Long
    Dim i As Long
    Dim bRet As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swSketchSeg = swSelMgr.GetSelectedObject5(1)
    Set swCurve = swSketchSeg.GetCurve
    bRet = swCurve.GetEndParams(nStartParam, nEndParam, bIsClosed, bIsPeriodic)
    vStart = swCurve.Evaluate(nStartParam)
    vEnd = swCurve.Evaluate(nEndParam)
    ExtractFields vStart(6), nStartSuccess, nDummy
    ExtractFields vEnd(6), nEndSuccess, nDummy

    Debug.Print "Length = " & swCurve.GetLength2(nStartParam, nEndParam) * 1000# & " mm"
    Debug.Print "  Start Pt      = (" & vStart(0) * 1000# & ", " & vStart(1) * 1000# & ", " & vStart(2) * 1000# & ") mm"
```

```
Debug.Print "    tangent    = (" & vStart(3) & ", " & vStart(4) & ", " & vStart(5) & ")"
Debug.Print "    End    Pt    = (" & vEnd(0) * 1000# & ", " & vEnd(1) * 1000# & ", " & vEnd(2) * 1000# & ") mm"
Debug.Print "    tangent    = (" & vEnd(3) & ", " & vEnd(4) & ", " & vEnd(5) & ")"
```

```
End Sub
```

Get Structural Member Body Sketch Segments Example (C#)

This example shows how to get the sketch segments that define the path of a body in a structural member.

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Gets the feature definition of Structural Member1.
// 2. Selects body, Structural Member1[1].
// 3. Gets the number of sketch segments that define the path of the
//    selected body.
// 4. Examine the Immediate window.
//
// NOTE: Because the part is used elsewhere, do not save changes.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace GetSketchSegments_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;

        bool boolstatus;
        Feature swFeat;
        StructuralMemberFeatureData featData;
        SelectionMgr selMan;
        Body2 swBody;
        object[] vSeg;

        public void Main()
        {
            Part = (ModelDoc2)swApp.ActiveDoc;
            boolstatus = Part.Extension.SelectByID("Structural Member1", "BODYFEATURE", 0, 0, 0,
false, 0, null, 0);
            selMan = (SelectionMgr)Part.SelectionManager;
            swFeat = (Feature)selMan.GetSelectedObject6(1, -1);
            featData = (StructuralMemberFeatureData)swFeat.GetDefinition();

            Part.ClearSelection2(true);

            // Select a body in Structural Member1
            boolstatus = Part.Extension.SelectByID("Structural Member1[1]", "SOLIDBODY", 0, 0, 0,
false, 0, null, 0);
            swBody = (Body2)selMan.GetSelectedObject6(1, -1);

            // Get the sketch segments that define the path of the selected body
            vSeg = (object[])featData.GetSketchSegments(swBody);
            Debug.Print(" Number of sketch segments: " + vSeg.Length);

        }

        public SldWorks swApp;
    }
}
```


Get Structural Member Body Sketch Segments Example (VBA)

This example shows how to get the sketch segments that define the path of a body in a structural member.

```
'-----  
' Preconditions:  
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the feature definition of Structural Member1.  
' 2. Selects body, Structural Member1[1].  
' 3. Gets the number of sketch segments that define the path of the  
'    selected body.  
' 4. Examine the Immediate window.  
'  
' NOTE: Because the part is used elsewhere, do not save changes.  
'-----  
  
Dim swApp As SldWorks.SldWorks  
Dim Part As SldWorks.ModelDoc2  
Dim boolstatus As Boolean  
  
Dim swFeat As SldWorks.Feature  
Dim featData As SldWorks.StructuralMemberFeatureData  
Dim selMan As SldWorks.SelectionMgr  
Dim swBody As SldWorks.Body2  
Dim vSeg As Variant  
  
Option Explicit  
Sub main()  
    Set swApp = Application.SldWorks  
  
    Set Part = swApp.ActiveDoc  
    boolstatus = Part.Extension.SelectByID2("Structural Member1", "BODYFEATURE", 0, 0, 0, False, 0,  
Nothing, 0)  
    Set selMan = Part.SelectionManager  
    Set swFeat = selMan.GetSelectedObject6(1, -1)  
    Set featData = swFeat.GetDefinition  
  
    Part.ClearSelection2 True  
  
    ' Select a body in Structural Member1  
    boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0, 0, 0, False, 0,  
Nothing, 0)  
    Set swBody = selMan.GetSelectedObject6(1, -1)  
  
    ' Get the sketch segments that define the path of the selected body  
    vSeg = featData.GetSketchSegments(swBody)  
    Debug.Print " Number of sketch segments: " & UBound(vSeg) + 1  
  
End Sub
```

Get Structural Member Body Sketch Segments Example (VB.NET)

This example shows how to get the sketch segments that define the path of a body in a structural member.

```

' -----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Gets the feature definition of Structural Member1.
' 2. Selects body, Structural Member1[1].
' 3. Gets the number of sketch segments that define the path of the
'    selected body.
' 4. Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do not save changes.
' -----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2
    Dim boolstatus As Boolean

    Dim swFeat As Feature
    Dim featData As StructuralMemberFeatureData
    Dim selMan As SelectionMgr
    Dim swBody As Body2
    Dim vSeg As Object

    Sub main()

        Part = swApp.ActiveDoc
        boolstatus = Part.Extension.SelectByID2("Structural Member1", "BODYFEATURE", 0, 0, 0, False,
0, Nothing, 0)
        selMan = Part.SelectionManager
        swFeat = selMan.GetSelectedObject6(1, -1)
        featData = swFeat.GetDefinition

        Part.ClearSelection2(True)

        ' Select a body in Structural Member1
        boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0, 0, 0, False,
0, Nothing, 0)
        swBody = selMan.GetSelectedObject6(1, -1)

        ' Get the sketch segments that define the path of the selected body
        vSeg = featData.GetSketchSegments(swBody)
        Debug.Print(" Number of sketch segments: " & UBound(vSeg) + 1)

    End Sub

    Public swApp As SldWorks

End Class

```

Get Style Spline Curve Type Example (C#)

This example shows how to create a style spline and get its type of curve.

```
//-----
// Preconditions:
// 1. Verify that the specified part template exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens a new part document.
// 2. Creates a style spline.
// 3. Selects the style spline.
// 4. Gets whether the selection is a style spline
//    and, if so, gets its curve type.
// 5. Examine the Immediate window.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SketchManager swSketchManager = default(SketchManager);
            SketchSegment swSketchSegment = default(SketchSegment);
            SketchSpline swSketchSpline = default(SketchSpline);
            SelectionMgr swSelectionMgr = default(SelectionMgr);
            object pointArray = null;
            double[] points = null;
            bool status = false;

            swModel = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SolidWorks\\SOLIDWORKS 2016\\templates\\part.prtdot", 0, 0, 0);

            // Create style spline
            Array.Resize(ref points, 33);
            points[0] = -0.068952134919552;
            points[1] = 0.00871923799128056;
            points[2] = 0;
            points[3] = -0.0563242730011457;
            points[4] = 0.0185409083722633;
            points[5] = 0;
            points[6] = -0.0418924308086813;
            points[7] = 0.00871923799128056;
            points[8] = 0;
            points[9] = -0.0204451097726579;
            points[10] = 0.0243537336997836;
            points[11] = 0;
            points[12] = 0.00621370983286659;
            points[13] = -0.0125276407920698;
            points[14] = 0;
            points[15] = 0.0244539548261202;
            points[16] = -0.00450995068514512;
            points[17] = 0;
            points[18] = 0.0330729716910642;
            points[19] = 0.00631393095920317;
            points[20] = 0;
            points[21] = 0.048306582894221;
            points[22] = 0.0117258717813773;
            points[23] = 0;
            points[24] = 0.05852913778055;
            points[25] = -0.00611348870653004;
            points[26] = 0;
            points[27] = 0.0653441743714359;
            points[28] = -0.0107236605180117;
            points[29] = 0;
            points[30] = -999999999;
            points[31] = -999999999;
            points[32] = -999999969;
            pointArray = points;
            swSketchManager = (SketchManager)swModel.SketchManager;
            swSketchSegment = (SketchSegment)swSketchManager.CreateSpline2((pointArray), true);
            swModel.ClearSelection2(true);
            swSketchManager.InsertSketch(true);
        }
    }
}
```

```

        // Get whether selection is style spline and, if so, get its curve type
        swModelDocExt = (ModelDocExtension)swModel.Extension;
        status = swModelDocExt.SelectByID2("Spline1@Sketch1", "EXTSKETCHSEGMENT", -0.0311890911939585, 0.01229425181
44824, 0, false, 0, null, 0);
        swSelectionMgr = (SelectionMgr)swModel.SelectionManager;
        swSketchSpline = (SketchSpline)swSelectionMgr.GetSelectedObject6(1, -1);
        status = swSketchSpline.IsStyleSpline;
        Debug.Print("Is the selection a style spline? " + status);
        if (status)
        {
            Debug.Print("Style spline curve
type (3 = swStyleSplineCurveType_e.BSpline_Degree7): " + swSketchSpline.CurveType);
        }

    }

    /// <summary>
    /// The SldWorks swApp variable is pre-assigned for you.
    /// </summary>
    public SldWorks swApp;
}

```

Get Style Spline Curve Type Example (VBA)

This example shows how to create a style spline and get its type of curve.

```
'-----  
' Preconditions:  
' 1. Verify that the specified part template exists.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Opens a new part document.  
' 2. Creates a style spline.  
' 3. Selects the style spline.  
' 4. Gets whether the selection is a style spline  
'    and, if so, gets its curve type.  
' 5. Examine the Immediate window.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim swSketchManager As SldWorks.SketchManager  
Dim swSketchSegment As SldWorks.SketchSegment  
Dim swSketchSpline As SldWorks.SketchSpline  
Dim swSelectionMgr As SldWorks.SelectionMgr  
Dim pointArray As Variant  
Dim points() As Double  
Dim status As Boolean  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2016\templates\part.prtdot", 0, 0, 0)  
  
    ' Create style spline  
    ReDim points(0 To 32) As Double  
    points(0) = -0.068952134919552  
    points(1) = 8.71923799128056E-03  
    points(2) = 0  
    points(3) = -5.63242730011457E-02  
    points(4) = 1.85409083722633E-02  
    points(5) = 0  
    points(6) = -4.18924308086813E-02  
    points(7) = 8.71923799128056E-03  
    points(8) = 0  
    points(9) = -2.04451097726579E-02  
    points(10) = 2.43537336997836E-02  
    points(11) = 0  
    points(12) = 6.21370983286659E-03  
    points(13) = -1.25276407920698E-02  
    points(14) = 0  
    points(15) = 2.44539548261202E-02  
    points(16) = -4.50995068514512E-03  
    points(17) = 0  
    points(18) = 3.30729716910642E-02  
    points(19) = 6.31393095920317E-03  
    points(20) = 0  
    points(21) = 0.048306582894221  
    points(22) = 1.17258717813773E-02  
    points(23) = 0  
    points(24) = 0.05852913778055  
    points(25) = -6.11348870653004E-03  
    points(26) = 0  
    points(27) = 6.53441743714359E-02  
    points(28) = -1.07236605180117E-02  
    points(29) = 0  
    points(30) = -999999999  
    points(31) = -999999999  
    points(32) = -999999969  
    pointArray = points  
    Set swSketchManager = swModel.SketchManager  
    Set swSketchSegment = swSketchManager.CreateSpline2((pointArray), True)
```

```

swModel.ClearSelection2 True
swSketchManager.InsertSketch True

' Get whether selection is style spline and, if so, get its curve type
Set swModelDocExt = swModel.Extension
status = swModelDocExt.SelectByID2("Spline1@Sketch1", "EXTSKETCHSEGMENT", -3.11890911939585E-02, 1.22942518144824E-
02, 0, False, 0, Nothing, 0)
Set swSelectionMgr = swModel.SelectionManager
Set swSketchSpline = swSelectionMgr.GetSelectedObject6(1, -1)
status = swSketchSpline.IsStyleSpline
Debug.Print "Is the selection a style spline? " & status
If status Then
    Debug.Print "Style spline curve type (3 = swStyleSplineCurveType_e.BSpline_Degree7): " &
swSketchSpline.CurveType
End If

End Sub

```

Get Style Spline Curve Type Example (VB.NET)

This example shows how to create a style spline and get its type of curve.

```

'-----
' Preconditions:
' 1. Verify that the specified part template exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens a new part document.
' 2. Creates a style spline.
' 3. Selects the style spline.
' 4. Gets whether the selection is a style spline
'    and, if so, gets its curve type.
' 5. Examine the Immediate window.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()
        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swSketchManager As SketchManager
        Dim swSketchSegment As SketchSegment
        Dim swSketchSpline As SketchSpline
        Dim swSelectionMgr As SelectionMgr
        Dim pointArray As Object
        Dim points() As Double
        Dim status As Boolean

        swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2016\templates\part.prtdot", 0, 0, 0)

        ' Create style spline
        ReDim points(0 To 32)
        points(0) = -0.068952134919552
        points(1) = 0.00871923799128056
        points(2) = 0
        points(3) = -0.0563242730011457
        points(4) = 0.0185409083722633
        points(5) = 0
        points(6) = -0.0418924308086813
        points(7) = 0.00871923799128056
        points(8) = 0
        points(9) = -0.0204451097726579
        points(10) = 0.0243537336997836
        points(11) = 0
        points(12) = 0.00621370983286659
        points(13) = -0.0125276407920698
        points(14) = 0
        points(15) = 0.0244539548261202
        points(16) = -0.00450995068514512
        points(17) = 0
        points(18) = 0.0330729716910642
        points(19) = 0.00631393095920317
        points(20) = 0
        points(21) = 0.048306582894221
        points(22) = 0.0117258717813773
        points(23) = 0
        points(24) = 0.05852913778055
        points(25) = -0.00611348870653004
        points(26) = 0
        points(27) = 0.0653441743714359
        points(28) = -0.0107236605180117
        points(29) = 0
        points(30) = -999999999
        points(31) = -999999999
        points(32) = -999999969
        pointArray = points
        swSketchManager = swModel.SketchManager
        swSketchSegment = swSketchManager.CreateSpline2((pointArray), True)
        swModel.ClearSelection2(True)
        swSketchManager.InsertSketch(True)

        ' Get whether selection is style spline and, if so, get its curve type
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SelectByID2("Spline1@Sketch1", "EXTSKETCHSEGMENT", -0.0311890911939585, 0.012294251814482
4, 0, False, 0, Nothing, 0)
        swSelectionMgr = swModel.SelectionManager
    
```

```

        swSketchSpline = swSelectionMgr.GetSelectedObject6(1, -1)
        status = swSketchSpline.IsStyleSpline
        Debug.Print("Is the selection a style spline? " & status)
        If status Then
            Debug.Print("Style spline curve
type (3 = swStyleSplineCurveType_e.BSpline_Degree7): " & swSketchSpline.CurveType)
        End If

    End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Get Surface-finish Image Path and File Name Example (C#)

This example shows how to set and get the path and file name of the image used for an appearance's surface finish.

```
//-----
// Preconditions:
// 1. Verify that:
//    * specified model document exists.
//    * specified materials and surface-finish image files exist.
//    * a ray-trace rendering engine is loaded.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens the specified model document.
// 2. Applies the specified appearance and surface finish to the model.
// 3. Gets the file name of the first appearance and surface finish
//    applied to the model.
// 4. If prompted to use perspective views in rendering, click
//    Continue without Camera or Perspective.
// 5. Renders the model.
// 6. Examine the Final Render and Immediate windows.
//
// NOTES:
// * Rendering can take several minutes to complete.
// * Because the model is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace FileNameCSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SelectionMgr swSelectionMgr = default(SelectionMgr);
            Component2 swComponent = default(Component2);
            Configuration swConfiguration = default(Configuration);
            RenderMaterial swRenderMaterial = default(RenderMaterial);
            RayTraceRenderer swRayTraceRenderer = default(RayTraceRenderer);
            int nbrRenderMaterials = 0;
            string fileName = null;
            int warnings = 0;
            int errors = 0;
            bool status = false;
            string materialName = null;
            string[] displayStateNames = null;
            int materialID1 = 0;
            int materialID2 = 0;

            //Open part document
            fileName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS
2018\\samples\\tutorial\\tolanalyst\\offset\\bushing.sldprt";
            swModel = (ModelDoc2)swApp.OpenDoc6(fileName, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOp
enDocOptions_Silent, "", ref errors, ref warnings);

            //Access the ray-trace rendering engine
            //swRayTraceRenderer = (RayTraceRenderer)swApp.GetRayTraceRenderer();

            //Create material for the appearance
            materialName = "C:\\Program Files\\SolidWorks Corp\\SolidWorks\\data\\graphics\\Materials\\metal\\bronze\\ca
st bronze.p2m";
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            swRenderMaterial = (RenderMaterial)swModelDocExt.CreateRenderMaterial(materialName);

            //Verify that RealView Graphics is set
            Debug.Print("RealView Graphics set? " + swModelDocExt.ViewDisplayRealView);

            //Add an appearance
            status = swModelDocExt.SelectByID2("bushing.SLDPRT", "COMPONENT", 0, 0, 0, false, 0, null, 0);
            swSelectionMgr = (SelectionMgr)swModel.SelectionManager;
            swComponent = (Component2)swSelectionMgr.GetSelectedObject6(1, -1);
            status = swRenderMaterial.AddEntity(swComponent);
            Debug.Print("Appearance added to model? " + status);

            //Add surface finish
            fileName = "C:\\Program Files\\SolidWorks Corp\\SolidWorks\\data\\Images\\textures\\metal\\cast\\cast_bump.j
pg";
        }
    }
}
```

```

swRenderMaterial.BumpTextureFilename = fileName;

swModel.ClearSelection2(true);

//Get the names of display states
//Add the appearance to all of the display states
swConfiguration = (Configuration)swModel.GetActiveConfiguration();
displayStateNames = (String[])swConfiguration.GetDisplayStates();
status = swModelDocExt.AddDisplayStateSpecificRenderMaterial(swRenderMaterial, (int)swDisplayStateOpts_e.swAllDisplayState, displayStateNames, out materialID1, out materialID2);

//Get the number of appearances
nbrRenderMaterials = swModelDocExt.GetRenderMaterialsCount2((int)swDisplayStateOpts_e.swAllDisplayState, displayStateNames);

//If one or more appearances are applied to the model,
//then get the file names of the first appearance and surface finish applied
if (nbrRenderMaterials > 0)
{
    Debug.Print("First appearance's file name: " + swRenderMaterial.FileName);
    Debug.Print("Surface finish's file name: " + swRenderMaterial.BumpTextureFilename);
    Debug.Print("Fixed aspect ratio? " + swRenderMaterial.FixedAspectRatio);
    Debug.Print("Width: " + swRenderMaterial.Width * 1000 + "mm");
    Debug.Print("Height: " + swRenderMaterial.Height * 1000 + "mm");
    if (swRenderMaterial.DoubleSided == false)
    {
        swRenderMaterial.DoubleSided = true;
    }
    Debug.Print("Doubled-sided? " + swRenderMaterial.DoubleSided);
    swRenderMaterial.RoundSharpEdges = 0.005;
    Debug.Print("Round sharp edges value: " + swRenderMaterial.RoundSharpEdges * 1000 + "mm");

    //Display final render window
    status = swRayTraceRenderer.InvokeFinalRender();
}
else
{
    Debug.Print("No appearances applied to this model.");
}
}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
}
}

```

Get Surface-finish Image Path and File Name Example (VBA)

This example shows how to set and get the path and file name of the image used for an appearance's surface finish.

```
'-----
' Preconditions:
' 1. Verify that:
'   * specified model document exists.
'   * specified materials and surface-finish image exist.
'   * a ray-trace rendering engine is loaded.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified model document.
' 2. Applies the specified appearance and surface finish to the model.
' 3. Gets the file names of the first appearance and surface finish
'   applied to the model.
' 4. If prompted to use perspective views in rendering, click
'    Continue without Camera or Perspective.
' 5. Renders the model.
' 6. Examine the Final Render and Immediate windows.
'
' NOTES:
' * Rendering can take several minutes to complete.
' * Because the model is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swComponent As SldWorks.Component2
Dim swConfiguration As SldWorks.Configuration
Dim swRenderMaterial As SldWorks.RenderMaterial
Dim swRayTraceRenderer As SldWorks.RayTraceRenderer
Dim nbrRenderMaterials As Long
Dim fileName As String
Dim warnings As Long
Dim errors As Long
Dim status As Boolean
Dim materialName As String
Dim displayStateNames() As String
Dim materialID1 As Long
Dim materialID2 As Long

Sub main()

    Set swApp = Application.SldWorks
    'Open model document
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\tolanalyst\offset\bushing.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "",
    errors, warnings)

    'Access the ray-trace rendering engine
    'Set swRayTraceRenderer = swApp.GetRayTraceRenderer()

    'Create material for the appearance
    materialName = "C:\Program Files\SolidWorks Corp\SolidWorks\data\graphics\Materials\metal\bronze\cast bronze.p2m"
    Set swModelDocExt = swModel.Extension
    Set swRenderMaterial = swModelDocExt.CreateRenderMaterial(materialName)

    'Verify that RealView Graphics is set
    Debug.Print "RealView Graphics set? " & swModelDocExt.ViewDisplayRealView

    'Add an appearance
    status = swModelDocExt.SelectByID2("bushing.SLDPRT", "COMPONENT", 0, 0, 0, False, 0, Nothing, 0)
    Set swSelectionMgr = swModel.SelectionManager
    Set swComponent = swSelectionMgr.GetSelectedObject6(1, -1)
    status = swRenderMaterial.AddEntity(swComponent)
    Debug.Print "Appearance added to model? " & status

    'Add surface finish
    fileName = "C:\Program Files\SolidWorks Corp\SolidWorks\data\Images\textures\metal\cast\cast_bump.jpg"
```

```

swRenderMaterial.BumpTextureFilename = fileName

swModel.ClearSelection2 True

'Get the names of display states
'Add the appearance to all of the display states
Set swConfiguration = swModel.GetActiveConfiguration
displayStateNames = swConfiguration.GetDisplayStates
status = swModelDocExt.AddDisplayStateSpecificRenderMaterial(swRenderMaterial,
swDisplayStateOpts_e.swAllDisplayState, displayStateNames, materialID1, materialID2)

'Get the number of appearances
nbrRenderMaterials = swModelDocExt.GetRenderMaterialsCount2(swDisplayStateOpts_e.swAllDisplayState,
displayStateNames)

'If one or more appearances are applied to the model,
'then get the file names of the first appearance and surface finish applied
If nbrRenderMaterials > 0 Then
    Debug.Print "First appearance's file name: " & swRenderMaterial.fileName
    Debug.Print " Surface finish's file name: " & swRenderMaterial.BumpTextureFilename
    Debug.Print "Fixed aspect ratio? " & swRenderMaterial.FixedAspectRatio
    Debug.Print " Width: " & swRenderMaterial.Width * 1000# & "mm"
    Debug.Print " Height: " & swRenderMaterial.Height * 1000# & "mm"
    If Not swRenderMaterial.DoubleSided Then
        swRenderMaterial.DoubleSided = True
    End If
    Debug.Print "Double-sided? " & swRenderMaterial.DoubleSided
    swRenderMaterial.RoundSharpEdges = 0.005
    Debug.Print "Round sharp edges value: " & swRenderMaterial.RoundSharpEdges * 1000# & "mm"

    'Display final render window
    status = swRayTraceRenderer.InvokeFinalRender
Else
    Debug.Print "No appearances applied to this model."
End If

End Sub

```

Get_Surface-finish_Image_Path_and_Filename_Example_VBNET (Topic)

SOLIDWORKS API Help

Get Surface-finish Image Path and File Name Example (VB.NET)

This example shows how to set and get the path and file name of the image used for an appearance's surface finish.

```
'-----  
' Preconditions:  
' 1. Verify that:  
'    * specified model document exists.  
'    * specified materials and surface-finish image exist.  
'    * a ray-trace rendering engine is loaded.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Opens the specified model document.  
' 2. Applies the specified appearance and surface finish to the model.  
' 3. Gets the file names of the first appearance and surface finish  
'    applied to the model.  
' 4. If prompted to use perspective views in rendering, click  
'    Continue without Camera or Perspective.  
' 5. Renders the model.  
' 6. Examine the Final Render and Immediate windows.  
'  
' NOTES:  
' * Rendering can take several minutes to complete.  
' * Because the model is used elsewhere, do not save changes.  
'-----
```

```
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System.Runtime.InteropServices  
Imports System  
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Public Sub main()
```

```
        Dim swModel As ModelDoc2  
        Dim swModelDocExt As ModelDocExtension  
        Dim swSelectionMgr As SelectionMgr  
        Dim swComponent As Component2  
        Dim swConfiguration As Configuration  
        Dim swRenderMaterial As RenderMaterial  
        Dim swRayTraceRenderer As RayTraceRenderer  
        Dim nbrRenderMaterials As Integer  
        Dim fileName As String  
        Dim warnings As Integer  
        Dim errors As Integer  
        Dim status As Boolean  
        Dim materialName As String  
        Dim displayStateNames() As String  
        Dim materialID1 As Integer  
        Dim materialID2 As Integer
```

```
        'Open model document  
        fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS  
2018\samples\tutorial\toanalyst\offset\bushing.sldprt"  
        swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "",  
errors, warnings)
```

```
        'Access the ray-trace rendering engine  
        swRayTraceRenderer = swApp.GetRayTraceRenderer()
```

```
        'Create material for the appearance  
        materialName = "C:\Program Files\SolidWorks Corp\SolidWorks\data\graphics\Materials\metal\bronze\cast bronze.p2m"  
    ;
```

```
        swModelDocExt = swModel.Extension  
        swRenderMaterial = swModelDocExt.CreateRenderMaterial(materialName)
```

```
        'Verify that RealView Graphics is set  
        Debug.Print("RealView Graphics set? " & swModelDocExt.ViewDisplayRealView)
```

```
        'Add an appearance  
        status = swModelDocExt.SelectByID2("bushing.SLDPR", "COMPONENT", 0, 0, 0, False, 0, Nothing, 0)  
        swSelectionMgr = swModel.SelectionManager  
        swComponent = swSelectionMgr.GetSelectedObject6(1, -1)  
        status = swRenderMaterial.AddEntity(swComponent)  
        Debug.Print("Appearance added to model? " & status)
```

```
        'Add surface finish  
        fileName = "C:\Program Files\SolidWorks Corp\SolidWorks\data\Images\textures\metal\cast\cast_bump.jpg"  
        swRenderMaterial.BumpTextureFilename = fileName
```

```
        swModel.ClearSelection2(True)
```

```

'Get the names of display states
'Add the appearance to all of the display states
swConfiguration = swModel.GetActiveConfiguration
displayStateNames = swConfiguration.GetDisplayStates
status = swModelDocExt.AddDisplayStateSpecificRenderMaterial(swRenderMaterial, swDisplayStateOpts_e.swAllDisplay
State, displayStateNames, materialID1, materialID2)

'Get the number of appearances
nbrRenderMaterials = swModelDocExt.GetRenderMaterialsCount2(swDisplayStateOpts_e.swAllDisplayState, displayState
Names)

'If one or more appearances are applied to the model,
'then get the file names of the first appearance and surface finish applied
If nbrRenderMaterials > 0 Then
    Debug.Print("First appearance's file name: " & swRenderMaterial.FileName)
    Debug.Print(" Surface finish's file name: " & swRenderMaterial.BumpTextureFilename)
    Debug.Print("Fixed aspect ratio? " & swRenderMaterial.FixedAspectRatio)
    Debug.Print(" Width: " & swRenderMaterial.Width * 1000.0# & "mm")
    Debug.Print(" Height: " & swRenderMaterial.Height * 1000.0# & "mm")
    If Not swRenderMaterial.DoubleSided Then
        swRenderMaterial.DoubleSided = True
    End If
    Debug.Print("Double-sided? " & swRenderMaterial.DoubleSided)
    swRenderMaterial.RoundSharpEdges = 0.005
    Debug.Print("Round sharp edges value: " & swRenderMaterial.RoundSharpEdges * 1000.0# & "mm")

    'Display final render window
    status = swRayTraceRenderer.InvokeFinalRender
Else
    Debug.Print("No appearances applied to this model.")
End If

End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Get Surface Trim Feature Data Example (C#)

This example shows how to get surface trim feature data.

```
//-----  
// Preconditions:  
// 1. Open a part document with a surface trim feature.  
// 2. Select the surface trim feature.  
// 3. Open the Immediate window.  
//  
// Postconditions:  
// 1. Gets the type of surface trim feature.  
// 2. Examine the Immediate window.  
//-----  
  
using Microsoft.VisualBasic;  
  
using System;  
  
using System.Collections;  
  
using System.Collections.Generic;  
  
using System.Data;  
  
using System.Diagnostics;  
  
using SolidWorks.Interop.sldworks;  
  
using SolidWorks.Interop.swconst;  
  
using System.Runtime.InteropServices;  
  
namespace SurfaceTrimFeatureData_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
  
        ModelDoc2 swModel;  
  
        SelectionMgr swSelMgr;  
  
        Feature swFeat;  
  
        SurfaceTrimFeatureData swSurfTrimFeat;  
  
        int surftrimtype;  
  
        public void Main()  
        {  
            swModel = (ModelDoc2)swApp.ActiveDoc;  
  
            swSelMgr = (SelectionMgr)swModel.SelectionManager;  
  
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1,  
-1);  
  
            swSurfTrimFeat = (SurfaceTrimFeatureData)swFeat.GetDefinition();  
  
            surftrimtype = swSurfTrimFeat.GetType();  
  
            Debug.Print("Surface trim type (swSurfaceTrimType_e): " + surftrimtype);  
        }  
    }  
}
```

```
}
```

```
public SldWorks swApp;
```

```
}
```

```
}
```

Get Surface Trim Feature Data Example (VBA)

This example shows how to get surface trim feature data.

```
'-----  
' Preconditions:  
' 1. Open a part document with a surface trim feature.  
' 2. Select the surface trim feature.  
' 3. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the type of surface trim feature.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swFeat As SldWorks.Feature  
Dim swSurfTrimFeat As SldWorks.SurfaceTrimFeatureData  
Dim surftrimtype As Long  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swSurfTrimFeat = swFeat.GetDefinition  
  
    surftrimtype = swSurfTrimFeat.GetType  
    Debug.Print "Surface trim type (swSurfaceTrimType_e): " & surftrimtype  
End Sub
```

Get Surface Trim Feature Data Example (VB.NET)

This example shows how to get surface trim feature data.

```
' .....  
' Preconditions:  
' 1. Open a part document with a surface trim feature.  
' 2. Select the surface trim feature.  
' 3. Open the Immediate window.  
' .....  
' Postconditions:  
' 1. Gets the type of surface trim feature.  
' 2. Examine the Immediate window.  
' .....
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System.Runtime.InteropServices
```

```
Imports System
```

```
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2
```

```
    Dim swSelMgr As SelectionMgr
```

```
    Dim swFeat As Feature
```

```
    Dim swSurfTrimFeat As SurfaceTrimFeatureData
```

```
    Dim surftrimtype As Integer
```

```
    Sub main()
```

```
        swModel = swApp.ActiveDoc
```

```
        swSelMgr = swModel.SelectionManager
```

```
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
```

```
        swSurfTrimFeat = swFeat.GetDefinition
```

```
        surftrimtype = swSurfTrimFeat.GetType
```

```
        Debug.Print("Surface trim type (swSurfaceTrimType_e):  
" & surftrimtype)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Surface Type Example (VBA)

This example shows how to get the type of surface.

```
'-----  
' Preconditions:  
' 1. Open a part.  
' 2. Select a face on the part.  
'  
' Postconditions: Examine the Immediate window.  
'-----  
Option Explicit  
  
Public Enum swSurfaceTypes_e  
    PLANE_TYPE = 4001  
    CYLINDER_TYPE = 4002  
    CONE_TYPE = 4003  
    SPHERE_TYPE = 4004  
    TORUS_TYPE = 4005  
    BSURF_TYPE = 4006  
    BLEND_TYPE = 4007  
    OFFSET_TYPE = 4008  
    EXTRU_TYPE = 4009  
    SREV_TYPE = 4010  
End Enum  
  
Sub main()  
  
    Dim swSurfaceTypes(10) As String  
    swSurfaceTypes(1) = "PLANE"  
    swSurfaceTypes(2) = "CYLINDER"  
    swSurfaceTypes(3) = "CONE"  
    swSurfaceTypes(4) = "SPHERE"  
    swSurfaceTypes(5) = "TORUS"  
    swSurfaceTypes(6) = "BSURF"  
    swSurfaceTypes(7) = "BLEND"  
    swSurfaceTypes(8) = "OFFSET"  
    swSurfaceTypes(9) = "EXTRU"  
    swSurfaceTypes(10) = "SREV"  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swFace As SldWorks.Face2  
    Dim swSurf As SldWorks.Surface  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFace = swSelMgr.GetSelectedObject5(1)  
    Set swSurf = swFace.GetSurface  
    Debug.Print "File = " & swModel.GetPathName  
    Debug.Print " Identity = " & swSurfaceTypes(swSurf.Identity - 4000)  
  
End Sub
```

Get Table-driven Pattern Properties Example (VBA)

This example shows how to get the properties of a table-driven pattern.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\tablepattern.sldprt.
' 2. Select TPattern1.
' 3. Open the Immediate window.
'
' Postconditions: Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do not save
' changes.
'-----
```

Option Explicit

```
' TablePatternFeatureData::GetReferencePointType
'      0 = closed curve
'      1 = sketch point
'      2 = vertex
Sub ProcessCoordSys(swApp As SldWorks.SldWorks, swModel As SldWorks.ModelDoc2, swCoordSys As SldWorks.Feature)

    Dim swDocExt As SldWorks.ModelDocExtension
    Dim swXform As SldWorks.MathTransform

    Set swDocExt = swModel.Extension
    Set swXform = swDocExt.GetCoordinateSystemTransformByName(swCoordSys.Name)

    Debug.Print " " & swCoordSys.Name
    Debug.Print "      Origin = (" & -1# * swXform.ArrayData(9) * 1000# & ", " & -1# * swXform.ArrayData(10) * 1000# & ", " & -1# * swXform.ArrayData(11) * 1000# & ") mm"
    Debug.Print "      Rot1  = (" & swXform.ArrayData(0) & ", " & swXform.ArrayData(1) & ", " & swXform.ArrayData(2) &
    ") "
    Debug.Print "      Rot2  = (" & swXform.ArrayData(3) & ", " & swXform.ArrayData(4) & ", " & swXform.ArrayData(5) &
    ") "
    Debug.Print "      Rot3  = (" & swXform.ArrayData(6) & ", " & swXform.ArrayData(7) & ", " & swXform.ArrayData(8) &
    ") "
    Debug.Print "      Trans = (" & swXform.ArrayData(9) * 1000# & ", " & swXform.ArrayData(10) * 1000# & ", " &
    swXform.ArrayData(11) * 1000# & ") mm"
    Debug.Print "      Scale = " & swXform.ArrayData(12)

End Sub

Sub main()
```

```
    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swFeat As SldWorks.Feature
    Dim swTableFeatData As SldWorks.TablePatternFeatureData
    Dim swRefPt As SldWorks.Vertex
    Dim swCoordSys As SldWorks.Feature 'Object
    Dim vBasePt As Variant
    Dim vFace As Variant
    Dim vFeat As Variant
    Dim vPt As Variant
    Dim nPtType As Long
    Dim vRefPtParam As Variant
    Dim bRet As Boolean
    Dim i As Long

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject5(1)
    Set swTableFeatData = swFeat.GetDefinition

    ' Roll back to get the reference point
    bRet = swTableFeatData.AccessSelections(swModel, Nothing)
    Set swRefPt = swTableFeatData.ReferencePoint
    Set swCoordSys = swTableFeatData.CoordinateSystem
    vBasePt = swTableFeatData.GetBasePoint
```

```

vFace = swTableFeatData.PatternFaceArray
vFeat = swTableFeatData.PatternFeatureArray
vPt = swTableFeatData.PointArray

Debug.Print swFeat.Name
Debug.Print "   Coord sys           = " & swCoordSys.Name

ProcessCoordSys swApp, swModel, swCoordSys

Debug.Print "   FaceCount           = " & swTableFeatData.GetPatternFaceCount
Debug.Print "   FeatureCount        = " & swTableFeatData.GetPatternFeatureCount
Debug.Print "   PointCount           = " & swTableFeatData.GetPointCount
Debug.Print "   ReferencePointType      = " & swTableFeatData.GetReferencePointType
Debug.Print "   BasePt                  = (" & vBasePt(0) * 1000# & ", " & vBasePt(1) * 1000# & ", " & vBasePt(2) *
1000# & ") mm"
Debug.Print "   GeometryPattern        = " & swTableFeatData.GeometryPattern
Debug.Print "   UseCentroid            = " & swTableFeatData.UseCentroid
If Not swRefPt Is Nothing Then
    ' Is null if centroid used
    vRefPtParam = swRefPt.GetPoint
    Debug.Print "   RefPt                  = (" & vRefPtParam(0) * 1000# & ", " & vRefPtParam(1) * 1000# & ", " &
vRefPtParam(2) * 1000# & ") mm"
End If
    ' Roll forward
    swTableFeatData.ReleaseSelectionAccess
End Sub

```

Get Table Anchor Example (C#)

This example shows how to get the anchor for a selected bill of materials table.

```
//-----  
// Preconditions:  
// 1. Open a drawing document that has a bill of materials.  
// 2. Expand the Sheet Format node in the FeatureManager design tree and  
//    select the bill of materials anchor feature.  
//  
// Postconditions: Inspect the Immediate window.  
//-----  
  
using Microsoft.VisualBasic;  
using System;  
using System.Collections;  
using System.Collections.Generic;  
using System.Data;  
using System.Diagnostics;  
using SolidWorks.Interop.sldworks;  
using SolidWorks.Interop.swconst;  
using System.Runtime.InteropServices;  
namespace TableAnchor_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
  
        ModelDoc2 swModel;  
        Feature swFeat;  
        SelectionMgr swSelMgr;  
        TableAnchor swTableAnchor;  
  
        public void Main()  
        {  
            swModel = (ModelDoc2)swApp.ActiveDoc;  
            swSelMgr = (SelectionMgr)swModel.SelectionManager;  
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, 0);  
            swTableAnchor = (TableAnchor)swFeat.GetSpecificFeature2();  
  
            // Type of table anchor as defined in swTableAnnotationType_e  
            Debug.Print("Type of table anchor = " + swTableAnchor.Type);  
        }  
  
        public SldWorks swApp;  
    }  
}
```

Get Table Anchor Example (VBA)

This example shows how to get the anchor for a selected bill of materials table.

```
'-----  
' Preconditions:  
' 1. Open a drawing document that has a bill of materials.  
' 2. Expand the Sheet Format node in the FeatureManager design tree and  
'    select the bill of materials anchor feature.  
'  
' Postconditions: Inspect the Immediate window.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swFeat As SldWorks.Feature  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swTableAnchor As SldWorks.TableAnchor  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject6(1, 0)  
    Set swTableAnchor = swFeat.GetSpecificFeature2  
  
    ' Type of table anchor as defined in swTableAnnotationType_e  
    Debug.Print "Type of table anchor = " & swTableAnchor.Type  
  
End Sub
```

Get Table Anchor Example (VB.NET)

This example shows how to get the anchor for a selected bill of materials table.

```
'-----  
' Preconditions:  
' 1. Open a drawing document that has a bill of materials.  
' 2. Expand the Sheet Format node in the FeatureManager design tree and  
'    select the bill of materials anchor feature.  
'  
' Postconditions: Inspect the Immediate window.  
'-----
```

```
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System.Runtime.InteropServices  
Imports System  
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2  
    Dim swFeat As Feature  
    Dim swSelMgr As SelectionMgr  
    Dim swTableAnchor As TableAnchor
```

```
    Sub main()
```

```
        swModel = swApp.ActiveDoc  
        swSelMgr = swModel.SelectionManager  
        swFeat = swSelMgr.GetSelectedObject6(1, 0)  
        swTableAnchor = swFeat.GetSpecificFeature2
```

```
        ' Type of table anchor as defined in swTableAnnotationType_e  
        Debug.Print("Type of table anchor = " & swTableAnchor.Type)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Table Annotation and Contents Example (VBA)

This example shows how to get the table annotation and its contents.

```
'-----  
' Preconditions:  
' 1. Open public_documents\samples\tutorial\api\2012-sm.slddrw.  
' 2. Open the Immediate window.  
'  
' Postconditions: Examine the Immediate window.  
'-----
```

Option Explicit

```
Sub ProcessTable(swApp As SldWorks.SldWorks, swModel As SldWorks.ModelDoc2, swTable As SldWorks.TableAnnotation)
```

```
    Dim swAnn As SldWorks.Annotation  
    Dim nNumCol As Long  
    Dim nNumRow As Long  
    Dim sRowStr As String  
    Dim i As Long  
    Dim j As Long
```

```
    Set swAnn = swTable.GetAnnotation  
    nNumCol = swTable.ColumnCount  
    nNumRow = swTable.RowCount
```

```
    ' Show the name and type of table  
    Debug.Print "      " & swAnn.GetName & " <" & swTable.Type & ">"
```

```
    ' Get the table contents  
    For i = 0 To nNumRow - 1  
        sRowStr = "      ["  
        For j = 0 To nNumCol - 1  
            sRowStr = sRowStr & swTable.DisplayedText2(i, j, True) & ","  
        Next j  
        ' Show the table contents  
        Debug.Print Left(sRowStr, Len(sRowStr) - 1) & "]"  
    Next i
```

```
End Sub
```

```
Sub main()
```

```
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swDraw As SldWorks.DrawingDoc  
    Dim swView As SldWorks.View  
    Dim swTable As SldWorks.TableAnnotation  
    Dim bRet As Boolean
```

```
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel
```

```
    Debug.Print "File = " & swModel.GetPathName
```

```
    ' Get the first view  
    Set swView = swDraw.GetFirstView  
    Do While Not swView Is Nothing  
        ' Show the name of the view  
        Debug.Print "    " & swView.Name  
        ' Get the first table annotation for this view  
        Set swTable = swView.GetFirstTableAnnotation2  
        Do While Not swTable Is Nothing  
            ProcessTable swApp, swModel, swTable  
            ' Get next table annotation for this view  
            Set swTable = swTable.GetNext  
        Loop  
        ' Get the next view  
        Set swView = swView.GetNextView  
    Loop
```

```
End Sub
```

Get Tangent Faces Example (VBA)

This examples shows how to get faces tangent to the selected face.

```
'-----  
' Preconditions:  
' 1. Open a part or assembly.  
' 2. Select a face tangent to other faces.  
' 3. Open the Immediate window.  
,  
' Postconditions:  
' 1. Gets all tangent faces to the selected face.  
' 2. Examine the Immediate window and graphics area.  
'-----  
Option Explicit  
  
Sub ProcessSelections(swApp As SldWorks.SldWorks, swModel As SldWorks.ModelDoc2)  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim nSelCount As Long  
    Dim i As Long  
  
    Set swSelMgr = swModel.SelectionManager  
    nSelCount = swSelMgr.GetSelectedObjectCount  
    For i = 1 To nSelCount  
        Debug.Print "    Selected type of tangent entity(" & i & ") (2 = face) = " & swSelMgr.GetSelectedObjectType2(i)  
    Next i  
End Sub  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim bRe As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Debug.Print "Selected type of entity before calling IModelDoc2::SelectTangency (2 = face):"  
    ProcessSelections swApp, swModel  
    swModel.SelectTangency  
    Debug.Print "Selected type of tangent entities after calling IModelDoc2::SelectTangency (2 = face):"  
    ProcessSelections swApp, swModel  
  
End Sub
```

Get Tangent Edges of Bendlines Example (C#)

This example shows how to get each bendline's visible tangent edges in a flat-pattern view in a drawing of a sheet metal part.

```
//-----  
// Preconditions:  
// 1. Open a SOLIDWORKS Design drawing of a sheet metal part  
//    that has a flat-pattern view with  
//    bend lines with tangent edges.  
// 2. Open the Immediate window.  
//  
// Preconditions: Examine the Immediate window.  
//-----  
using SolidWorks.Interop.sldworks;  
  
using SolidWorks.Interop.swconst;  
using System;  
using System.Diagnostics;  
namespace GetBendlinesTangentEdges_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
        public void Main()  
        {  
            ModelDoc2 swModel = default(ModelDoc2);  
            DrawingDoc swDrawing = default(DrawingDoc);  
            Sheet swSheet = default(Sheet);  
            View swView = default(View);  
            long nbrBendlines = 0;  
            object[] BendlinesArr = null;  
            long nbrRelatedTangentEdges = 0;  
            object RelatedTangentEdgesArr = null;  
            long i = 0;  
            SketchSegment swSketchSegment = default(SketchSegment);  
  
            swModel = (ModelDoc2)swApp.ActiveDoc;  
            swDrawing = (DrawingDoc)swModel;  
  
            // Get drawing sheet  
            swSheet = (Sheet)swDrawing.GetCurrentSheet();  
  
            // Get name of drawing sheet  
            Debug.Print("Name of drawing sheet: " + swSheet.GetName());  
  
            // Get number of views on drawing sheet  
            Debug.Print(" Number of drawing views on drawing sheet: " + swDrawing.GetViewCount());  
  
            // First view is the current drawing sheet  
            swView = (View)swDrawing.GetFirstView();  
            Debug.Print(" First drawing view is the current drawing sheet, so...");  
  
            // Get first drawing view on drawing sheet
```

```

        swView = (View)swView.GetNextView();
        while ((swView != null))
        {
            Debug.Print(" Get next drawing view on drawing sheet...");

            // Get first true drawing view on current drawing sheet
            Debug.Print(" Is this drawing view a flat-pattern view? " +
swView.IsFlatPatternView());

            // If this drawing view is a flat pattern view, then
            // get its bendlines and their related tangent edges
            if (swView.IsFlatPatternView())
            {
                // Get number of bendlines in the drawing view
                nbrBendlines = swView.GetBendLineCount();
                BendlinesArr = new Object[nbrBendlines];
                Debug.Print(" Number of bendlines in this drawing view: " + nbrBendlines);
                if ((nbrBendlines > 0))
                {
                    BendlinesArr = (Object[])swView.GetBendLines();
                    for (i = 0; i <= BendlinesArr.GetUpperBound(0); i++)
                    {
                        swSketchSegment = (SketchSegment)BendlinesArr[i];
                        Debug.Print(" Is bend line " + i + " really a bend line? " +
swSketchSegment.IsBendLine());
                        // Get the number of tangent lines related to
                        // the bendline
                        nbrRelatedTangentEdges =
swView.GetRelatedTangentEdgeCount(BendlinesArr[i]);
                        Debug.Print(" Number of tangent edges for Bendline " + i + ": " +
nbrRelatedTangentEdges);
                        // Get the tangent lines related to the bendline
                        if ((nbrRelatedTangentEdges > 0))
                        {
                            RelatedTangentEdgesArr =
swView.GetRelatedTangentEdges(BendlinesArr[i]);
                            nbrRelatedTangentEdges = nbrRelatedTangentEdges - 1;
                        }
                    }
                }
            }
            // Get next drawing view
            swView = (View)swView.GetNextView();
        }
    }
    public SldWorks swApp;
}
}

```

Get Tangent Edges of Bendlines Example (VBA)

This example shows how to get each bendline's visible tangent edges in a flat-pattern view in a drawing of a sheet metal part.

```
'-----
' Preconditions:
' 1. Open a SOLIDWORKS Design drawing of a sheet metal part
'    that has a flat-pattern view with
'    bend lines with tangent edges.
' 2. Open the Immediate window.
'
' Preconditions: Examine the Immediate window.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swDrawing As SldWorks.DrawingDoc
Dim swSheet As SldWorks.Sheet
Dim swView As SldWorks.View
Dim nbrBendlines As Long
Dim BendlinesArr As Variant
Dim nbrRelatedTangentEdges As Long
Dim RelatedTangentEdgesArr As Variant
Dim i As Long
Dim swSketchSegment As SldWorks.SketchSegment

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swDrawing = swModel

    ' Get drawing sheet
    Set swSheet = swDrawing.GetCurrentSheet

    ' Get name of drawing sheet
    Debug.Print "Name of drawing sheet: " & swSheet.GetName

    ' Get number of views on drawing sheet
    Debug.Print "    Number of drawing views on drawing sheet: " & swDrawing.GetViewCount

    ' First view is the current drawing sheet
    Set swView = swDrawing.GetFirstView
    Debug.Print "    First drawing view is the current drawing sheet, so..."

    ' Get first drawing view on drawing sheet
    Set swView = swView.GetNextView

    While Not swView Is Nothing
        Debug.Print "        Get next drawing view on drawing sheet..."

        ' Get first true drawing view on current drawing sheet
        Debug.Print "            Is this drawing view a flat-pattern view? " & swView.IsFlatPatternView

        ' If this drawing view is a flat pattern view, then
        ' get its bendlines and their related tangent edges
        If swView.IsFlatPatternView Then

            ' Get number of bendlines in the drawing view
            nbrBendlines = swView.GetBendLineCount
            Debug.Print "                Number of bendlines in this drawing view: " & nbrBendlines
            If (nbrBendlines > 0) Then
                BendlinesArr = swView.GetBendLines
                For i = 0 To UBound(BendlinesArr)
                    Set swSketchSegment = BendlinesArr(i)
                    Debug.Print "                    Is bend line " & i & " really a bend line? " & swSketchSegment.IsBendLine

                    ' Get the number of tangent lines related to
                    ' the bendline
                    nbrRelatedTangentEdges = swView.GetRelatedTangentEdgeCount(BendlinesArr(i))
                    Debug.Print "                        Number of tangent edges for Bendline " & i & ": " &
nbrRelatedTangentEdges
                Next i
            End If
        End If

        ' Get next drawing view on drawing sheet
        Set swView = swView.GetNextView
    End While
End Sub
```

```
        ' Get the tangent lines related to the bendline
        If (nbrRelatedTangentEdges > 0) Then
            RelatedTangentEdgesArr = swView.GetRelatedTangentEdges(BendlinesArr(i))
            nbrRelatedTangentEdges = nbrRelatedTangentEdges - 1
        End If
    Next i
End If
' Get next drawing view
Set swView = swView.GetNextView
Wend
End Sub
```

Get Tangent Edges of Bendlines Example (VB.NET)

This example shows how to get each bendline's visible tangent edges in a flat-pattern view in a drawing of a sheet metal part.

```
'-----  
' Preconditions:  
' 1. Open a SOLIDWORKS Design drawing of a sheet metal part  
'    that has a flat-pattern view with  
'    bend lines with tangent edges.  
' 2. Open the Immediate window.  
'  
' Preconditions: Examine the Immediate window.  
'-----  
Imports SolidWorks.Interop.sldworks  
  
Imports SolidWorks.Interop.swconst  
Imports System  
Imports System.Diagnostics  
  
Partial Class SolidWorksMacro  
  
    Public Sub main()  
  
        Dim swModel As ModelDoc2  
        Dim swDrawing As DrawingDoc  
        Dim swSheet As Sheet  
        Dim swView As View  
        Dim nbrBendlines As Long  
        Dim BendlinesArr As Object  
        Dim nbrRelatedTangentEdges As Long  
        Dim RelatedTangentEdgesArr As Object  
        Dim i As Long  
        Dim swSketchSegment as SketchSegment  
  
        swModel = swApp.ActiveDoc  
        swDrawing = swModel  
  
        ' Get drawing sheet  
        swSheet = swDrawing.GetCurrentSheet  
        ' Get name of drawing sheet  
        Debug.Print("Name of drawing sheet: " & swSheet.GetName)  
        ' Get number of views on drawing sheet  
        Debug.Print("  Number of drawing views on drawing sheet: " & swDrawing.GetViewCount)  
        ' First view is the current drawing sheet  
        swView = swDrawing.GetFirstView  
        Debug.Print("    First drawing view is the current drawing sheet, so...")  
        ' Get first drawing view on drawing sheet  
        swView = swView.GetNextView  
        While Not swView Is Nothing  
            Debug.Print("          Get next drawing view on drawing sheet...")
```

```

        ' Get first true drawing view on current drawing sheet
        Debug.Print("                Is this drawing view a flat-pattern view? " &
swView.IsFlatPatternView)
        ' If this drawing view is a flat pattern view, then
        ' get its bendlines and their related tangent edges
        If swView.IsFlatPatternView Then
            ' Get number of bendlines in the drawing view
            nbrBendlines = swView.GetBendLineCount
            Debug.Print("                Number of bendlines in this drawing view: " & nbrBendlines)
            If (nbrBendlines > 0) Then
                BendlinesArr = swView.GetBendLines
                For i = 0 To UBound(BendlinesArr)
                    swSketchSegment = BendlinesArr(i)
                    Debug.Print("                Is bend line " & i & " really a bend line? " &
swSketchSegment.IsBendLine)
                        ' Get the number of tangent lines related to
                        ' the bendline
                        nbrRelatedTangentEdges = swView.GetRelatedTangentEdgeCount(BendlinesArr(i))
                        Debug.Print("                Number of tangent edges for Bendline " & i & ": " &
nbrRelatedTangentEdges)
                            ' Get the tangent lines related to the bendline
                            If (nbrRelatedTangentEdges > 0) Then
                                RelatedTangentEdgesArr = swView.GetRelatedTangentEdges(BendlinesArr(i))
                                nbrRelatedTangentEdges = nbrRelatedTangentEdges - 1
                            End If
                        Next i
                    End If
                End If
                ' Get next drawing view
                swView = swView.GetNextView
            End While

        End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Get Sheet Metal Feature Data Example (C#)

This example shows how to get the sheet metal feature data from a sheet metal part created in SOLIDWORKS 2013 or later.

```
//-----
// Preconditions:
// 1. Open a sheet metal part created in SOLIDWORKS 2013 or later.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Inspect the Immediate window to see whether a fixed face reference is selected.
// 2. Inspect the selection in the graphics window.
// 3. Press F5 to finish the macro.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace GetTemplateSheetMetal_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelExt = default(ModelDocExtension);
            SelectData swSelData = null;
            Feature swSheetMetalTemplFeature = default(Feature);
            SheetMetalFeatureData swSheetMetal = default(SheetMetalFeatureData);
            Face2 swFixedFace = default(Face2);
            Entity swEntity = default(Entity);
            bool bRet = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swModelExt = swModel.Extension;

            swSheetMetalTemplFeature = (Feature)swModelExt.GetTemplateSheetMetal();
            swSheetMetal = (SheetMetalFeatureData)swSheetMetalTemplFeature.GetDefinition();

            // Roll back the model
            bRet = swSheetMetal.AccessSelections(swModel, null);
            swFixedFace = (Face2)swSheetMetal.FixedReference;

            if ((swFixedFace != null))
            {
                Debug.Print("A fixed face or fixed edge is selected.");
                swEntity = (Entity)swFixedFace;
                bRet = swEntity.Select4(false, swSelData);
                System.Diagnostics.Debugger.Break();
            }
            else
            {
                // A fixed face or fixed edge is not required,
                // so Null is a valid value
                Debug.Print("There is no fixed face or fixed edge in this sheet metal part.");
            }

            // Cancel changes and roll forward the model
            swSheetMetal.ReleaseSelectionAccess();
        }

        public SldWorks swApp;
```

} }